
ALGOQUEST

Complete Source Code Directory & Documentation

Project	Algoquest Game
Purpose	Innovation Challenge
Date	June 2026
Total Files	36 Source Files

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```
1  const express = require('express');
2  const cors = require('cors');
3  const { Pool } = require('pg');
4  const bcrypt = require('bcrypt');
5  const jwt = require('jsonwebtoken');
6
7  const app = express();
8
9  // 1. Use the Port Render assigns, or 5000 if local
10 const port = process.env.PORT || 5000;
11 const JWT_SECRET = process.env.JWT_SECRET || "a_super_secret_key_that_should_be_in_an_env_file";
12
13 // 2. Configure Database Connection (Smart Switch)
14 const isProduction = process.env.NODE_ENV === 'production' || process.env.DATABASE_URL;
15
16 const pool = new Pool(
17   isProduction
18   ? {
19     connectionString: process.env.DATABASE_URL,
20     ssl: {
21       rejectUnauthorized: false,
22     },
23   }
24   : {
25     user: 'postgres',
26     host: 'localhost',
27     database: 'algoquest_db',
28     password: '61212611414142002',
29     port: 5432,
30   }
31 );
32
33 app.use(cors());
34 app.use(express.json());
35
36 // --- Authentication Middleware ---
37 const authenticateToken = (req, res, next) => {
38   const authHeader = req.headers['authorization'];
39   const token = authHeader && authHeader.split(' ')[1];
40   if (token == null) return res.sendStatus(401);
41
42   jwt.verify(token, JWT_SECRET, (err, user) => {
43     if (err) return res.sendStatus(403);
44     req.user = user;
45     next();
46   });
47 };
48
49 // --- Auth Routes ---
50 app.post('/api/register', async (req, res) => {
51   const { name, email, password } = req.body;
52   try {
53     const salt = await bcrypt.genSalt(10);
54     const hashedPassword = await bcrypt.hash(password, salt);
55     const newUser = await pool.query(
56       "INSERT INTO users (name, email, password) VALUES ($1, $2, $3) RETURNING *",
57       [name, email, hashedPassword]
58     );
59     res.json(newUser.rows[0]);
60   } catch (err) {
61     console.error(err.message);
62     res.status(500).send("Server error");
63   }
64 });
65
66 app.post('/api/login', async (req, res) => {
67   const { email, password } = req.body;
68   try {
```



```

69     const user = await pool.query("SELECT * FROM users WHERE email = $1", [email]);
70     if (user.rows.length === 0) return res.status(401).json("Invalid credential");
71     const validPassword = await bcrypt.compare(password, user.rows[0].password);
72     if (!validPassword) return res.status(401).json("Invalid credential");
73
74     const token = jwt.sign({ id: user.rows[0].id }, JWT_SECRET);
75     res.json({ status: 'ok', token });
76   } catch (err) {
77     console.error(err.message);
78     res.status(500).send("Server error");
79   }
80 });
81
82 // --- Get User Info ---
83 app.get('/api/user/me', authenticateToken, async (req, res) => {
84   try {
85     const user = await pool.query("SELECT id, name, email FROM users WHERE id = $1", [
86       req.user.id,
87     ]);
88     if (user.rows.length === 0) {
89       return res.status(404).json("User not found");
90     }
91     res.json(user.rows[0]);
92   } catch (err) {
93     console.error(err.message);
94     res.status(500).send("Server error");
95   }
96 });
97
98 // --- PROGRESS API FOR HUB ---
99 app.get('/api/progress', authenticateToken, async (req, res) => {
100   try {
101     // 1. Total score grouping
102     const scoresQuery = await pool.query(
103       `SELECT mission, SUM(score) as total_score
104        FROM mission_scores
105        WHERE user_id = $1
106        GROUP BY mission`,
107       [req.user.id]
108     );
109
110     // 2. Pull badges per mission
111     const badgeQuery = await pool.query(
112       `SELECT mission, badge FROM user_badges WHERE user_id = $1`,
113       [req.user.id]
114     );
115
116     // Group badges by mission
117     const missionBadges = {};
118     badgeQuery.rows.forEach(row => {
119       const missionKey = `mission${row.mission}`;
120       if (!missionBadges[missionKey]) missionBadges[missionKey] = [];
121       missionBadges[missionKey].push(row.badge);
122     });
123
124     // 3. Build formatted response
125     const formattedData = scoresQuery.rows.map(row => {
126       const missionKey = `mission${row.mission}`;
127       return {
128         mission_id: missionKey,
129         score: parseInt(row.total_score),
130         badges: missionBadges[missionKey] || []
131       };
132     });
133
134     res.json(formattedData);
135   } catch (err) {
136     console.error(err.message);
137     res.status(500).send("Server error");
138   }
139 });
140
141 // --- MISSION SUBMISSION LOGIC ---
142 // --- NEW: Submit Attempt (Trust Frontend Logic) ---
143 app.post("/api/mission/submit", authenticateToken, async (req, res) => {
144   try {
145     const { mission, phase, isCorrect, score, badge } = req.body;

```

```

146     const userId = req.user.id;
147
148     // 1. Log the attempt (for history)
149     const attemptCountResult = await pool.query(
150       `SELECT COUNT(*) AS count FROM mission_attempts WHERE user_id=$1 AND mission=$2 AND phase=$3`,
151       [userId, mission, phase]
152     );
153     const attemptNumber = Number(attemptCountResult.rows[0].count) + 1;
154
155     // Use score from frontend or default to 0
156     const finalScore = score !== undefined ? score : 0;
157
158     await pool.query(
159       `INSERT INTO mission_attempts
160       (user_id, mission, phase, attempt_number, correct, score)
161       VALUES ($1, $2, $3, $4, $5, $6)`,
162       [userId, mission, phase, attemptNumber, isCorrect, finalScore]
163     );
164
165     // 2. Update Best Score (Keep Highest)
166     if (isCorrect) {
167       await pool.query(
168         `INSERT INTO mission_scores (user_id, mission, phase, score)
169         VALUES ($1, $2, $3, $4)
170         ON CONFLICT (user_id, mission, phase)
171         DO UPDATE SET score = GREATEST(mission_scores.score, EXCLUDED.score),
172         updated_at = now()`,
173         [userId, mission, phase, finalScore]
174       );
175     }
176
177     // 3. Award Badge (TRUST THE FRONTEND)
178     // If the frontend sends a badge string (e.g., "Master Algoritma"), save it.
179     // We append the mission ID to make it unique per mission if needed,
180     // or just save it as is if your frontend handles unique names.
181     if (badge) {
182       // Optional: You can prefix it if you want easier filtering later, e.g. `${mission}-${badge}`
183       // For now, we save exactly what the frontend sends.
184       await pool.query(
185         `INSERT INTO user_badges (user_id, mission, badge)
186         VALUES ($1, $2, $3)
187         ON CONFLICT (user_id, badge) DO NOTHING`,
188         [userId, mission, badge]
189       );
190     }
191
192     res.json({ status: "ok", message: "Saved", attemptNumber });
193
194   } catch (err) {
195     console.error("MISSION SUBMIT ERROR:", err);
196     res.status(500).json({ error: "Server error" });
197   }
198 });
199
200 app.listen(port, () => {
201   console.log(`Server is running on port ${port}`);
202 });

```

backend\package.json

19 lines

0.36 KB

JSON

```
1  {
2    "name": "backend",
3    "version": "1.0.0",
4    "description": "",
5    "main": "index.js",
6    "scripts": {
7      "test": "echo \"Error: no test specified\" && exit 1"
8    },
9    "keywords": [],
10   "author": "",
11   "license": "ISC",
12   "dependencies": {
13     "bcrypt": "^6.0.0",
14     "cors": "^2.8.5",
15     "express": "^5.1.0",
16     "jsonwebtoken": "^9.0.2",
17     "pg": "^8.16.3"
18   }
19 }
```

```

1  /* --- Global Reset and Main App Layout --- */
2  * { box-sizing: border-box; }
3
4  html, body, #root {
5      margin: 0;
6      padding: 0;
7      width: 100%;
8      height: 100%;
9      font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
10     -webkit-font-smoothing: antialiased;
11     -moz-osx-font-smoothing: grayscale;
12     /* Base Background: Blue/Purple Gradient */
13     background: linear-gradient(135deg, #1e3a8a 0%, #5b21b6 100%);
14 }
15
16 .App {
17     text-align: center;
18     width: 100%;
19     height: 100vh;
20     overflow: hidden;
21     background-color: transparent;
22 }
23
24 /* --- Core Screen and Header Styles (Combined) --- */
25 .splash-screen, .App-header {
26     width: 100%;
27     height: 100%;
28     display: flex;
29     flex-direction: column;
30     align-items: center;
31     justify-content: center; /* Default for splash, overridden for header */
32     color: white;
33 }
34
35 /* Splash Screen Specifics */
36 .splash-screen {
37     position: fixed; top: 0; left: 0;
38     background: linear-gradient(135deg, #1e3a8a 0%, #5b21b6 100%);
39     z-index: 1000; color: #79f8f8;
40     transition: opacity 0.5s ease-out;
41 }
42 .splash-screen.fade-out { opacity: 0; }
43 .splash-screen img { width: 40vmin; animation: none; border-radius: 10px; }
44
45 /* App Header Specifics */
46 .App-header {
47     background-color: rgba(0, 0, 0, 0.6); /* Default Dark overlay for GAME SCREENS */
48     justify-content: flex-start; /* Default align to top */
49     font-size: calc(8px + 1vmin);
50     padding: 20px;
51     overflow-y: auto;
52     position: relative; /* Anchor for absolute children */
53 }
54 .App-header.no-overlay, .App-header.no-overlay-auth {
55     background-color: transparent !important;
56 }
57 .App-header.no-overlay-auth {
58     justify-content: center !important;
59 }
60 .App-header h1 {
61     font-size: 2.5vmin;
62     color: #79f8f8;
63     text-shadow: 0 0 5px rgba(121, 248, 248, 0.5);
64 }
65
66 /* --- Button Consolidation (Primary and Mission) --- */
67 .primary-button, .mission-card button, .auth-form button {
68     width: 100%;

```

```

69     padding: 10px;
70     font-size: 1.9vmin;
71     background-color: #1e90ff;
72     border: none;
73     border-radius: 4px;
74     cursor: pointer;
75     font-weight: bold;
76     color: white;
77     transition: background-color 0.2s;
78 }
79 .primary-button { margin-top: 20px; }
80 .mission-card button { width: 100%; margin-top: 10px; font-size: 1.6vmin; }
81 .auth-form button { width: 90%; font-size: 1.9vmin; }
82
83 .primary-button:hover, .mission-card button:hover, .auth-form button:hover {
84     background-color: #1c7ad0;
85 }
86
87 /* --- Auth Page Styles (Login/Register) --- */
88 .auth-logo {
89     max-width: 350px; width: 40vmin; height: auto;
90     margin-top: 50px; margin-bottom: -50px;
91     filter: drop-shadow(0 0 10px rgba(121, 248, 248, 0.8));
92 }
93 .auth-container { display: flex; justify-content: space-around; width: 80%; max-width: 900px; }
94 .auth-form {
95     border: 1px solid #79f8f8; border-radius: 8px; padding: 20px; width: 45%;
96     background-color: rgba(40, 44, 52, 0.9); box-shadow: 0 0 10px rgba(121, 248, 248, 0.4);
97 }
98 .auth-form div { margin-bottom: 15px; text-align: center; }
99 .auth-form label { display: block; margin-bottom: 5px; font-size: 1.8vmin; color: #79f8f8; }
100 .auth-form input {
101     width: 90%; padding: 8px; font-size: 1.8vmin; border: none; border-radius: 4px;
102     background-color: #3a3f47; color: white; margin: 0 auto; display: block;
103 }
104 .auth-form p { font-size: 1.6vmin; margin-top: 10px; }
105 .auth-form p a { color: #79f8f8; font-weight: bold; text-decoration: none; }
106 .auth-form p a:hover { text-decoration: underline; }
107
108 /* --- Game Hub/Container --- */
109 .game-container {
110     width: 100%; max-width: 1200px; margin: 10px auto; padding: 15px;
111     background-color: rgba(40, 44, 52, 0.9); border-radius: 8px; flex-grow: 1;
112     display: flex; flex-direction: column; box-shadow: 0 0 15px rgba(0, 0, 0, 0.5);
113 }
114 .progress-bar-container {
115     width: 100%; max-width: 800px; height: 25px; background-color: #3a3f47;
116     border-radius: 15px; overflow: hidden; margin: 10px auto; border: 1px solid #79f8f8;
117     box-shadow: 0 0 8px rgba(121, 248, 248, 0.5);
118 }
119 .progress-fill {
120     height: 100%; background: linear-gradient(90deg, #1e90ff 0%, #79f8f8 100%);
121     transition: width 0.5s ease-in-out; display: flex; align-items: center; justify-content: center;
122     font-weight: bold; font-size: 1.6vmin; color: #1a1a2e; text-shadow: 0 0 1px white;
123 }
124
125 /* Mission Cards */
126 .hub-mission-list { display: flex; justify-content: space-around; flex-wrap: wrap; margin-top: 20px; }
127 .mission-card {
128     border: 1px solid #79f8f8; border-radius: 8px; padding: 20px; margin: 10px;
129     width: 300px; background-color: rgba(26, 26, 46, 0.9); box-shadow: 0 0 15px rgba(121, 248, 248, 0.2);
130     transition: transform 0.2s;
131 }
132 .mission-card:hover { transform: translateY(-5px) scale(1.02); }
133 .mission-card h3 { color: #79f8f8; margin-top: 0; font-size: 2vmin; }
134 .mission-card p { font-size: 1.5vmin; line-height: 1.5; }
135 .status { font-weight: bold; font-size: 1.5vmin; }
136 .status-pending { color: #f39c12; }
137 .status-complete { color: #2ecc71; }
138
139 /* --- Content Screens (Welcome, HowToPlay) --- */
140 .content-screen { display: flex; align-items: center; justify-content: space-around; padding: 20px; }
141 .content-text { width: 60%; text-align: left; line-height: 1.6; font-size: 1.8vmin; }
142 .content-text h2 { color: #79f8f8; font-size: 2.2vmin; }
143 .content-text ul { list-style-type: '🎮'; padding-left: 30px; }
144 .content-image { width: 30%; }
145 .content-image img { max-width: 100%; height: auto; background-color: transparent; }

```

```

146
147 /* --- Mission Layouts --- */
148 .mission-layout {
149     display: flex; justify-content: space-between;
150     width: 100%; height: 100%; overflow-y: auto;
151 }
152 .dnd-item {
153     border: 1px solid #79f8f8; padding: 10px; margin: 5px 0; background-color: #3a3f47;
154     border-radius: 4px; cursor: grab; font-size: 1.6vmin; color: white;
155 }
156 .dnd-item-overlay {
157     border: 1px solid #79f8f8; padding: 10px; margin: 5px 0; background-color: #4a5059;
158     border-radius: 4px; cursor: grabbing; font-size: 1.6vmin; box-shadow: 0 0 15px rgba(121, 248, 248, 0.8);
159 }
160 .dnd-column, .dnd-column-split, .flowchart-pseudocode, .quiz-answers, .sorter-box, .pseudocode-box {
161     border-radius: 8px; padding: 10px; background-color: rgba(0, 0, 0, 0.2);
162 }
163 .dnd-column, .dnd-column-split {
164     border: 2px dashed #79f8f8; min-height: 300px;
165 }
166 .dnd-column { width: 23%; }
167 .dnd-column-split { width: 48%; }
168 .dnd-column h4, .dnd-column-split h4 { margin-top: 0; color: #79f8f8; }
169 .dnd-column.available, .dnd-column-split.available { border: 1px solid #555; background-color: #282c34; }
170
171 /* Drag & Drop Containers */
172 .dnd-container { display: flex; justify-content: space-around; width: 100%; }
173 .dnd-container-split { display: flex; justify-content: space-around; margin-bottom: 20px; width: 100%; }
174
175 /* Table and Quiz Styles */
176 .styled-table { width: 100%; border-collapse: collapse; margin: 20px 0; font-size: 1.6vmin; border: 1px solid #555; }
177 .styled-table th, .styled-table td { border: 1px solid #555; padding: 10px; text-align: left; }
178 .styled-table thead th { background-color: #4a5059; color: #79f8f8; }
179 .styled-table tbody tr:nth-child(even) { background-color: #3a3f47; }
180
181 .table-input, .quiz-input {
182     background-color: #3a3f47; color: white; border: 2px solid #79f8f8;
183     border-radius: 4px; padding: 5px; font-size: 1.6vmin; text-align: center;
184 }
185 .table-input { width: 50px; }
186 .quiz-input { font-family: monospace; width: 200px; }
187
188 .choice-button {
189     display: block; width: 100%; padding: 12px; margin: 8px 0; font-size: 1.6vmin;
190     text-align: left; background-color: #4a5059; border: 1px solid #555; color: white;
191     border-radius: 4px; cursor: pointer; transition: background-color 0.2s, box-shadow 0.2s;
192 }
193 .choice-button:hover { background-color: #5a6069; }
194 .choice-button.selected {
195     background-color: #1e90ff; border-color: #fff; color: white; font-weight: bold;
196     box-shadow: 0 0 8px rgba(121, 248, 248, 0.8);
197 }
198
199 /* --- Algorithm Building (Pembinaan) Styles --- */
200 .sorter-container { display: flex; justify-content: space-around; width: 100%; }
201 .sorter-box { border: 1px solid #555; width: 48%; }
202 .sorter-item {
203     display: flex; justify-content: space-between; align-items: center; margin: 5px 0;
204     padding: 8px; background-color: #3a3f47; border: 1px solid #555; border-radius: 4px;
205     font-size: 1.6vmin;
206 }
207 .sorter-item button {
208     background-color: #1e90ff; border: none; color: white; font-weight: bold; cursor: pointer;
209     border-radius: 4px; padding: 2px 6px; margin-left: 5px; width: auto;
210 }
211 .sorter-item button:disabled { background-color: #555; color: #999; cursor: not-allowed; }
212
213 /* Flowchart Matcher (M2) */
214 .flowchart-container { display: flex; justify-content: space-around; width: 100%; }
215 .flowchart-pseudocode { border: 1px solid #555; width: 40%; min-height: 400px; background-color: #282c34; }
216 .flowchart-boxes { width: 55%; }
217 .flowchart-box {
218     border: none; padding: 0; width: 100%; min-height: 0;
219     background-color: transparent; border-radius: 0; margin-bottom: 10px;
220 }
221 .flowchart-box h4 { margin: 0 0 5px 0; color: #79f8f8; font-size: 1.5vmin; }
222

```

```

223 /* Fill-in-the-blanks (M3) */
224 .quiz-container { display: flex; justify-content: space-around; }
225 .quiz-pseudocode {
226     border: 1px solid #79f8f8; padding: 20px; font-family: monospace; text-align: left;
227     width: 55%; background-color: #282c34; font-size: 1.8vmin; line-height: 2;
228 }
229 .quiz-answers { border: 1px solid #79f8f8; width: 40%; background-color: #282c34; }
230 .quiz-answers code { display: block; background-color: #3a3f47; padding: 8px; margin: 5px 0; border-radius:
231
232 /* --- Debugging (Penyahpepijat) Styles --- */
233 .debug-container { display: flex; justify-content: space-around; width: 100%; }
234 .pseudocode-box {
235     border: 1px solid #79f8f8; padding: 15px; font-family: monospace; text-align: left;
236     width: 48%; background-color: #282c34; font-size: 1.6vmin; line-height: 1.8;
237 }
238 .pseudocode-box .error { color: #ff6b6b; font-weight: bold; }
239 .debug-answer-box { border: 2px dashed #79f8f8; padding: 15px; width: 48%; background-color: rgba(0, 0, 0,
240 .order-input {
241     width: 40px; margin-right: 10px; text-align: center; font-size: 1.6vmin;
242     padding: 5px; border-radius: 4px; border: 1px solid #79f8f8; background-color: #3a3f47; color: white;
243 }
244 .output-table { width: 100%; font-size: 1.5vmin; margin-top: 10px; border: 1px solid #79f8f8; border-colla
245 .output-table th { color: #79f8f8; background-color: #3a3f47; }
246 .output-table td { border: 1px solid #555; padding: 8px; }
247 .output-table .fail { background-color: #5a2e2e; }
248 .output-table .pass { background-color: #2e5a3e; }
249
250 .debug-answer-box {
251     border: 2px dashed #79f8f8;
252     padding: 15px;
253     width: 48%;
254     background-color: rgba(0, 0, 0, 0.2);
255     text-align: left; /* everything left by default */
256 }
257
258 /* only the heading centered */
259 .debug-answer-box .debug-title {
260     text-align: center;
261     font-weight: bold;
262 }
263
264
265 /* Make both titles look the same */
266 .pseudocode-box h4,
267 .debug-answer-box h4 {
268     text-align: center;
269     font-weight: bold;
270     font-size: 2.5vmin; /* tweak this if one looked larger/smaller before */
271     margin: 0 0 10px 0;
272 }
273
274
275
276 /* --- Mission Complete Screen --- */
277 .complete-screen {
278     display: flex; flex-direction: column; align-items: center;
279     justify-content: center; text-align: center; padding: 40px;
280 }
281 .complete-screen h2 { font-size: 3vmin; color: #2ecc71; margin-bottom: 10px; }
282 .complete-screen h3 { font-size: 4vmin; color: #79f8f8; margin: 10px 0; }
283 .complete-screen p { font-size: 1.8vmin; line-height: 1.6; }
284
285 /* --- Welcome Page Highlights/Text --- */
286 .welcome-highlight {
287     text-align: center; font-weight: bold; font-size: 1.2em; margin-bottom: 20px; color: #79f8f8;
288 }
289 .welcome-highlight p { margin: 5px 0; }
290 .welcome-highlight.super-bold-large {
291     font-size: 2.5em; font-weight: 900; color: #79f8f8;
292     text-shadow: 0 0 10px rgba(121, 248, 248, 0.7);
293 }
294 .welcome-highlight.normal-welcome-text {
295     font-size: 0.7em; font-weight: normal; opacity: 0.8; color: white; text-shadow: none;
296 }
297 .white-text, .white-text h2, .white-text p { color: white; }
298
299 /* --- Global Alignment --- */

```

```

300 .center-full-width { width: 100%; text-align: center; margin-top: 20px; }
301
302 /* --- Background Images (Cleaned) --- */
303 .bg-intro-image, .bg-auth-image {
304     background-repeat: no-repeat; background-position: center center; background-attachment: fixed;
305     background-size: cover; background-blend-mode: overlay; background-color: rgba(0, 0, 0, 0.4);
306 }
307 .bg-auth-image { background-image: url('../public/background5.jpg'); }
308 .bg-intro-image { background-image: url('../public/background6.jpg'); }
309
310 /* --- Robot Sidebar and Bubble (Mission Screen) --- */
311 .robot-sidebar {
312     flex: 0 0 25%;
313
314     --robot-width: 130px;
315     --robot-bottom-gap: 70px;
316     --bubble-robot-gap: 80px;
317     --bubble-max-width: 320px;
318     width: 20%; display: block; position: relative; padding: 20px 10px; min-height: 260px;
319 }
320 .robot-sidebar img {
321     position: absolute; bottom: var(--robot-bottom-gap); left: 50%;
322     transform: translateX(-50%); width: var(--robot-width); height: auto;
323     display: block; z-index: 10; order: 1; margin: 0;
324 }
325 .robot-chat-bubble {
326     position: absolute; left: 50%; transform: translateX(-50%);
327     bottom: calc(var(--robot-bottom-gap) + var(--robot-width) + var(--bubble-robot-gap));
328     max-width: var(--bubble-max-width); width: 180px; background-color: #4a5059;
329     border-radius: 10px; padding: 12px 14px; font-size: 1.6vmin; line-height: 1.5;
330     text-align: left; border: 2px solid #79f8f8; z-index: 20; box-sizing: border-box;
331     transition: all 160ms ease; box-shadow: 0 6px 18px rgba(0,0,0,0.25);
332 }
333 .robot-chat-bubble::after {
334     content: ''; position: absolute; bottom: -8px; left: 50%; transform: translateX(-50%);
335     width: 0; height: 0; border: 8px solid transparent; border-top: 8px solid #79f8f8; z-index: 19;
336 }
337
338 /* --- Robot Auth Screen Bubble --- */
339 .auth-robot-image {
340     position: absolute; bottom: 0; right: 0; width: 20vmin; max-width: 200px;
341     transform: scaleX(-1);
342 }
343 .robot-speech-container {
344     position: absolute; bottom: 25vmin; right: 5%; width: 250px; z-index: 50;
345 }
346 .robot-speech-bubble {
347     background-color: #4a5059; border: 2px solid #79f8f8; border-radius: 10px;
348     padding: 15px; text-align: left; font-size: 1.6vmin; color: white;
349     box-shadow: 0 0 10px rgba(121, 248, 248, 0.4);
350 }
351 .robot-speech-bubble::after {
352     content: ''; position: absolute; bottom: -10px; right: 15px; width: 0; height: 0;
353     border: 10px solid transparent; border-top: 10px solid #79f8f8;
354 }
355 .auth-speech-bubble span { border-right: none !important; }
356
357 /* --- Animations --- */
358 .typewriter {
359     overflow: hidden; white-space: nowrap;
360     animation: typing 2.5s steps(30, end), blink 0.5s step-end infinite;
361     animation-fill-mode: forwards; border-right: 2px solid rgba(121, 248, 248, 0.7);
362 }
363 .message-reveal { opacity: 0; animation: fadeIn 0.2s ease-in forwards; animation-delay: 0.1s; }
364 @keyframes fadeIn { from { opacity: 0; } to { opacity: 1; } }
365
366 /* --- How To Play Layout --- */
367 .how-to-play-layout {
368     display: flex; flex-direction: column; justify-content: flex-start;
369     height: 100%; width: 100%; align-items: center; padding: 20px 0 0 0;
370 }
371 .how-to-play-layout .card-row-wrapper {
372     display: flex; justify-content: space-between; width: 100%; max-width: 1000px;
373     margin-bottom: 20px; gap: 20px; flex-grow: 1;
374 }
375 .how-to-play-layout .primary-button-wrapper { margin-top: auto; padding-top: 20px; }
376 .how-to-play-layout ul { list-style-type: none; padding-left: 0; margin: 5px 0; }

```



```

377 .how-to-play-layout ul li {
378     padding: 5px 0; border-bottom: 1px dashed rgba(121, 248, 248, 0.2); color: white;
379 }
380 .how-to-play-layout ul li::before {
381     content: '▶'; color: #1e90ff; font-weight: bold; margin-right: 8px; font-size: 1.2em;
382 }
383
384 /* --- Settings Menu --- */
385 .settings-icon, .settings-menu button { transition: transform 0.2s, background-color 0.2s; }
386 .settings-icon {
387     position: absolute; top: 20px; left: 20px; width: 42px; height: 42px;
388     background-color: rgba(40, 44, 52, 0.9); border-radius: 50%; cursor: pointer; z-index: 900;
389     border: 2px solid #79f8f8; display: flex; align-items: center; justify-content: center;
390     filter: drop-shadow(0 0 6px #00eaff); animation: settingsGlow 1.8s infinite ease-in-out;
391 }
392 .settings-wrapper { position: relative; display: flex; flex-direction: column; align-items: center; margin:
393 .settings-icon:hover { transform: rotate(25deg) scale(1.1); }
394 .settings-icon img { width: 60%; }
395
396 @keyframes settingsGlow {
397     0% { filter: drop-shadow(0 0 4px #00f0ff); }
398     50% { filter: drop-shadow(0 0 12px #79f8f8); }
399     100% { filter: drop-shadow(0 0 4px #00f0ff); }
400 }
401
402 .settings-tooltip {
403     visibility: hidden; opacity: 0; position: absolute; top: 50%; left: 60px;
404     transform: translateY(-50%); background: rgba(0,0,0,0.8); padding: 5px 10px;
405     border-radius: 6px; color: #79f8f8; font-size: 14px; transition: opacity 0.3s ease;
406     white-space: nowrap; pointer-events: none;
407 }
408 .settings-wrapper:hover .settings-tooltip { visibility: visible; opacity: 1; }
409
410 .settings-backdrop {
411     position: fixed; top: 0; left: 0; width: 100%; height: 100%;
412     background-color: rgba(0, 0, 0, 0.8); z-index: 999;
413     display: flex; align-items: center; justify-content: center;
414 }
415 .settings-menu {
416     background-color: #282c34; border: 3px solid #79f8f8; border-radius: 10px;
417     padding: 30px; width: 350px; box-shadow: 0 0 30px rgba(121, 248, 248, 0.9);
418     text-align: center;
419 }
420 .settings-menu button { width: 100%; margin: 10px 0; padding: 12px; font-size: 1.8vmin; font-weight: bold;
421
422 /* --- Flowchart (Carta Alir) Shapes --- */
423 .flowchart-shape {
424     min-width: 140px; min-height: 70px; display: flex; justify-content: center;
425     align-items: center; text-align: center; padding: 8px; box-sizing: border-box;
426     border: 2px solid #00ffff; background-color: transparent; border-style: solid;
427     outline: none; background: #3498db; color: #fff; font-weight: bold;
428 }
429
430 /* Flowchart - remove drag-and-drop item look */
431 .flowchart-shape .dnd-item, .flowchart-shape .dnd-item-overlay {
432     background: none; border: none; box-shadow: none; padding: 0; margin: 0;
433     width: 90%; height: auto; text-align: center;
434 }
435
436 /* Shape-specific styles */
437 .flowchart-shape-oval { border-radius: 999px; }
438 .flowchart-shape-rectangle { border-radius: 8px; }
439 .flowchart-shape-parallelogram { transform: skewX(-20deg); }
440 .flowchart-shape-parallelogram .dnd-item { transform: skewX(20deg); }
441
442 .flowchart-shape-diamond {
443     width: 100px; height: 100px; transform: rotate(10deg); display: flex;
444     align-items: center; justify-content: center; margin: 0 auto;
445 }
446 .flowchart-shape-diamond .dnd-item { transform: rotate(0deg) scaleY(1.18); }
447
448 /* Arrows & Labels */
449 .flowchart-arrow-row {
450     display: flex; justify-content: center; align-items: center;
451     gap: 6px; font-weight: bold; color: #fff;
452 }
453 .flowchart-arrow-row span.arrow {

```

```

454     display: inline-block; width: 0; height: 0;
455     border-left: 6px solid transparent; border-right: 6px solid transparent;
456     border-top: 10px solid #fff;
457 }
458 .flowchart-arrow-row span.label-yes { font-weight: bold; margin-right: 4px; }
459
460 .mission-content {
461     flex: 1; /* remaining space = 75% */
462     padding: 20px;
463     box-sizing: border-box;
464     overflow-x: auto; /* in case wide content */
465 }
466
467 .confetti-container {
468     position: absolute;
469     top: 0;
470     left: 50%;
471     width: 0;
472     height: 0;
473     pointer-events: none;
474 }
475 /* ===== MOBILE MODE ===== */
476 .mobile-view .flowchart-container {
477     flex-direction: column !important;
478     align-items: center;
479 }
480
481 .mobile-view .flowchart-pseudocode,
482 .mobile-view .flowchart-boxes {
483     width: 100% !important;
484     max-width: 100%;
485 }
486
487 .mobile-view .flowchart-diagram {
488     transform: scale(0.9);
489 }
490
491 /* Disable absolute arrows on mobile */
492 .mobile-view .branch-arrows {
493     display: none !important;
494 }
495
496 /* Allow horizontal scroll when needed */
497 .mobile-view .flowchart-boxes {
498     overflow-x: auto;
499 }

```

```

1  import React, { useState, useEffect, useCallback } from 'react';
2  import './App.css';
3  import { Routes, Route, Navigate, useLocation, useNavigate } from 'react-router-dom';
4  // NOTE: These components must exist in your project structure for this code to run
5  import Login from './components/Login';
6  import Register from './components/Register';
7  import Game from './components/Game';
8  import SettingsMenu from './components/SettingsMenu';
9  import RobotMessage from './components/RobotMessage';
10 import SplashScreen from './components/SplashScreen'; // ✨ NEW IMPORT
11 import useDeviceType from './hooks/useDeviceType';
12
13 function AppLayout({ children }) {
14   const { isMobile } = useDeviceType();
15
16   return (
17     <div className={isMobile ? 'mobile-view' : 'desktop-view'}>
18       {children}
19     </div>
20   );
21 }
22
23
24 function App() {
25   const location = useLocation();
26   const navigate = useNavigate();
27
28   const [token, setToken] = useState(localStorage.getItem('token'));
29   const [user, setUser] = useState(null);
30   const [isLoading, setIsLoading] = useState(true);
31   const [showSettings, setShowSettings] = useState(false);
32   const [robotMessage, setRobotMessage] = useState("Sila log masuk atau daftar untuk memulihkan Kampus Di
33
34   // ✨ NEW: Splash Screen State
35   const [showSplash, setShowSplash] = useState(true);
36
37   // ✨ NEW: Global audio mute state
38   const [isMuted, setIsMuted] = useState(true);
39
40   // ✨ NEW: Toggle mute handler
41   const toggleMute = useCallback(() => {
42     setIsMuted(prev => !prev);
43   }, []);
44
45   useEffect(() => {
46     // REMOVE old splash timeout
47     // Now completely controlled by the SplashScreen itself
48
49     if (location.pathname === '/login' || location.pathname === '/register') {
50       setRobotMessage("Sila log masuk atau daftar untuk memulihkan Kampus Digital.");
51     }
52
53     const fetchUser = async () => {
54       if (token) {
55         try {
56           const response = await fetch('https://algoquest-api.onrender.com/api/user/me', {
57             headers: { 'Authorization': `Bearer ${token}` }
58           });
59
60           if (response.ok) {
61             const data = await response.json();
62             setUser(data);
63           } else {
64             localStorage.removeItem('token');
65             setToken(null);
66           }
67         } catch (err) {
68           console.error("Failed to fetch user", err);

```

```

69         }
70     }
71     setIsLoading(false);
72 };
73
74     fetchUser();
75 }, [token, location.pathname]);
76
77 const handleLogout = () => {
78     localStorage.removeItem('token');
79     setToken(null);
80     setUser(null);
81     navigate('/login', { replace: true });
82 };
83
84 const handleReturnToHub = () => {
85     setShowSettings(false);
86     navigate('/game', { state: { screen: 'hub' }, replace: true });
87 };
88
89 // 1. FINAL BACKGROUND LOGIC
90 const getBackgroundStyle = () => {
91     const gameState = location.state?.screen;
92
93     if (location.pathname === '/login' || location.pathname === '/register') {
94         return {
95             backgroundImage: `url('/background5.jpg')`,
96             backgroundRepeat: 'no-repeat',
97             backgroundPosition: 'center center',
98             backgroundAttachment: 'fixed',
99             backgroundSize: 'cover',
100         };
101     }
102
103     if (token && (gameState === 'welcome' || gameState === 'howToPlay' || gameState === 'hub')) {
104         return {
105             backgroundImage: `url('/background6.jpg')`,
106             backgroundRepeat: 'no-repeat',
107             backgroundPosition: 'center center',
108             backgroundAttachment: 'fixed',
109             backgroundSize: 'cover',
110         };
111     }
112
113     if (token && location.pathname === '/game') {
114         return {
115             backgroundImage: `url('/background7.jpg')`,
116             backgroundRepeat: 'no-repeat',
117             backgroundPosition: 'center center',
118             backgroundAttachment: 'fixed',
119             backgroundSize: 'cover',
120         };
121     }
122
123     return {};
124 };
125
126 const isAuthScreen = location.pathname === '/login' || location.pathname === '/register';
127
128 // 2. HEADER CLASS LOGIC
129 const getHeaderClass = () => {
130     const gameState = location.state?.screen;
131     const isGamePath = location.pathname === '/game';
132
133     if (
134         isAuthScreen ||
135         (token && (gameState === 'welcome' || gameState === 'howToPlay' || gameState === 'hub'))
136     ) {
137         return 'no-overlay-auth center-content';
138     }
139
140     if (token && isGamePath) {
141         return 'no-overlay';
142     }
143
144     return '';
145 };

```

```

146 // 🌟 NEW: Splash Screen Rendering
147
148 if (showSplash) {
149   return (
150     <SplashScreen
151       onFinish={() => setShowSplash(false)}
152     />
153   );
154 }
155
156 if (isLoading) {
157   return (
158     <div className="App">
159       <header className="App-header"><h1>Loading...</h1></header>
160     </div>
161   );
162 }
163
164 const headerContent = token ? (
165   <>
166     <h1>ALGOQUEST</h1>
167     <hr style={{ width: '90%', borderColor: '#79f8f8' }} />
168   </>
169 ) : (
170   
171 );
172
173 return (
174   <div className="App" style={getBackgroundStyle()}>
175
176     {/🌟 GLOBAL AUDIO PLAYER */}
177     <audio id="background-music" autoPlay loop muted={isMuted}>
178       <source src="/bgsound.mp3" type="audio/mpeg" />
179       Your browser does not support the audio element.
180     </audio>
181
182     {/SETTINGS MODAL */}
183     {showSettings && token && (
184       <SettingsMenu
185         onClose={() => setShowSettings(false)}
186         onLogout={handleLogout}
187         onReturnToHub={handleReturnToHub}
188         isMuted={isMuted}
189         onToggleMute={toggleMute}
190       />
191     )}
192
193     <header className={`App-header ${getHeaderClass()}`}>
194
195       {token && (
196         <div
197           className="settings-icon"
198           onClick={() => setShowSettings(true)}
199           title="Settings"
200         >
201           
202         </div>
203       )}
204
205       {headerContent}
206
207       {isAuthScreen && (
208         <>
209           
214         </>
215       )}
216
217       <Routes>
218         {token ? (
219           <>
220             <Route path="/game" element={<Game user={user} onLogout={handleLogout} />} />
221             <Route path="/hub" element={<Navigate to="/game" state={{ screen: 'hub' }} replace />} />
222             <Route path="*" element={<Navigate to="/game" />} />

```

```

223         </>
224     ) : (
225         <>
226             <Route path="/login" element={
227                 <div className="auth-container" style={{ flexGrow: 1, alignItems: 'center'
228                     <Login />
229                 </div>
230             } />
231             <Route path="/register" element={
232                 <div className="auth-container" style={{ flexGrow: 1, alignItems: 'center'
233                     <Register />
234                 </div>
235             } />
236             <Route path="*" element={<Navigate to="/login" replace />} />
237         </>
238     )}
239 </Routes>
240 </header>
241 </div>
242 );
243 }
244
245 export default App;

```

```
1 // audioUtils.js
2
3 // Create audio objects once (ensure these files are in your public folder)
4 const clickAudio = new Audio('/click.mp3');
5 const hoverAudio = new Audio('/hover.mp3');
6
7 // Optional: adjust volumes
8 clickAudio.volume = 1.0; // Full volume (change if you want)
9 hoverAudio.volume = 0.4; // Softer hover sound
10
11 /**
12  * Plays the click sound.
13  * Uses async/await to properly handle the Promise returned by play()
14  * so that if the browser blocks playback (no user interaction yet),
15  * the rejection is caught and does NOT cause an uncaught runtime error.
16  */
17 export const playClickSound = async () => {
18   try {
19     clickAudio.currentTime = 0;
20     await clickAudio.play();
21   } catch (e) {
22     // Playback blocked or failed - ignore silently
23     // console.warn("Click sound blocked:", e);
24   }
25 };
26
27 /**
28  * Plays the hover sound.
29  * Also uses async/await to catch the play() Promise rejection,
30  * preventing "Uncaught (in promise)" runtime errors.
31  */
32 export const playHoverSound = async () => {
33   try {
34     hoverAudio.currentTime = 0;
35     await hoverAudio.play();
36   } catch (e) {
37     // Playback blocked or failed - ignore silently
38     // console.warn("Hover sound blocked:", e);
39   }
40 };
```

```

1  import React, { useState, useEffect } from 'react';
2  import { useLocation, useNavigate } from 'react-router-dom';
3  import Welcome from './Welcome';
4  import HowToPlay from './HowToPlay';
5  import Hub from './Hub';
6  import Mission1 from './Mission1';
7  import Mission2 from './Mission2';
8  import Mission3 from './Mission3';
9
10 function Game({ user, onLogout }) {
11   const navigate = useNavigate();
12   const location = useLocation();
13
14   // 1. Initialize state from URL history (for persistence across refreshes)
15   const [gameState, setGameState] = useState(location.state?.screen || 'welcome');
16
17   // 2. Update the URL state every time the gameState changes
18   useEffect(() => {
19     // Only update if the screen state has actually changed
20     if (location.state?.screen !== gameState) {
21       // IMPORTANT: We pass 'user' in the state so Hub.js can read location.state.user
22       navigate(location.pathname, {
23         replace: true,
24         state: { screen: gameState, user: user }
25       });
26     }
27   }, [gameState, navigate, location.pathname, location.state, user]);
28
29   const handleSetGameState = (newState) => {
30     setGameState(newState);
31   };
32
33   // 3. Central function to send score to the backend (Render URL)
34   const handleMissionComplete = async (missionId, score) => {
35     const token = localStorage.getItem('token');
36     try {
37       await fetch('https://algoquest-api.onrender.com/api/progress', {
38         method: 'POST',
39         headers: {
40           'Content-Type': 'application/json',
41           'Authorization': `Bearer ${token}`
42         },
43         body: JSON.stringify({ missionId, score })
44       });
45     } catch (err) {
46       console.error("Failed to save progress:", err);
47     }
48     // Always return to the hub after completion
49     setGameState('hub');
50   };
51
52   const renderGameState = () => {
53     switch (gameState) {
54       case 'welcome':
55         return <Welcome user={user} onContinue={() => handleSetGameState('howToPlay')} />;
56       case 'howToPlay':
57         return <HowToPlay onContinue={() => handleSetGameState('hub')} />;
58       case 'hub':
59         return (
60           <Hub
61             onLogout={onLogout}
62             onStartMission1={() => handleSetGameState('mission1')}
63             onStartMission2={() => handleSetGameState('mission2')}
64             onStartMission3={() => handleSetGameState('mission3')}
65           />
66         );
67       case 'mission1':
68         return <Mission1 onMissionComplete={(score) => handleMissionComplete('mission1', score)} />;

```



```

69     case 'mission2':
70       return <Mission2 onMissionComplete={score => handleMissionComplete('mission2', score)} />;
71     case 'mission3':
72       return <Mission3 onMissionComplete={score => handleMissionComplete('mission3', score)} />;
73     default:
74       return <div>Loading...</div>;
75   }
76 };
77
78 // 4. Wrap the rendered component in the game container
79 return (
80   <div className="game-container">
81     {renderGameState()}
82   </div>
83 );
84 }
85
86 export default Game;

```

```

1  import React, { useEffect, useState } from 'react';
2
3  const hoverAudio = new Audio('/hover.mp3');
4  hoverAudio.volume = 0.4;
5
6  const playHoverSound = () => {
7    try { hoverAudio.currentTime = 0; hoverAudio.play(); } catch (e) { }
8  };
9
10 function HowToPlay({ onContinue }) {
11   const [animateCards, setAnimateCards] = useState(false);
12   const [animateButton, setAnimateButton] = useState(false);
13
14   useEffect(() => {
15     const timer1 = setTimeout(() => setAnimateCards(true), 200);
16     const timer2 = setTimeout(() => setAnimateButton(true), 800);
17     return () => {
18       clearTimeout(timer1);
19       clearTimeout(timer2);
20     };
21   }, []);
22
23   const handleClick = () => onContinue();
24
25   const cardBaseStyle = {
26     flex: '1 1 400px',
27     backgroundColor: '#282c34',
28     border: '1px solid #79f8f8',
29     borderRadius: '12px',
30     padding: '25px',
31     boxShadow: '0 0 10px rgba(121,248,248,0.2)',
32     textAlign: 'left',
33     transition: 'all 0.4s ease',
34     cursor: 'pointer',
35   };
36
37   return (
38     <div style={{
39       display: 'flex',
40       flexDirection: 'column',
41       justifyContent: 'center',
42       alignItems: 'center',
43       minHeight: '80vh',
44       padding: '20px',
45       gap: '60px',
46       backgroundColor: '#1f2228',
47       color: 'fff',
48       textAlign: 'center'
49     }}>
50
51     <div style={{
52       display: 'flex',
53       flexWrap: 'wrap',
54       justifyContent: 'center',
55       gap: '30px',
56       width: '100%',
57       maxWidth: '1200px'
58     }}>
59       { /* Left Card */ }
60       <div style={{
61         ...cardBaseStyle,
62         transform: animateCards ? 'translateY(0)' : 'translateY(50px)',
63         opacity: animateCards ? 1 : 0,
64         transition: 'all 0.8s ease-out',
65       }}
66       onMouseEnter={e => e.currentTarget.style.boxShadow = '0 0 12px #79f8f8, 0 0 15px #00e5ff'}
67       onMouseLeave={e => e.currentTarget.style.boxShadow = '0 0 10px rgba(121,248,248,0.2)'}
68     >

```

```

69     <h2 style={{ color: '#79f8f8', marginBottom: '15px' }}>🧠 Apa itu AlgoQuest?</h2>
70     <p>AlgoQuest ialah permainan serius untuk melatih pemikiran algoritma melalui tugas dan misi</p>
71     <p style={{ fontWeight: 'bold', marginTop: '15px' }}>Setiap misi menguji 4 kemahiran DA</p>
72     <ul style={{ marginLeft: '20px', lineHeight: '1.5' }}>
73         <li>Penguraian - Pecahkan masalah kepada bahagian kecil</li>
74         <li>Pengabstrakan - Kenal pasti maklumat penting</li>
75         <li>Pembinaan Algoritma - Susun langkah penyelesaian</li>
76         <li>Penyahpepijatan - Kesan dan betulkan ralat</li>
77     </ul>
78     <p style={{ fontWeight: 'bold', color: '#79f8f8', marginTop: '10px' }}>
79         🎮 Objektif: Selesaikan 3 misi dan kumpul lencana pencapaian!
80     </p>
81 </div>
82
83 /* Right Card */
84 <div style={{
85     ...cardBaseStyle,
86     transform: animateCards ? 'translateY(0)' : 'translateY(50px)',
87     opacity: animateCards ? 1 : 0,
88     transition: 'all 0.8s ease-out 0.2s'
89 }}>
90     onMouseEnter={e => e.currentTarget.style.boxShadow = '0 0 12px #79f8f8, 0 0 15px #00e5ff'}
91     onMouseLeave={e => e.currentTarget.style.boxShadow = '0 0 10px rgba(121,248,248,0.2)'}
92 >
93     <h2 style={{ color: '#79f8f8', marginBottom: '15px' }}>🎮 Cara Bermain</h2>
94     <ul style={{ marginLeft: '20px', lineHeight: '1.6' }}>
95         <li>Pilih misi (Unit Akademik, Unit Peperiksaan, Unit Kewangan)</li>
96         <li>Baca naratif dan arahan yang diberikan oleh Robot Algo.</li>
97         <li>Lengkapkan tugas pada setiap fasa (Penguraian, Pengabstrakan, Pembinaan Algoritma)</li>
98         <li>Tekan butang simpan untuk menyimpan jawapan.</li>
99         <li>Setiap jawapan betul akan memberi anda mata.</li>
100        <li>Tekan hantar untuk mengetahui markah, ganjaran dan lencana untuk setiap misi.</li>
101    </ul>
102 </div>
103 </div>
104
105 /* Continue Button */
106 <button
107     onClick={handleButtonClick}
108     onMouseEnter={playHoverSound}
109     style={{
110         background: 'linear-gradient(135deg, #79f8f8, #00e5ff)',
111         color: '#000',
112         fontWeight: 'bold',
113         fontSize: '1.2rem',
114         padding: '12px 30px',
115         borderRadius: '10px',
116         border: 'none',
117         cursor: 'pointer',
118         boxShadow: '0 4px 10px rgba(0,0,0,0.3)',
119         transition: 'transform 0.2s ease, box-shadow 0.3s ease, opacity 0.6s',
120         opacity: animateButton ? 1 : 0,
121         transform: animateButton ? 'scale(1)' : 'scale(0.8)',
122         animation: animateButton ? 'pulse 1.5s infinite alternate' : 'none'
123     }}>
124 >
125     Seterusnya
126 </button>
127
128 /* Inline keyframes for pulse */
129 <style>{`
130     @keyframes pulse {
131         0% { transform: scale(1); }
132         50% { transform: scale(1.05); }
133         100% { transform: scale(1); }
134     }
135 `}</style>
136 </div>
137 );
138 }
139
140 export default HowToPlay;

```

```

1  import React, { useState, useEffect } from 'react';
2  import { useLocation, useNavigate } from 'react-router-dom';
3  import { playClickSound, playHoverSound } from '../audioUtils';
4
5  function Hub({ onStartMission1, onStartMission2, onStartMission3, onLogout }) {
6      const location = useLocation();
7      const navigate = useNavigate();
8
9      const [user, setUser] = useState(location.state?.user || null);
10     const [progress, setProgress] = useState({});
11     const [progressPercent, setProgressPercent] = useState(0);
12
13     const handleMissionStart = (startFunction) => {
14         try { playClickSound(); } catch (e) {}
15         if (typeof startFunction === 'function') startFunction();
16     };
17
18     useEffect(() => {
19         const fetchUserAndProgress = async () => {
20             const token = localStorage.getItem('token');
21             if (!token) return;
22
23             if (!user) {
24                 try {
25                     const res = await fetch('https://algoquest-api.onrender.com/api/user/me', {
26                         headers: { 'Authorization': `Bearer ${token}` },
27                     });
28                     if (res.ok) setUser(await res.json());
29                     else {
30                         localStorage.removeItem('token');
31                         navigate('/login', { replace: true });
32                     }
33                 } catch (err) { console.error('Failed to fetch user:', err); }
34             }
35
36             try {
37                 const response = await fetch('https://algoquest-api.onrender.com/api/progress', {
38                     headers: { 'Authorization': `Bearer ${token}` },
39                 });
40                 if (!response.ok) throw new Error('Could not fetch progress');
41                 const data = await response.json();
42                 const progressData = {};
43                 for (const record of data) {
44                     progressData[record.mission_id] = {
45                         score: record.score
46                     };
47                 }
48                 setProgress(progressData);
49                 setProgressPercent(Math.round((Object.keys(progressData).length / 3) * 100));
50             } catch (err) { console.error(err.message); }
51         };
52
53         fetchUserAndProgress();
54     }, [user, navigate]);
55
56     const getScore = (missionId) => progress[missionId]?.score ?? '-';
57     const isMissionCompleted = (missionId) => progress[missionId]?.score !== undefined;
58     const getStatus = (missionId) => isMissionCompleted(missionId)
59         ? { text: 'Selesai', color: '#2ecc71' }
60         : { text: 'Belum Selesai', color: '#f39c12' };
61
62     const userName = user?.name || 'Pemain!';
63
64     const renderMissionCard = (title, description, missionId, onStart) => {
65         const status = getStatus(missionId);
66
67         return (
68             <div className="mission-card" style={{ textAlign: 'center', marginBottom: '20px' }} onMouseEnter

```

```

69     <h3>{title}</h3>
70     <p>{description}</p>
71     <p>🏆 Markah: {getScore(missionId)}</p>
72
73     <div style={{
74         display: 'flex', alignItems: 'center', gap: 8,
75         fontWeight: 'bold', fontSize: '1.5vmin', justifyContent: 'center', margin: '10px 0'
76     }}>
77         <span style={{
78             width: 16, height: 16, borderRadius: '50%',
79             backgroundColor: status.color, display: 'inline-block',
80             boxShadow: `0 0 6px ${status.color}`
81         }}></span>
82         <span>{status.text}</span>
83     </div>
84
85     <button onClick={() => handleMissionStart(onStart)} onMouseEnter={playHoverSound}>
86         Mulakan Misi
87     </button>
88 </div>
89 );
90 };
91
92 return (
93     <div className="game-container">
94         <h2>SELAMAT DATANG KE KAMPUS DIGITAL, {userName}</h2>
95
96         <div className="progress-bar-container">
97             <div className="progress-fill" style={{ width: `${progressPercent}%` }}>
98                 Bar Kemajuan: {progressPercent}%
99             </div>
100         </div>
101
102         <hr />
103
104         <div className="hub-mission-list">
105             {renderMissionCard(
106                 'UNIT AKADEMIK',
107                 'Bantu sistem mendaftar pelajar baharu mengikut urutan langkah yang betul.',
108                 'mission1',
109                 () => onStartMission1()
110             )}
111             {renderMissionCard(
112                 'UNIT PEPERIKSAAN',
113                 'Semak dan sahkan pengiraan markah pelajar bagi memastikan keputusan tepat.',
114                 'mission2',
115                 () => onStartMission2()
116             )}
117             {renderMissionCard(
118                 'UNIT KEWANGAN',
119                 'Bantu sistem mengira yuran pelajar berulang mengikut bilangan kursus yang diambil.',
120                 'mission3',
121                 () => onStartMission3()
122             )}
123         </div>
124
125         <hr style={{ marginTop: '20px' }} />
126
127         <button
128             onClick={() => handleMissionStart(onLogout)}
129             onMouseEnter={playHoverSound}
130             className="primary-button"
131             style={{ width: 300, margin: 'auto', backgroundColor: '#c74f4f' }}
132         >
133             Tamat Permainan
134         </button>
135     </div>
136 );
137 }
138
139 export default Hub;

```

```
1 import React, { useState, useEffect } from 'react';
2 import { Link } from 'react-router-dom';
3
4 // Define audio object outside the component for efficiency
5 const hoverAudio = new Audio('/hover.mp3');
6 hoverAudio.volume = 0.4;
7
8 const playHoverSound = () => {
9   try {
10     hoverAudio.currentTime = 0;
11     hoverAudio.play();
12   } catch (e) { }
13 };
14
15 const INITIAL_MOTIVATION = "Masukkan ID anda dan bantu Algo memulihkan Kampus Digital!";
16 const ERROR_MESSAGE = "Kata laluan salah. Sila cuba lagi!";
17
18 function Login() {
19   const [email, setEmail] = useState('');
20   const [password, setPassword] = useState('');
21   const [robotMessage, setRobotMessage] = useState(INITIAL_MOTIVATION);
22
23   // Set the initial message when the component mounts
24   useEffect(() => {
25     setRobotMessage(INITIAL_MOTIVATION);
26   }, []);
27
28   const handleSubmit = async (e) => {
29     e.preventDefault();
30
31     setRobotMessage("Sedang menyemak identiti..."); // Message while loading
32
33     try {
34       const response = await fetch('https://algoquest-api.onrender.com/api/login', {
35         method: 'POST',
36         headers: { 'Content-Type': 'application/json' },
37         body: JSON.stringify({ email, password }),
38       });
39
40       const data = await response.json();
41
42       if (data.token) {
43         localStorage.setItem('token', data.token);
44         window.location.href = '/game';
45       } else {
46         // Show error message from the robot
47         setRobotMessage(ERROR_MESSAGE);
48         // Reset the message after 3 seconds
49         setTimeout(() => setRobotMessage(INITIAL_MOTIVATION), 3000);
50       }
51     } catch (err) {
52       console.error(err);
53       setRobotMessage("Gagal menghubungi pelayan. Pastikan pelayan Node.js berjalan.");
54       setTimeout(() => setRobotMessage(INITIAL_MOTIVATION), 4000);
55     }
56   };
57
58   return (
59     <div className="auth-form">
60       {/* 1. Robot Chat Bubble Structure */}
61       <div className="robot-speech-container">
62         <div className="robot-speech-bubble">
63           {/* 2. Apply the simple CSS fade-in animation */}
64           <span className="message-reveal">{robotMessage}</span>
65         </div>
66       </div>
67
68       <h2>Log Masuk</h2>
```

```

69     <form onSubmit={handleSubmit}>
70       <div>
71         <label>ID Pengguna (Emel): </label>
72         <input type="email" placeholder="Masukkan emel anda" value={email} onChange={(e) => setEmail(e.target.value)} />
73       </div>
74       <div>
75         <label>Kata Laluan: </label>
76         <input type="password" placeholder="Masukkan kata laluan" value={password} onChange={(e) => setPassword(e.target.value)} />
77       </div>
78       <button type="submit" onMouseEnter={playHoverSound}>Log Masuk</button>
79     </form>
80     <p>Belum ada akaun? <Link to="/register">Daftar di sini</Link></p>
81   </div>
82 );
83 }
84
85 export default Login;

```

```

1  import React, { useState } from 'react';
2  import MissionLayout from './MissionLayout';
3  import Mission1_Penguraian from './Mission1_Penguraian';
4  import Mission1_Pengabstrakan from './Mission1_Pengabstrakan';
5  import Mission1_Pembinaan from './Mission1_Pembinaan';
6  import Mission1_Penyahpepijat from './Mission1_Penyahpepijat';
7  import Mission1_Complete from './Mission1_Complete';
8
9  const ROBOT_TEXTS = {
10   penguraian: "Hai! Mari kita bantu Kampus Digital mengenal pasti semula Input, Proses, dan Output dalam mc
11   pengabstrakan: "Klik dan seret maklumat penting ke ruangan Maklumat Penting. Kemudian, susun langkah penc
12   pembinaan: "Klik anak panah ➡ untuk pindahkan langkah ke kotak algoritma. Bila semua langkah dah pindah
13   penyahpepijat: "Hmm... nampaknya sistem cetak slip dulu sebelum daftar kursus. Susun semula langkah denga
14   complete: "Kerja yang bagus! Sistem pendaftaran pelajar telah dipulihkan. Anda boleh kembali ke Kampus Di
15 };
16
17 function Mission1({ onMissionComplete }) {
18   const [missionPhase, setMissionPhase] = useState('penguraian');
19   const [totalScore, setTotalScore] = useState(0);
20   const [robotText, setRobotText] = useState(ROBOT_TEXTS['penguraian']);
21   const [isCorrect, setIsCorrect] = useState(null);
22   const [earnedBadges, setEarnedBadges] = useState([]);
23
24   // -----
25   // Glow & feedback function
26   // -----
27   const onFeedback = (text, duration = 3000, correctState = null) => {
28     setRobotText(text);
29
30     // Set robot glow: true=green, false=red, null=normal
31     setIsCorrect(correctState);
32
33     setTimeout(() => {
34       setIsCorrect(null);
35       setRobotText(ROBOT_TEXTS[missionPhase]);
36     }, duration);
37   };
38
39   // -----
40   // Unified Backend Submit
41   // -----
42   const submitPhaseToBackend = async (phaseName, isCorrectAnswer, score, badge) => {
43     const token = localStorage.getItem('token');
44     if (!token) return;
45
46     try {
47       await fetch("https://algoquest-api.onrender.com/api/mission/submit", {
48         method: "POST",
49         headers: {
50           "Content-Type": "application/json",
51           "Authorization": `Bearer ${token}`
52         },
53         body: JSON.stringify({
54           mission: 1,
55           phase: phaseName,
56           isCorrect: isCorrectAnswer,
57           score: score,
58           badge: badge
59         })
60       });
61     } catch (e) {
62       console.error("Submit mission data failed", e);
63     }
64   };
65
66   // -----
67   // Phase Completion Handler
68   // -----

```



```

69 const handlePhaseComplete = (nextPhase, scoreEarned, badgeEarned = null) => {
70   const isCorrectAnswer = scoreEarned > 0;
71
72   // Send to backend
73   submitPhaseToBackend(missionPhase, isCorrectAnswer, scoreEarned, badgeEarned);
74
75   setTotalScore(prev => prev + scoreEarned);
76
77   if (badgeEarned) {
78     setEarnedBadges(prev => [...prev, badgeEarned]);
79   }
80
81   // Update robot glow for phase completion
82   setIsCorrect(isCorrectAnswer);
83
84   setMissionPhase(nextPhase);
85   setRobotText(ROBOT_TEXTS[nextPhase]);
86 };
87
88 // -----
89 // Render Current Phase
90 // -----
91 const renderPhase = () => {
92   switch (missionPhase) {
93     case 'penguraian':
94       return (
95         <Mission1_Penguraian
96           onContinue={(score) => handlePhaseComplete('pengabstrakan', score)}
97           onFeedback={onFeedback}
98         />
99       );
100     case 'pengabstrakan':
101       return (
102         <Mission1_Pengabstrakan
103           onContinue={(score) => handlePhaseComplete('pembinaan', score)}
104           onFeedback={onFeedback}
105         />
106       );
107     case 'pembinaan':
108       return (
109         <Mission1_Pembinaan
110           onContinue={(score, badge) => handlePhaseComplete('penyahpepijat', score, badge)}
111           onFeedback={onFeedback}
112         />
113       );
114     case 'penyahpepijat':
115       return (
116         <Mission1_Penyahpepijat
117           onContinue={(score, badge) => handlePhaseComplete('complete', score, badge)}
118           onFeedback={onFeedback}
119         />
120       );
121     case 'complete':
122       return (
123         <Mission1_Complete
124           score={totalScore}
125           badges={earnedBadges}
126           onContinue={() => onMissionComplete(totalScore, earnedBadges)}
127         />
128       );
129     default:
130       return <div>Memuatkan...</div>;
131   }
132 };
133
134 return (
135   <MissionLayout
136     robotText={robotText}
137     isCorrect={isCorrect} // ✅ pass correct/wrong status for glow
138   >
139     {renderPhase()}
140   </MissionLayout>
141 );
142 }
143
144 export default Mission1;

```



```

1  import React from 'react';
2
3  function Mission1_Complete({ score, badges, onContinue }) {
4    // 1. Ensure badges is always an array
5    const displayBadges = Array.isArray(badges) ? badges : [];
6
7    // 2. Fix: Use the score directly (it's already a number)
8    const finalScore = score || 0;
9
10   return (
11     <div
12       style={{
13         width: '100%',
14         height: '100%',
15         display: 'flex',
16         justifyContent: 'center',
17         alignItems: 'center',
18         animation: 'fadeIn 1s ease-in'
19       }}
20     >
21       <div className="complete-screen" style={{ textAlign: 'center', animation: 'fadeInUp 1s ease-out' }}>
22         <h2>Tahniah! Anda Selesai Unit Akademik! 🎉</h2>
23         <p>Sistem pendaftaran pelajar kini berfungsi dengan betul.</p>
24         <hr style={{ width: '80%' }} />
25
26         { /* Display the score directly */ }
27         <h3>Markah Misi: {finalScore} / 100</h3>
28
29         {displayBadges.length > 0 && (
30           <div className="badges-container" style={{ margin: '15px 0' }}>
31             <h4>🏆 Lencana Diperolehi:</h4>
32             <div style={{ display: 'flex', flexWrap: 'wrap', gap: '10px', justifyContent: 'center' }}>
33               {displayBadges.map((badge, idx) => (
34                 <div
35                   key={idx}
36                   style={{
37                     position: 'relative',
38                     display: 'inline-flex',
39                     alignItems: 'center',
40                     justifyContent: 'center',
41                     padding: '6px 16px',
42                     fontWeight: 'bold',
43                     fontSize: '0.95rem',
44                     color: '#fff',
45                     background: 'linear-gradient(135deg, #FFD700, #FFC200, #FFB000)',
46                     borderRadius: '9999px',
47                     boxShadow: '0 2px 6px rgba(0,0,0,0.3)',
48                     overflow: 'hidden',
49                     textAlign: 'center',
50                     cursor: 'default',
51                     transition: 'transform 0.3s ease, box-shadow 0.3s ease'
52                   }}
53                   className="badge"
54                 >
55                   <span style={{ display: 'inline-block', width: '16px', height: '16px', marginRight: '8px' }}>
56                     <span>{badge}</span>
57                   { /* Shine Animation */ }
58                   <span
59                     style={{
60                       position: 'absolute',
61                       top: '-50%',
62                       left: '-50%',
63                       width: '200%',
64                       height: '200%',
65                       background: 'linear-gradient(120deg, rgba(255,255,255,0.2) 0%, rgba(255,255,255,0) 60%',
66                       transform: 'rotate(25deg)',
67                       pointerEvents: 'none',

```

```

68         animation: 'shine 2s infinite',
69         zIndex: 0
70     }}
71     </span>
72 </div>
73 ))}
74 </div>
75 </div>
76 )}
77
78 <hr style={{ width: '80%' }} />
79 <button
80     onClick={onContinue}
81     className="primary-button"
82     style={{
83         width: '300px',
84         marginTop: '20px',
85         transition: 'transform 0.2s ease, background-color 0.2s ease'
86     }}
87     onMouseEnter={e => (e.currentTarget.style.transform = 'scale(1.05)'')}
88     onMouseLeave={e => (e.currentTarget.style.transform = 'scale(1)'')}
89 >
90     Kembali ke Kampus Digital
91 </button>
92
93 <style>{`
94     @keyframes shine {
95         0% { transform: rotate(25deg) translateX(-100%); }
96         50% { transform: rotate(25deg) translateX(100%); }
97         100% { transform: rotate(25deg) translateX(-100%); }
98     }
99     @keyframes fadeIn { from { opacity: 0; } to { opacity: 1; } }
100     @keyframes fadeInUp { from { opacity: 0; transform: translateY(20px); } to { opacity: 1; transform: translateY(0); } }
101     .badge:hover { transform: scale(1.1); box-shadow: 0 4px 12px rgba(0,0,0,0.4); }
102 `}</style>
103 </div>
104 </div>
105 );
106 }
107
108 export default Mission1_Complete;

```

```
1 import React, { useState } from 'react';
2
3 const initialAvailableSteps = [
4   { id: 's1', content: 'Cetak Slip Pendaftaran' },
5   { id: 's2', content: 'Masukkan Nama Pelajar, No Pendaftaran,\n Program, Semester, Kursus' },
6   { id: 's3', content: 'Daftar kursus' },
7   { id: 's4', content: 'Tamat' },
8   { id: 's5', content: 'Mula' },
9 ];
10
11 function Mission1_Pembinaan({ onContinue, onFeedback, onBadgeEarned }) {
12   const [availableSteps, setAvailableSteps] = useState(initialAvailableSteps);
13   const [orderedSteps, setOrderedSteps] = useState([]);
14   const [isCorrect, setIsCorrect] = useState(false);
15
16   // Scoring State
17   const [earnedScore, setEarnedScore] = useState(0);
18   const [attempts, setAttempts] = useState(0); // Track attempts locally
19   const [isSubmitting, setIsSubmitting] = useState(false);
20
21   const moveStep = (stepToMove) => {
22     setAvailableSteps(prev => prev.filter(step => step.id !== stepToMove.id));
23     setOrderedSteps(prev => [...prev, stepToMove]);
24     setIsCorrect(false);
25   };
26
27   const moveUp = (index) => {
28     if (index === 0) return;
29     setOrderedSteps(prev => {
30       const newOrder = [...prev];
31       [newOrder[index - 1], newOrder[index]] = [newOrder[index], newOrder[index - 1]];
32       return newOrder;
33     });
34     setIsCorrect(false);
35   };
36
37   const moveDown = (index) => {
38     if (index === orderedSteps.length - 1) return;
39     setOrderedSteps(prev => {
40       const newOrder = [...prev];
41       [newOrder[index], newOrder[index + 1]] = [newOrder[index + 1], newOrder[index]];
42       return newOrder;
43     });
44     setIsCorrect(false);
45   };
46
47   // -----
48   // Updated: checkAnswer & submitPhase
49   // -----
50   const submitPhase = async (score, badge = null) => {
51     const token = localStorage.getItem('token');
52     if (!token) return;
53
54     try {
55       await fetch('https://algoquest-api.onrender.com/api/mission/submit', {
56         method: 'POST',
57         headers: {
58           'Content-Type': 'application/json',
59           'Authorization': `Bearer ${token}`
60         },
61         body: JSON.stringify({
62           mission: 1,
63           phase: 'pembinaan',
64           isCorrect: true,
65           score: score,
66           badge: badge
67         })
68       });
69     } catch (error) {
70       console.error('Error submitting mission:', error);
71     }
72   };
73 }
```

```

68     });
69     } catch (err) {
70       console.error("Mission submit failed:", err);
71       onFeedback('⚠️ Ralat menghantar markah ke pelayan.', 3000, false);
72     }
73   });
74
75   const checkAnswer = async () => {
76     const correctOrder = ['s5', 's2', 's3', 's1', 's4'];
77     const playerOrder = orderedSteps.map(step => step.id);
78     const ok = JSON.stringify(playerOrder) === JSON.stringify(correctOrder);
79
80     if (!ok) {
81       setAttempts(prev => prev + 1);
82       setIsCorrect(false);
83       onFeedback('❌ 😞 Masih ada langkah yang tersilap! (-5 Markah)', 3000, false);
84       return;
85     }
86
87     const calculatedScore = Math.max(5, 25 - (attempts * 5));
88     const badgeEarned = attempts === 0 ? 'Master Algoritma' : null;
89
90     setEarnedScore(calculatedScore);
91     setIsCorrect(true);
92
93     await submitPhase(calculatedScore, badgeEarned);
94
95     let msg = `✅ 🎉 Hebat! Algoritma kamu betul!`;
96     if (badgeEarned) msg += ' 🏆 Lencana Master Algoritma diperolehi!';
97     onFeedback(msg, 3000, true);
98   });
99
100   const reset = () => {
101     setAvailableSteps(initialAvailableSteps);
102     setOrderedSteps([]);
103     setIsCorrect(false);
104     setAttempts(0);
105     onFeedback('🔄 Susunan algoritma telah direset. Cuba bina semula dari awal.', 2000, null);
106   });
107
108   const handleNext = () => {
109     if (!isCorrect) return;
110     const badge = attempts === 0 ? 'Master Algoritma' : null;
111
112     // ✅ Notify Hub immediately about the earned badge
113     if (badge && typeof onBadgeEarned === 'function') {
114       onBadgeEarned('mission1', badge);
115     }
116
117     onContinue(earnedScore, badge);
118   };
119
120
121   return (
122     <div>
123       <h3>TAHAP 3: PEMBINAAN ALGORITMA</h3>
124       <p>
125         Sistem Akademik memerlukan algoritma baharu bagi memastikan proses pendaftaran pelajar berjalan der
126         Tugas anda ialah menyusun langkah pendaftaran dalam urutan yang betul dalam penghasilan pseudokod
127       </p>
128       <hr />
129
130       <div className="sorter-container">
131         <div className="sorter-box" style={{ backgroundColor: '#282c34' }}>
132           <h4>SENARAI LANGKAH</h4>
133           {availableSteps.map((step) => (
134             <div key={step.id} className="sorter-item">
135               <span style={step.id === 's2' ? { textAlign: 'left', whiteSpace: 'pre-line', display: 'block' } : {}}>
136                 {step.content}
137               </span>
138               <button onClick={() => moveStep(step)}>🔄</button>
139             </div>
140           ))}
141         </div>
142
143         <div className="sorter-box" style={{ backgroundColor: 'rgba(0,0,0,0.2)' }}>
144           <h4>SUSUNAN ALGORITMA</h4>

```

```

145         {orderedSteps.map((step, index) => (
146             <div key={step.id} className="sorter-item">
147                 <span style={step.id === 's2' ? { textAlign: 'left', whiteSpace: 'pre-line', display: 'block'
148                     {step.content}}
149                 </span>
150                 <div>
151                     <button onClick={() => moveUp(index)} disabled={index === 0}></button>
152                     <button onClick={() => moveDown(index)} disabled={index === orderedSteps.length - 1}></b
153                 </div>
154             </div>
155         ))}
156     </div>
157 </div>
158
159 <hr />
160
161 <div style={{ display: 'flex', justifyContent: 'flex-end', gap: 10, marginTop: 20 }}>
162     <button onClick={reset} className="primary-button" disabled={isSubmitting}>
163         Buat Semula
164     </button>
165
166     <button onClick={checkAnswer} className="primary-button" disabled={isSubmitting}>
167         {isSubmitting ? 'Menghantar...' : 'Semak Jawapan'}
168     </button>
169
170     <button
171         className="primary-button"
172         style={{
173             backgroundColor: isCorrect ? '#2ecc71' : '#999',
174             cursor: isCorrect ? 'pointer' : 'not-allowed',
175         }}
176         disabled={!isCorrect}
177         onClick={handleNext}
178     >
179         Seterusnya
180     </button>
181 </div>
182
183 </div>
184 );
185 }
186
187 export default Mission1_Pembinaan;

```

```

1  import React, { useState } from 'react';
2  import { DndContext, DragOverlay, closestCenter } from '@dnd-kit/core';
3  import { SortableContext, arrayMove, verticalListSortingStrategy, useSortable } from '@dnd-kit/sortable';
4  import { useDroppable } from '@dnd-kit/core';
5  import { CSS } from '@dnd-kit/utilities';
6
7  // --- Reusable Sortable Item ---
8  function SortableItem({ id, content }) {
9    const { attributes, listeners, setNodeRef, transform, transition } = useSortable({ id });
10   const style = { transform: CSS.Transform.toString(transform), transition };
11   return (
12     <div
13       ref={setNodeRef}
14       style={style}
15       {...attributes}
16       {...listeners}
17       className="dnd-item"
18     >
19       {content}
20     </div>
21   );
22 }
23
24 // --- Reusable Draggable Item (for the overlay) ---
25 function DraggableItem({ id, content }) {
26   return (
27     <div className="dnd-item-overlay">
28       {content}
29     </div>
30   );
31 }
32
33 // --- Reusable Droppable Column (using new class) ---
34 function DroppableSortableColumn({ id, title, items }) {
35   const { setNodeRef } = useDroppable({ id });
36
37   return (
38     <div
39       ref={setNodeRef}
40       className={`dnd-column-split ${id.startsWith('available') ? 'available' : ''}`}
41     >
42       <h4>{title.toUpperCase()}</h4>
43       <SortableContext items={items.map(i => i.id)} strategy={verticalListSortingStrategy}>
44         {items.map(item => (
45           <SortableItem key={item.id} id={item.id} content={item.content} />
46         ))}
47       </SortableContext>
48     </div>
49   );
50 }
51
52 // --- Initial states so BUAT SEMULA can reset cleanly ---
53 const INITIAL_MAKLUMAT = {
54   available: [
55     { id: 'm-1', content: 'Nama Pelajar' },
56     { id: 'm-2', content: 'No Telefon' },
57     { id: 'm-3', content: 'Semester' },
58     { id: 'm-4', content: 'Alamat' },
59     { id: 'm-5', content: 'Program' },
60     { id: 'm-6', content: 'Masa' },
61     { id: 'm-7', content: 'Kursus' },
62     { id: 'm-8', content: 'No Pendaftaran' },
63   ],
64   pending: [],
65 };
66
67 const INITIAL_LANGKAH = {

```



```

68     available_langkah: [
69       { id: '1-1', content: 'Daftar Kursus' },
70       { id: '1-2', content: 'Cetak Slip Pendaftaran' },
71       { id: '1-3', content: 'Isi Maklumat Pelajar' },
72     ],
73     ordered_langkah: [],
74   };
75
76   // simple deep clone helper
77   const clone = (obj) => JSON.parse(JSON.stringify(obj));
78
79   // --- Main Component ---
80   function Mission1_Pengabstrakan({ onContinue, onFeedback }) {
81     const [maklumatItems, setMaklumatItems] = useState(() => clone(INITIAL_MAKLUMAT));
82     const [langkahItems, setLangkahItems] = useState(() => clone(INITIAL_LANGKAH));
83     const [activeId, setActiveId] = useState(null);
84     const [isCorrect, setIsCorrect] = useState(false);
85
86     // Scoring State
87     const [earnedScore, setEarnedScore] = useState(0);
88     const [attempts, setAttempts] = useState(0); // Track attempts locally
89     const [isSubmitting, setIsSubmitting] = useState(false);
90
91     function findContainer(itemId) {
92       if (!itemId) return null;
93       if (Object.keys(maklumatItems).includes(itemId)) return itemId;
94       if (Object.keys(langkahItems).includes(itemId)) return itemId;
95       for (const container of Object.keys(maklumatItems)) {
96         if (maklumatItems[container].find(item => item.id === itemId)) return container;
97       }
98       for (const container of Object.keys(langkahItems)) {
99         if (langkahItems[container].find(item => item.id === itemId)) return container;
100       }
101       return null;
102     }
103
104     function handleDragStart(event) {
105       setActiveId(event.active.id);
106     }
107
108     function handleDragEnd(event) {
109       const { active, over } = event;
110       if (!over) {
111         setActiveId(null);
112         return;
113       }
114       const activeContainer = findContainer(active.id);
115       const overContainer = findContainer(over.id);
116       if (!activeContainer || !overContainer) {
117         setActiveId(null);
118         return;
119       }
120
121       // Reset correct state if user moves items after checking
122       setIsCorrect(false);
123
124       if (Object.keys(maklumatItems).includes(activeContainer)) {
125         setItemsState(maklumatItems, setMaklumatItems, active, over);
126       } else {
127         setItemsState(langkahItems, setLangkahItems, active, over);
128       }
129       setActiveId(null);
130     }
131
132     function setItemsState(items, setItems, active, over) {
133       const activeId = active.id;
134       const overId = over.id;
135       const activeContainer = findContainer(activeId);
136       const overContainer = findContainer(overId) || overId;
137
138       if (activeContainer !== overContainer) {
139         setItems((prev) => {
140           const activeItems = prev[activeContainer];
141           const overItems = prev[overContainer];
142           const activeIndex = activeItems.findIndex((item) => item.id === activeId);
143           const [movedItem] = activeItems.splice(activeIndex, 1);
144           let overIndex;

```

```

145     const overItem = overItems.find(item => item.id === overId);
146     if (overItem) {
147         overIndex = overItems.findIndex((item) => item.id === overId);
148     } else {
149         overIndex = overItems.length;
150     }
151     overItems.splice(overIndex, 0, movedItem);
152     return { ...prev, [activeContainer]: [...activeItems], [overContainer]: [...overItems] };
153 });
154 } else {
155     setItems((prev) => {
156         const activeItems = prev[activeContainer];
157         const activeIndex = activeItems.findIndex((item) => item.id === active.id);
158         const overIndex = activeItems.findIndex((item) => item.id === overId);
159         return { ...prev, [activeContainer]: arrayMove(activeItems, activeIndex, overIndex) };
160     });
161 }
162 }
163
164 const checkAnswer = async () => {
165     const correctMaklumat = ['m-1', 'm-3', 'm-5', 'm-7', 'm-8'].sort();
166     const playerMaklumat = maklumatItems.penting.map(i => i.id).sort();
167     const isMaklumatCorrect = JSON.stringify(correctMaklumat) === JSON.stringify(playerMaklumat);
168     const correctLangkah = ['l-3', 'l-1', 'l-2'];
169     const playerLangkah = langkahItems.ordered_langkah.map(i => i.id);
170     const isLangkahCorrect = JSON.stringify(correctLangkah) === JSON.stringify(playerLangkah);
171
172     const ok = isMaklumatCorrect && isLangkahCorrect;
173
174     if (!ok) {
175         // Incorrect: Deduct marks locally
176         setAttempts(prev => prev + 1);
177         setIsCorrect(false);
178
179         let errorMsg = 'Hmm... ada langkah belum kena. ';
180         if (!isMaklumatCorrect) errorMsg += 'Semak bahagian MAKLUMAT. ';
181         if (!isLangkahCorrect) errorMsg += 'Semak bahagian MODEL LOGIK.';
182
183         // Robot Glow: Red ('false')
184         onFeedback(errorMsg, 3000, false);
185         return;
186     }
187
188     // Correct: Calculate final score for this session
189     const calculatedScore = Math.max(5, 25 - (attempts * 5));
190
191     setIsSubmitting(true);
192     const token = localStorage.getItem('token');
193
194     try {
195         await fetch('https://algoquest-api.onrender.com/api/mission/submit', {
196             method: 'POST',
197             headers: {
198                 'Content-Type': 'application/json',
199                 'Authorization': `Bearer ${token}`
200             },
201             body: JSON.stringify({
202                 mission: 1,
203                 phase: 'pengabstrakan',
204                 isCorrect: true,
205                 score: calculatedScore, // Send calculated score
206                 badge: null
207             })
208         });
209
210         setEarnedScore(calculatedScore);
211         setIsCorrect(true);
212         // Robot Glow: Green ('true')
213         onFeedback(`Hebat! Kamu berjaya membina model logik ringkas untuk sistem pendaftaran.`, 3000, true);
214
215     } catch (err) {
216         console.error("Error submitting:", err);
217         onFeedback(`⚠️ Ralat menghubungi pelayan.`, 3000, false);
218     } finally {
219         setIsSubmitting(false);
220     }
221 };

```

```

222
223   const handleReset = () => {
224     setMaklumatItems(clone(INITIAL_MAKLUMAT));
225     setLangkahItems(clone(INITIAL_LANGKAH));
226     setActiveId(null);
227     setIsCorrect(false);
228     setAttempts(0); // Reset attempts to 0 for fresh scoring
229     // Robot Glow: Neutral
230     onFeedback('🔄 Susunan telah direset. Cuba semula dengan teliti.', 2000, null);
231   };
232
233   const allItems = [...maklumatItems.available, ...maklumatItems.penting, ...langkahItems.available_langkah];
234   const activeItem = activeId ? allItems.find(i => i.id === activeId) : null;
235
236   return (
237     <div>
238       <h3>TAHAP 2: PENGABSTRAKAN</h3>
239       <p>Sistem Akademik menerima banyak maklumat pelajar tetapi hanya data penting diperlukan untuk penda
240 Tugas anda ialah memilih dan menyusun maklumat penting supaya proses pendaftaran berjalan dengan betul!
241 </p>
242       <hr />
243
244       <DndContext onDragStart={handleDragStart} onDragEnd={handleDragEnd} collisionDetection={closestCenter}>
245         <div className="dnd-container-split">
246           <DraggableSortColumn id="available" title="MAKLUMAT" items={maklumatItems.available} />
247           <DraggableSortColumn id="penting" title="MAKLUMAT PENTING" items={maklumatItems.penting} />
248         </div>
249
250         <div className="dnd-container-split">
251           <DraggableSortColumn id="available_langkah" title="LANGKAH MODEL (TIDAK TERSUSUN)" items={langkahItems.available_langkah} />
252           <DraggableSortColumn id="ordered_langkah" title="MODEL LOGIK (TERSUSUN)" items={langkahItems.ordered_langkah} />
253         </div>
254
255         <DragOverlay>
256           {activeItem ? <DraggableItem id={activeItem.id} content={activeItem.content} /> : null}
257         </DragOverlay>
258       </DndContext>
259
260       <hr />
261
262       <div style={{ display: 'flex', justifyContent: 'flex-end', gap: 10, marginTop: 20 }}>
263         <button className="primary-button" onClick={handleReset} disabled={isSubmitting}>
264           Buat Semula
265         </button>
266
267         <button className="primary-button" onClick={checkAnswer} disabled={isSubmitting}>
268           {isSubmitting ? 'Menghantar...' : 'Semak Jawapan'}
269         </button>
270
271         <button
272           className="primary-button"
273           style={{
274             backgroundColor: isCorrect ? '#2ecc71' : '#999',
275             cursor: isCorrect ? 'pointer' : 'not-allowed'
276           }}
277           disabled={!isCorrect}
278           onClick={isCorrect ? () => onContinue(earnedScore) : undefined}
279         >
280           Seterusnya
281         </button>
282       </div>
283     </div>
284   );
285 }
286
287 export default Mission1_Pengabstrakan;

```

```

1  import React, { useState } from 'react';
2  import { DndContext, DragOverlay, closestCenter } from '@dnd-kit/core';
3  import {
4    SortableContext,
5    arrayMove,
6    verticalListSortingStrategy,
7    useSortable
8  } from '@dnd-kit/sortable';
9  import { useDroppable } from '@dnd-kit/core';
10 import { CSS } from '@dnd-kit/utilities';
11
12 // --- Reusable Sortable Item ---
13 function SortableItem({ id, content }) {
14   const { attributes, listeners, setNodeRef, transform, transition } = useSortable({ id });
15   const style = { transform: CSS.Transform.toString(transform), transition };
16
17   return (
18     <div
19       ref={setNodeRef}
20       style={style}
21       {...attributes}
22       {...listeners}
23       className="dnd-item"
24     >
25       {content}
26     </div>
27   );
28 }
29
30 // --- Overlay Draggable Item ---
31 function DraggableItem({ content }) {
32   return <div className="dnd-item-overlay">{content}</div>;
33 }
34
35 // --- Droppable Column ---
36 function DroppableSortableColumn({ id, title, items }) {
37   const { setNodeRef } = useDroppable({ id });
38
39   return (
40     <div ref={setNodeRef} className={`dnd-column ${id}`>
41       <h4>{title}</h4>
42
43       <SortableContext items={items.map(i => i.id)} strategy={verticalListSortingStrategy}>
44         {items.map(item => (
45           <SortableItem key={item.id} id={item.id} content={item.content} />
46         ))}
47       </SortableContext>
48     </div>
49   );
50 }
51
52 const INITIAL = {
53   available: [
54     { id: 'i1', content: 'Nama Pelajar' },
55     { id: 'i2', content: 'No Pendaftaran' },
56     { id: 'i3', content: 'Program' },
57     { id: 'i4', content: 'Kursus' },
58     { id: 'i5', content: 'Cetak Slip Pendaftaran' },
59     { id: 'i6', content: 'Daftar kursus' },
60     { id: 'i7', content: 'Semester' }
61   ],
62   input: [],
63   proses: [],
64   output: []
65 };
66
67 const clone = obj => JSON.parse(JSON.stringify(obj));

```

```

68
69 function Mission1_Penguraian({ onContinue, onFeedback }) {
70   const [items, setItems] = useState(clone(INITIAL));
71   const [activeId, setActiveId] = useState(null);
72
73   // Scoring State
74   const [earnedScore, setEarnedScore] = useState(0);
75   const [attempts, setAttempts] = useState(0); // Tracks wrong attempts
76   const [isSubmitting, setIsSubmitting] = useState(false);
77
78   // --- find container of item ---
79   const findContainer = id => {
80     for (const key in items) {
81       if (items[key].find(i => i.id === id)) return key;
82     }
83     return null;
84   };
85
86   // --- Drag events ---
87   const handleDragStart = e => setActiveId(e.active.id);
88
89   const handleDragEnd = e => {
90     const { active, over } = e;
91     if (!over) return setActiveId(null);
92
93     const from = findContainer(active.id);
94     const to = findContainer(over.id) || over.id;
95
96     if (!from || !to) return;
97
98     if (from === to) {
99       // reorder
100       setItems(prev => {
101         const list = [...prev[from]];
102         const fromIndex = list.findIndex(i => i.id === active.id);
103         const toIndex = list.findIndex(i => i.id === over.id);
104         return {
105           ...prev,
106           [from]: arrayMove(list, fromIndex, toIndex)
107         };
108       });
109     } else {
110       // move between lists
111       setItems(prev => {
112         const source = [...prev[from]];
113         const target = [...prev[to]];
114         const moving = source.find(i => i.id === active.id);
115
116         return {
117           ...prev,
118           [from]: source.filter(i => i.id !== active.id),
119           [to]: [...target, moving]
120         };
121       });
122     }
123
124     setActiveId(null);
125   };
126
127   // --- CHECK Answer & SCORING ---
128   const checkAnswer = async () => {
129     const correct = {
130       input: ['i1', 'i2', 'i3', 'i4', 'i7'],
131       proses: ['i6'],
132       output: ['i5']
133     };
134
135     const ok =
136       JSON.stringify(items.input.map(i => i.id).sort()) === JSON.stringify(correct.input.sort()) &&
137       JSON.stringify(items.proses.map(i => i.id).sort()) === JSON.stringify(correct.proses.sort()) &&
138       JSON.stringify(items.output.map(i => i.id).sort()) === JSON.stringify(correct.output.sort());
139
140     if (ok) {
141       const finalScore = Math.max(5, 25 - (attempts * 5));
142       setEarnedScore(finalScore);
143       onFeedback('👏 Hebat! Semua komponen betul!', 3000, true);
144     } else {

```

```

145     setAttempts(prev => prev + 1);
146     onFeedback('😞 Masih ada komponen yang tersilap! (-5 Markah)', 3000, false);
147   }
148 };
149
150 // --- RESET ---
151 const handleReset = () => {
152   setItems(clone(INITIAL));
153   setAttempts(0);
154   onFeedback('👏 Reset berjaya. Markah kembali penuh.', 2000, null);
155 };
156
157 // --- SUBMIT TO BACKEND ---
158 const handleNext = async () => {
159   if (earnedScore === 0) return; // Only allow if scored
160
161   setIsSubmitting(true);
162   const token = localStorage.getItem('token');
163
164   try {
165     await fetch('https://algoquest-api.onrender.com/api/mission/submit', {
166       method: 'POST',
167       headers: {
168         'Content-Type': 'application/json',
169         'Authorization': `Bearer ${token}`
170       },
171       body: JSON.stringify({
172         mission: 1,
173         phase: 'penguraian',
174         isCorrect: true,
175         score: earnedScore,
176         badge: null
177       })
178     });
179
180     onContinue(earnedScore);
181
182   } catch (err) {
183     console.error("Error submitting:", err);
184   } finally {
185     setIsSubmitting(false);
186   }
187 };
188
189 const activeItem =
190   activeId &&
191   [...items.available, ...items.input, ...items.proses, ...items.output].find(
192     i => i.id === activeId
193   );
194
195 return (
196   <div>
197     <h3>TAHAP 1: PENGURAIAN</h3>
198     <p>Sistem Akademik sedang membina modul pendaftaran pelajar baharu. Namun, sistem masih belum dapat r
199
200     <DndContext
201       onDragStart={handleDragStart}
202       onDragEnd={handleDragEnd}
203       collisionDetection={closestCenter}
204     >
205       <div className="dnd-container">
206         <DraggableSortColumn id="available" title="Item Tersedia" items={items.available} />
207         <DraggableSortColumn id="input" title="Input" items={items.input} />
208         <DraggableSortColumn id="proses" title="Proses" items={items.proses} />
209         <DraggableSortColumn id="output" title="Output" items={items.output} />
210       </div>
211
212       <DragOverlay>
213         {activeItem ? <DraggableItem content={activeItem.content} /> : null}
214       </DragOverlay>
215     </DndContext>
216
217     <div style={{ display: 'flex', justifyContent: 'flex-end', gap: 10, marginTop: 20 }}>
218       <button className="primary-button" onClick={handleReset} disabled={isSubmitting}>
219         Buat Semula
220       </button>
221

```

```

222     <button
223       className="primary-button"
224       onClick={checkAnswer}
225       disabled={isSubmitting || earnedScore > 0}
226     >
227       Semak Jawapan
228     </button>
229
230     <button
231       className="primary-button"
232       style={{
233         backgroundColor: earnedScore > 0 ? '#2ecc71' : '#999',
234         cursor: earnedScore > 0 ? 'pointer' : 'not-allowed'
235       }}
236       disabled={earnedScore === 0 || isSubmitting}
237       onClick={handleNext}
238     >
239       {isSubmitting ? 'Menghantar...' : 'Seterusnya'}
240     </button>
241   </div>
242 </div>
243 );
244 }
245
246
247 export default Mission1_Penguraian;

```

```
1 import React, { useState } from 'react';
2
3 function Mission1_Penyahpepijat({ onContinue, onFeedback, onBadgeEarned }) {
4   const initialAnswers = {
5     step1: '', step2: '', step3: '', step4: '', step5: '',
6   };
7
8   const [answers, setAnswers] = useState(initialAnswers);
9   const [showNextButton, setShowNextButton] = useState(false);
10
11   // Scoring State
12   const [earnedScore, setEarnedScore] = useState(0);
13   const [attempts, setAttempts] = useState(0); // Track attempts locally
14   const [isSubmitting, setIsSubmitting] = useState(false);
15
16   const handleInputChange = (e) => {
17     const { name, value } = e.target;
18     setAnswers(prev => ({ ...prev, [name]: value }));
19     setShowNextButton(false);
20   };
21
22   const reset = () => {
23     setAnswers(initialAnswers);
24     setShowNextButton(false);
25     setAttempts(0); // Reset attempts for full marks
26     onFeedback('🔄 Reset berjaya. Markah kembali penuh.', 2000, null);
27   };
28
29   // -----
30   // Updated: checkAnswer & submitPhase
31   // -----
32   const submitPhase = async (score, badge = null) => {
33     const token = localStorage.getItem('token');
34     if (!token) return;
35
36     try {
37       await fetch('https://algoquest-api.onrender.com/api/mission/submit', {
38         method: 'POST',
39         headers: {
40           'Content-Type': 'application/json',
41           'Authorization': `Bearer ${token}`
42         },
43         body: JSON.stringify({
44           mission: 1,
45           phase: 'penyahpepijat',
46           isCorrect: true,
47           score: score,
48           badge: badge
49         })
50       });
51     } catch (err) {
52       console.error("Mission submit failed:", err);
53       onFeedback('⚠️ Ralat menghantar markah ke pelayan.', 3000, false);
54     }
55   };
56
57   const checkAnswer = async () => {
58     const isCorrect =
59       answers.step1 === '1' &&
60       answers.step2 === '2' &&
61       answers.step3 === '4' &&
62       answers.step4 === '3' &&
63       answers.step5 === '5';
64
65     if (!isCorrect) {
66       setAttempts(prev => prev + 1);
67       onFeedback('❌ Masih ada ralat. (-5 Markah)', 3000, false);
68     }
69   };
70 }
```



```

68     setShowNextButton(false);
69     return;
70 }
71
72 // Correct: Calculate Score (Start 25, minus 5 per retry, min 5)
73 const calculatedScore = Math.max(5, 25 - (attempts * 5));
74 const badgeEarned = attempts === 0 ? 'Master Pemulih Logik' : null;
75
76 setEarnedScore(calculatedScore);
77 setIsSubmitting(true);
78
79 await submitPhase(calculatedScore, badgeEarned);
80
81 let badgeMsg = badgeEarned ? '🏆 Lencana Master Pemulih Logik diperolehi!' : '';
82 onFeedback(✅ Hebat! Urutan betul.${badgeMsg}`, 3000, true);
83 setShowNextButton(true);
84 setIsSubmitting(false);
85 };
86
87 const handleNext = () => {
88   if (showNextButton) {
89     const badge = attempts === 0 ? 'Master Pemulih Logik' : null;
90
91     // ✅ Notify Hub immediately about the earned badge
92     if (badge && typeof onBadgeEarned === 'function') {
93       onBadgeEarned('mission1', badge);
94     }
95
96     onContinue(earnedScore, badge);
97   }
98 };
99
100
101 return (
102   <div>
103     <h3>TAHAP 4: PENYAHPEIJATAN</h3>
104     <p>Sistem Akademik mencetak slip pendaftaran sebelum pelajar mendaftar kursus. Susun semula langkah p
105   </p>
106   <hr />
107
108   <div className="debug-container">
109     <div className="pseudocode-box">
110       <h4 style={{ textAlign: 'center', fontWeight: 'bold' }}>
111         PSEUDOKOD (ralat disorot)
112       </h4>
113       <p>[1] MULA</p>
114       <p>[2] Masukkan Nama Pelajar, No Pendaftaran, Program, Semester, Kursus</p>
115       <p>[3] Cetak Slip Pendaftaran</p>
116       <p>[4] Daftar Kursus</p>
117       <p>[5] TAMAT</p>
118       <hr />
119       <p><strong>Masalah:</strong> Sistem mencetak slip sebelum pelajar mendaftar kursus.</p>
120       <p><strong>Output salah:</strong> Slip kosong tanpa maklumat kursus.</p>
121     </div>
122
123     <div className="debug-answer-box">
124       <h4 className="debug-title">SUSUNAN BETUL</h4>
125       <p className="debug-title">Tulis nombor urutan yang betul (1-5) di dalam kotak.</p>
126       {[ 'step1', 'step2', 'step3', 'step4', 'step5' ].map((step) => (
127         <div key={step} style={{ margin: '10px 0' }}>
128           <input
129             type="text"
130             name={step}
131             value={answers[step]}
132             onChange={handleInputChange}
133             className="order-input"
134           />
135           <label>
136             {step === 'step1' && ' MULA'}
137             {step === 'step2' && ' Masukkan Nama Pelajar, No Pendaftaran...'}
138             {step === 'step3' && ' Cetak Slip Pendaftaran'}
139             {step === 'step4' && ' Daftar Kursus'}
140             {step === 'step5' && ' TAMAT'}
141           </label>
142         </div>
143       ))}
144     </div>

```

```

145     </div>
146
147     <hr />
148     <div style={{ display: 'flex', justifyContent: 'flex-end', gap: 10, marginTop: 20 }}>
149       <button onClick={reset} className="primary-button" disabled={isSubmitting}>
150         Buat Semula
151       </button>
152
153       <button onClick={checkAnswer} className="primary-button" disabled={isSubmitting}>
154         {isSubmitting ? 'Menghantar...' : 'Semak Jawapan'}
155       </button>
156
157       <button
158         className="primary-button"
159         style={{
160           backgroundColor: showNextButton ? '#2ecc71' : '#999',
161           cursor: showNextButton ? 'pointer' : 'not-allowed',
162         }}
163         disabled={!showNextButton}
164         onClick={handleNext}
165       >
166         Hantar
167       </button>
168     </div>
169   </div>
170 );
171 }
172
173 export default Mission1_Penyahpepijat;

```

```
1 import React, { useState } from 'react';
2 import MissionLayout from './MissionLayout';
3 import Mission2_Penguraian from './Mission2_Penguraian';
4 import Mission2_Pengabstrakan from './Mission2_Pengabstrakan';
5 import Mission2_Pembinaan from './Mission2_Pembinaan';
6 import Mission2_Penyahpepijat from './Mission2_Penyahpepijat';
7 import Mission2_Complete from './Mission2_Complete';
8
9 function Mission2({ onMissionComplete }) {
10   const [missionPhase, setMissionPhase] = useState('penguraian');
11   const [totalMissionScore, setTotalMissionScore] = useState(0);
12   const [earnedBadges, setEarnedBadges] = useState([]);
13
14   // Global feedback state for Robot
15   const [feedbackText, setFeedbackText] = useState('');
16   const [isCorrect, setIsCorrect] = useState(null);
17
18   // --- API HELPERS ---
19   const savePhaseScore = async (phaseName, score) => {
20     const token = localStorage.getItem('token');
21     try {
22       await fetch('https://algoquest-api.onrender.com/api/mission/submit', {
23         method: 'POST',
24         headers: { 'Content-Type': 'application/json', 'Authorization': `Bearer ${token}` },
25         body: JSON.stringify({ mission: 2, phase: phaseName, isCorrect: score > 0, score: score, badge: null });
26       });
27     } catch (e) { console.error("Save failed", e); }
28   };
29
30   const saveBadge = async (badgeName) => {
31     const token = localStorage.getItem('token');
32     try {
33       await fetch('https://algoquest-api.onrender.com/api/mission/submit', {
34         method: 'POST',
35         headers: { 'Content-Type': 'application/json', 'Authorization': `Bearer ${token}` },
36         body: JSON.stringify({ mission: 2, phase: missionPhase, isCorrect: true, score: 0, badge: badgeName });
37       });
38     } catch (e) { console.error("Badge save failed", e); }
39   };
40
41   // --- FEEDBACK HANDLER ---
42   const handleFeedback = (message, duration = 3000, correctState = null) => {
43     setFeedbackText(message);
44
45     // Pass true/false/null for robot glow
46     setIsCorrect(correctState);
47
48     setTimeout(() => {
49       setFeedbackText('');
50       setIsCorrect(null);
51     }, duration);
52   };
53
54   // --- PHASE COMPLETION HANDLER ---
55   const handlePhaseComplete = (nextPhase, scoreEarned, badgeEarned = null) => {
56     const isCorrectAnswer = scoreEarned > 0;
57
58     setTotalMissionScore(prev => prev + scoreEarned);
59     if (badgeEarned) {
60       setEarnedBadges(prev => [...prev, badgeEarned]);
61     }
62
63     // Save data to backend
64     savePhaseScore(missionPhase, scoreEarned);
65     if (badgeEarned) saveBadge(badgeEarned);
66
67     // Set robot glow for phase completion
68     setIsCorrect(isCorrectAnswer);
```

```

69     setMissionPhase(nextPhase);
70     setFeedbackText('');
71 };
72
73
74 const robotTexts = {
75   penguraian: "Kenal pasti data yang diperlukan, proses utama, dan hasil akhir. Seret kad ke kotak Input,
76   pengabstrakan: "Tumpukan kepada data yang mempengaruhi keputusan pelajar. Pilih maklumat yang penting u
77   pembinaan: "Padankan setiap baris pseudokod dengan simbol carta alir yang betul. Seret langkah pseudokod
78   penyahpepijat: "Sistem peperiksaan mengalami ralat! Semak pseudokod dan output, kenal pasti ralat logik
79   complete: "Hebat! Sistem peperiksaan kini bebas ralat. Anda boleh kembali ke Kampus Digital."
80 };
81
82 const renderPhase = () => {
83   switch (missionPhase) {
84     case 'penguraian':
85       return (
86         <Mission2_Penguraian
87           onContinue={(score) => handlePhaseComplete('pengabstrakan', score)}
88           onFeedback={handleFeedback}
89         />
90       );
91     case 'pengabstrakan':
92       return (
93         <Mission2_Pengabstrakan
94           onContinue={(score) => handlePhaseComplete('pembinaan', score)}
95           onFeedback={handleFeedback}
96         />
97       );
98     case 'pembinaan':
99       return (
100         <Mission2_Pembinaan
101           onContinue={(score, badge) => handlePhaseComplete('penyahpepijat', score, badge)}
102           onFeedback={handleFeedback}
103         />
104       );
105     case 'penyahpepijat':
106       return (
107         <Mission2_Penyahpepijat
108           onContinue={(score, badge) => handlePhaseComplete('complete', score, badge)}
109           onFeedback={handleFeedback}
110         />
111       );
112     case 'complete':
113       return (
114         <Mission2_Complete
115           score={totalMissionScore}
116           badges={earnedBadges}
117           onContinue={() => onMissionComplete(totalMissionScore, earnedBadges)}
118         />
119       );
120     default:
121       return <div>Memuatkan...</div>;
122   }
123 };
124
125 return (
126   <MissionLayout
127     robotText={feedbackText || robotTexts[missionPhase]}
128     isCorrect={isCorrect} // ✅ robot glow based on correctness
129   >
130     {renderPhase()}
131   </MissionLayout>
132 );
133 }
134
135 export default Mission2;

```

```

1  import React from 'react';
2
3  function Mission2_Complete({ score, badges, onContinue }) {
4    // 1. Ensure badges is always an array
5    const displayBadges = Array.isArray(badges) ? badges : [];
6
7    // 2. Fix: Use the score directly (it's already a number)
8    const finalScore = score || 0;
9
10   return (
11     <div
12       style={{
13         width: '100%',
14         height: '100%',
15         display: 'flex',
16         justifyContent: 'center',
17         alignItems: 'center',
18         animation: 'fadeIn 1s ease-in'
19       }}
20     >
21       <div className="complete-screen" style={{ textAlign: 'center', animation: 'fadeInUp 1s ease-out' }}>
22         <h2>Tahniah! Anda Selesai Unit Peperiksaan! 🎉</h2>
23         <p>Sistem pengiraan markah pelajar kini berfungsi dengan betul.</p>
24         <hr style={{ width: '80%' }} />
25
26         { /* Display the score directly */ }
27         <h3>Markah Misi: {finalScore} / 100</h3>
28
29         {displayBadges.length > 0 && (
30           <div className="badges-container" style={{ margin: '15px 0' }}>
31             <h4>📋 Lencana Diperolehi:</h4>
32             <div style={{ display: 'flex', flexWrap: 'wrap', gap: '10px', justifyContent: 'center' }}>
33               {displayBadges.map((badge, idx) => (
34                 <div
35                   key={idx}
36                   style={{
37                     position: 'relative',
38                     display: 'inline-flex',
39                     alignItems: 'center',
40                     justifyContent: 'center',
41                     padding: '6px 16px',
42                     fontWeight: 'bold',
43                     fontSize: '0.95rem',
44                     color: '#fff',
45                     background: 'linear-gradient(135deg, #FFD700, #FFC200, #FFB000)',
46                     borderRadius: '9999px',
47                     boxShadow: '0 2px 6px rgba(0,0,0,0.3)',
48                     overflow: 'hidden',
49                     textAlign: 'center',
50                     cursor: 'default',
51                     transition: 'transform 0.3s ease, box-shadow 0.3s ease'
52                   }}
53                   className="badge"
54                 >
55                   <span style={{ display: 'inline-block', width: '16px', height: '16px', marginRight: '8px' }}>
56                     <span>{badge}</span>
57                   { /* Shine Animation */ }
58                   <span
59                     style={{
60                       position: 'absolute',
61                       top: '-50%',
62                       left: '-50%',
63                       width: '200%',
64                       height: '200%',
65                       background: 'linear-gradient(120deg, rgba(255,255,255,0.2) 0%, rgba(255,255,255,0) 60%',
66                       transform: 'rotate(25deg)',
67                       pointerEvents: 'none',

```

```

68         animation: 'shine 2s infinite',
69         zIndex: 0
70     }}
71     </span>
72 </div>
73 ))}
74 </div>
75 </div>
76 )}
77
78 <hr style={{ width: '80%' }} />
79 <p>Misi seterusnya menanti anda!</p>
80 <button
81     onClick={onContinue}
82     className="primary-button"
83     style={{
84         width: '300px',
85         marginTop: '20px',
86         transition: 'transform 0.2s ease, background-color 0.2s ease'
87     }}
88     onMouseEnter={e => (e.currentTarget.style.transform = 'scale(1.05)'')}
89     onMouseLeave={e => (e.currentTarget.style.transform = 'scale(1)'')}
90 >
91     Kembali ke Kampus Digital
92 </button>
93
94 <style>{`
95     @keyframes shine {
96         0% { transform: rotate(25deg) translateX(-100%); }
97         50% { transform: rotate(25deg) translateX(100%); }
98         100% { transform: rotate(25deg) translateX(-100%); }
99     }
100     @keyframes fadeIn { from { opacity: 0; } to { opacity: 1; } }
101     @keyframes fadeInUp { from { opacity: 0; transform: translateY(20px); } to { opacity: 1; transform: translateY(0); } }
102     .badge:hover { transform: scale(1.1); box-shadow: 0 4px 12px rgba(0,0,0,0.4); }
103 `}</style>
104 </div>
105 </div>
106 );
107 }
108
109 export default Mission2_Complete;

```

```

1  import React, { useState } from 'react';
2  import { DndContext, DragOverlay, closestCenter } from '@dnd-kit/core';
3  import {
4    SortableContext,
5    verticalListSortingStrategy,
6    useSortable,
7  } from '@dnd-kit/sortable';
8  import { CSS } from '@dnd-kit/utilities';
9  import { useDroppable } from '@dnd-kit/core';
10
11  // ----- Sortable item -----
12  // source: 'list' (pseudocode) or 'shape' (inside flowchart)
13  function SortableItem({ id, content, source = 'list' }) {
14    const { attributes, listeners, setNodeRef, transform, transition } =
15      useSortable({ id });
16
17    const isShape = source === 'shape';
18
19    const style = {
20      transform: CSS.Transform.toString(transform),
21      transition,
22      marginBottom: isShape ? '0' : '10px',
23      fontSize: isShape ? '0.8rem' : '1rem',
24      padding: isShape ? '4px' : '10px',
25      width: '100%',
26      display: 'flex',
27      justifyContent: 'center',
28      alignItems: 'center',
29      touchAction: 'none',
30    };
31
32    return (
33      <div
34        ref={setNodeRef}
35        style={style}
36        {...attributes}
37        {...listeners}
38        className="dnd-item flowchart-shape-rectangle"
39      >
40        {content}
41      </div>
42    );
43  }
44
45  // ----- Drag overlay -----
46  function DraggableItem({ id, content }) {
47    return (
48      <div
49        className="dnd-item-overlay flowchart-shape-rectangle"
50        style={{
51          boxShadow: '0 5px 15px rgba(0,0,0,0.25)',
52          fontSize: '0.9rem',
53          padding: '10px',
54        }}
55      >
56        {content}
57      </div>
58    );
59  }
60
61  // ----- Droppable flowchart shape (container) -----
62  function FlowchartShape({ id, shapeType, items }) {
63    const { setNodeRef, isOver } = useDroppable({ id });
64
65    let shapeClass = 'flowchart-shape-rectangle';
66    if (shapeType === 'oval') shapeClass = 'flowchart-shape-oval';
67    else if (shapeType === 'diamond') shapeClass = 'flowchart-shape-diamond';

```

```

68     else if (shapeType === 'parallelogram') shapeClass = 'flowchart-shape-parallelogram';
69
70     return (
71       <div
72         ref={setNodeRef}
73         className={`flowchart-shape ${shapeClass}`}
74         style={{
75           margin: '5px',
76           padding: '4px',
77           minWidth: '130px',
78           minHeight: '40px',
79           borderColor: '#00ffff',
80           backgroundColor: 'rgba(0,0,0,0.1)',
81           borderWidth: '2px',
82           borderStyle: 'solid',
83           outline: isOver ? '2px dashed #00ffff' : 'none',
84           position: 'relative',
85           display: 'flex',
86           alignItems: 'center',
87           justifyContent: 'center',
88           textAlign: 'center',
89         }}
90       >
91         <SortableContext
92           items={items.map((i) => i.id)}
93           strategy={verticalListSortingStrategy}
94         >
95           {items.length > 0 ? (
96             <div style={{ width: '100%', display: 'flex', justifyContent: 'center' }}>
97               <SortableItem id={items[0].id} content={items[0].content} source="shape" />
98             </div>
99           ) : (
100             <span style={{ opacity: 0.3, fontSize: '0.7rem', pointerEvents: 'none' }}>
101               Seret sini
102             </span>
103           )}
104         </SortableContext>
105       </div>
106     );
107   }
108
109   // ----- Main Component -----
110   function Mission2_Pembinaan({ onContinue, setRobotText, onBadgeEarned, onFeedback }) {
111     const [items, setItems] = useState({
112       available: [
113         { id: 's1', content: 'Mula' },
114         { id: 's7', content: 'Masukkan Markah PB, Markah PA' },
115         { id: 's2', content: 'Jika PB ≥ 50 ?' },
116         { id: 's3', content: 'Jika PA ≥ 50 ?' },
117         { id: 's5', content: 'Set Status = Lulus' },
118         { id: 's4', content: 'Set Status = Gagal' },
119         { id: 's6', content: 'Cetak Status' },
120         { id: 's8', content: 'Tamat' },
121       ],
122       oval1: [],
123       parallelogram1: [],
124       diamond1: [],
125       diamond2: [],
126       rectangle1: [],
127       rectangle2: [],
128       parallelogram2: [],
129       oval2: [],
130     });
131
132     const [activeId, setActiveId] = useState(null);
133     const [isCorrect, setIsCorrect] = useState(false);
134
135     // Scoring State
136     const [earnedScore, setEarnedScore] = useState(0);
137     const [attempts, setAttempts] = useState(0); // Track attempts locally
138     const [isSubmitting, setIsSubmitting] = useState(false);
139
140     function findContainer(itemId) {
141       if (!itemId) return null;
142       if (Object.prototype.hasOwnProperty.call(items, itemId)) return itemId;
143       for (const container of Object.keys(items)) {
144         if (items[container].find((item) => item.id === itemId)) return container;

```



```

145     }
146     return null;
147 }
148
149 const handleDragStart = (event) => {
150     setActiveId(event.active.id);
151     setIsCorrect(false);
152 };
153
154 const handleDragEnd = (event) => {
155     const { active, over } = event;
156     if (!over) {
157         setActiveId(null);
158         return;
159     }
160
161     const activeContainer = findContainer(active.id);
162     const overContainer = findContainer(over.id);
163
164     if (!activeContainer || !overContainer) {
165         setActiveId(null);
166         return;
167     }
168
169     if (activeContainer === overContainer && active.id === over.id) {
170         setActiveId(null);
171         return;
172     }
173
174     setItems((prev) => {
175         const newState = { ...prev };
176         const activeItems = [...prev[activeContainer]];
177         const overItems = [...prev[overContainer]];
178         const activeIndex = activeItems.findIndex((item) => item.id === active.id);
179         if (activeIndex === -1) return prev;
180         const [movedItem] = activeItems.splice(activeIndex, 1);
181
182         if (overContainer === 'available') {
183             const newAvailable = [...prev.available];
184             newAvailable.unshift(movedItem);
185             newState[activeContainer] = activeItems;
186             newState.available = newAvailable;
187             return newState;
188         }
189
190         if (overContainer !== 'available') {
191             const newOver = [...overItems];
192             if (newOver.length === 0) {
193                 newOver.push(movedItem);
194                 newState[activeContainer] = activeItems;
195                 newState[overContainer] = newOver;
196                 return newState;
197             } else {
198                 const [existing] = newOver.splice(0, 1);
199                 newOver.push(movedItem);
200                 const newActive = [...activeItems];
201                 newActive.push(existing);
202                 newState[activeContainer] = newActive;
203                 newState[overContainer] = newOver;
204                 return newState;
205             }
206         }
207         return prev;
208     });
209     setActiveId(null);
210 };
211
212 const handleReset = () => {
213     setItems({
214         available: [
215             { id: 's1', content: 'Mula' },
216             { id: 's7', content: 'Masukkan Markah PB, Markah PA' },
217             { id: 's2', content: 'Jika PB  $\geq$  50 ?' },
218             { id: 's3', content: 'Jika PA  $\geq$  50 ?' },
219             { id: 's5', content: 'Set Status = Lulus' },
220             { id: 's4', content: 'Set Status = Gagal' },
221             { id: 's6', content: 'Cetak Status' },

```

```

222     { id: 's8', content: 'Tamat' },
223   ],
224   oval1: [],
225   parallelogram1: [],
226   diamond1: [],
227   diamond2: [],
228   rectangle1: [],
229   rectangle2: [],
230   parallelogram2: [],
231   oval2: [],
232 });
233 setIsCorrect(false);
234 setAttempts(0); // Reset attempts
235 onFeedback?.(🔄 Carta alir telah direset. Coba susun semula langkah yang betul.', 2000, null);
236 };
237
238 const checkAnswer = async () => {
239   const correctOrder = {
240     oval1: 'Mula',
241     parallelogram1: 'Masukkan Markah PB, Markah PA',
242     diamond1: 'Jika PB ≥ 50 ?',
243     diamond2: 'Jika PA ≥ 50 ?',
244     rectangle1: 'Set Status = Lulus',
245     rectangle2: 'Set Status = Gagal',
246     parallelogram2: 'Cetak Status',
247     oval2: 'Tamat',
248   };
249
250   const isRight = Object.entries(correctOrder).every(([slot, text]) => {
251     return items[slot][0]?.content === text;
252   });
253
254   if (!isRight) {
255     setAttempts(prev => prev + 1);
256     setIsCorrect(false);
257     onFeedback?.(❌ Semak semula susunan dalam carta alir. (-5 Markah)', 3000, false);
258     return;
259   }
260
261   // Correct: Calculate Score
262   const calculatedScore = Math.max(5, 25 - (attempts * 5));
263
264   setIsSubmitting(true);
265   const token = localStorage.getItem('token');
266
267   try {
268     await fetch('https://algoquest-api.onrender.com/api/mission/submit', {
269       method: 'POST',
270       headers: {
271         'Content-Type': 'application/json',
272         'Authorization': `Bearer ${token}`
273       },
274       body: JSON.stringify({
275         mission: 2,
276         phase: 'pembinaan',
277         isCorrect: true,
278         score: calculatedScore,
279         // Award badge if correct on first attempt
280         badge: attempts === 0 ? 'Master Algoritma' : null
281       })
282     });
283
284     setEarnedScore(calculatedScore);
285     setIsCorrect(true);
286
287     let badgeMsg = '';
288     if (attempts === 0) {
289       badgeMsg = '🏆 Anda telah memperoleh lencana "Master Algoritma"';
290       onBadgeEarned?.('Master Algoritma');
291     }
292
293     onFeedback?.(✅ Struktur betul! PB dan PA disemak sebelum status ditetapkan.${badgeMsg}`, 3000, true);
294
295   } catch (err) {
296     console.error("Error submitting:", err);
297     onFeedback?.(⚠️ Ralat menghubungi pelayan.', 3000, false);
298   } finally {

```

```

299     setIsSubmitting(false);
300   }
301 };
302
303 const handleNext = () => {
304   if (!isCorrect) return;
305   // Pass score AND badge name to parent
306   const badge = attempts === 0 ? 'Master Algoritma' : null;
307   onContinue?.(earnedScore, badge);
308 };
309
310 const allItems = Object.values(items).flat();
311 const activeItem = activeId ? allItems.find((i) => i.id === activeId) : null;
312
313 return (
314   <div>
315     <h3>TAHAP 3: PEMBINAAN ALGORITMA</h3>
316     <p>Sistem Peperiksaan memerlukan algoritma keputusan lulus/gagal.
317 Tugas anda ialah menyusun aliran keputusan dan menukarkannya kepada Carta Alir.
318 </p>
319     <hr />
320
321     <DndContext
322       onDragStart={handleDragStart}
323       onDragEnd={handleDragEnd}
324       collisionDetection={closestCenter}
325     >
326       <div
327         className="flowchart-container"
328         style={{
329           display: 'flex',
330           gap: '40px',
331           justifyContent: 'center',
332           alignItems: 'stretch',
333           marginTop: '20px',
334         }}
335       >
336         {/* LEFT: Pseudokod list */}
337         <div
338           className="flowchart-pseudocode"
339           style={{
340             width: '320px',
341             backgroundColor: 'rgba(255,255,255,0.05)',
342             padding: '20px',
343             borderRadius: '8px',
344             display: 'flex',
345             flexDirection: 'column',
346           }}
347         >
348           <h4 style={{ marginTop: 0, textAlign: 'center', color: '#00ffff' }}>
349             PSEUDOKOD
350           </h4>
351           <SortableContext
352             items={items.available.map((i) => i.id)}
353             strategy={verticalListSortingStrategy}
354           >
355             <div style={{ flexGrow: 1 }}>
356               {items.available.map((item) => (
357                 <SortableItem
358                   key={item.id}
359                   id={item.id}
360                   content={item.content}
361                   source="list"
362                 />
363               ))}
364             </div>
365           </SortableContext>
366         </div>
367
368         {/* RIGHT: Carta Alir */}
369         <div
370           className="flowchart-boxes"
371           style={{
372             display: 'flex',
373             flexDirection: 'column',
374             justifyContent: 'center',
375             position: 'relative',

```

```

376     }}
377   >
378   <h4 style={{ marginTop: 0, textAlign: 'center', color: '#00ffff' }}>
379     CARTA ALIR
380   </h4>
381
382   <div
383     className="flowchart-diagram"
384     style={{
385       display: 'flex',
386       flexDirection: 'column',
387       alignItems: 'center',
388       position: 'relative',
389     }}
390   >
391     { /* Main vertical line */ }
392     <FlowchartShape id="oval1" shapeType="oval" items={items.oval1} />
393     <div style={{ margin: '2px 0', fontSize: '0.8rem' }}></div>
394
395     <FlowchartShape
396       id="parallelogram1"
397       shapeType="parallelogram"
398       items={items.parallelogram1}
399     />
400     <div style={{ margin: '2px 0', fontSize: '0.8rem' }}></div>
401
402     <FlowchartShape id="diamond1" shapeType="diamond" items={items.diamond1} />
403     <div
404       style={{
405         display: 'flex',
406         alignItems: 'center',
407         gap: '4px',
408         margin: '2px 0',
409         fontSize: '0.7rem',
410       }}
411     >
412       <span>Ya</span>
413       <span></span>
414     </div>
415
416     <FlowchartShape id="diamond2" shapeType="diamond" items={items.diamond2} />
417     <div
418       style={{
419         display: 'flex',
420         alignItems: 'center',
421         gap: '4px',
422         margin: '2px 0',
423         fontSize: '0.7rem',
424       }}
425     >
426       <span>Ya</span>
427       <span></span>
428     </div>
429
430     { /* Branch arrows from diamonds to rectangle2 */ }
431   <div style={{ position: 'relative', width: '100%', height: '0px' }}>
432     {(() => {
433       // === Arrow configuration ===
434       const baseLeftOffset = 70;
435       const arrowConfig = {
436         diamond1: { top: -200, horizontal: 80, vertical: 150 },
437         diamond2: { top: -70, horizontal: 80, vertical: 60 },
438       };
439
440       // Rectangle2 → Parallelogram2 arrow config
441       const rect2Arrow = {
442         top: 50,
443         down: 45,
444         left: 70,
445       };
446
447       return (
448         <>
449           { /* Diamond1 arrow */ }
450           <div
451             style={{
452               position: 'absolute',

```

```

453         top: `${arrowConfig.diamond1.top}px`,
454         left: `calc(50% + ${baseLeftOffset}px)`,
455         width: `${arrowConfig.diamond1.horizontal}px`,
456         height: '1px',
457         backgroundColor: 'fff',
458     }}
459 />
460 <div
461     style={{
462         position: 'absolute',
463         top: `${arrowConfig.diamond1.top}px`,
464         left: `calc(50% + ${baseLeftOffset + arrowConfig.diamond1.horizontal}px)`,
465         width: '1px',
466         height: `${arrowConfig.diamond1.vertical}px`,
467         backgroundColor: 'fff',
468     }}
469 />
470
471
472 <span
473     style={{
474         position: 'absolute',
475         top: `${arrowConfig.diamond1.top - 15}px`,
476         left: `calc(50% + ${baseLeftOffset + arrowConfig.diamond1.horizontal / 4}px)`,
477         fontSize: '0.7rem',
478         color: 'fff',
479     }}
480 >
481     Tidak
482 </span>
483
484 {/* Diamond2 arrow */}
485 <div
486     style={{
487         position: 'absolute',
488         top: `${arrowConfig.diamond2.top}px`,
489         left: `calc(50% + ${baseLeftOffset}px)`,
490         width: `${arrowConfig.diamond2.horizontal}px`,
491         height: '1px',
492         backgroundColor: 'fff',
493     }}
494 />
495 <div
496     style={{
497         position: 'absolute',
498         top: `${arrowConfig.diamond2.top}px`,
499         left: `calc(50% + ${baseLeftOffset + arrowConfig.diamond2.horizontal}px)`,
500         width: '1px',
501         height: `${arrowConfig.diamond2.vertical}px`,
502         backgroundColor: 'fff',
503     }}
504 />
505 {/* ↓ Arrow head (to Rectangle2) */}
506 <div
507     style={{
508         position: 'absolute',
509         top: `${arrowConfig.diamond2.top + arrowConfig.diamond2.vertical}px`,
510         left: `calc(50% + ${baseLeftOffset + arrowConfig.diamond2.horizontal - 4}px)`,
511         width: 0,
512         height: 0,
513         borderLeft: '4px solid transparent',
514         borderRight: '4px solid transparent',
515         borderTop: '8px solid #fff',
516     }}
517 />
518
519 <span
520     style={{
521         position: 'absolute',
522         top: `${arrowConfig.diamond2.top - 15}px`,
523         left: `calc(50% + ${baseLeftOffset + arrowConfig.diamond2.horizontal / 4}px)`,
524         fontSize: '0.7rem',
525         color: 'fff',
526     }}
527 >
528     Tidak
529 </span>

```

```

530
531 { /* Rectangle2 → Parallelogram2 */}
532 <div
533   style={{
534     position: 'absolute',
535     top: `${rect2Arrow.top}px`,
536     left: `calc(50% + ${baseLeftOffset + 80}px)`,
537     width: '1px',
538     height: `${rect2Arrow.down}px`,
539     backgroundColor: '#fff',
540   }}
541 />
542 <div
543   style={{
544     position: 'absolute',
545     top: `${rect2Arrow.top + rect2Arrow.down}px`,
546     left: `calc(50% + ${baseLeftOffset + 80 - rect2Arrow.left}px)`,
547     width: `${rect2Arrow.left}px`,
548     height: '1px',
549     backgroundColor: '#fff',
550   }}
551 />
552 { /* ← Arrow head (to Parallelogram2) */}
553 <div
554   style={{
555     position: 'absolute',
556     top: `${rect2Arrow.top + rect2Arrow.down - 4}px`,
557     left: `calc(50% + ${baseLeftOffset + 80 - rect2Arrow.left - 8}px)`,
558     width: 0,
559     height: 0,
560     borderTop: '4px solid transparent',
561     borderBottom: '4px solid transparent',
562     borderRight: '8px solid #fff',
563   }}
564 />
565 </div>
566 );
567 })()
568 </div>
569
570
571
572 { /* Rectangle1 in main line + Rectangle2 branching to the right */}
573 <div
574   style={{
575     position: 'relative',
576     width: '100%',
577     display: 'flex',
578     justifyContent: 'center',
579   }}
580 >
581   <FlowchartShape
582     id="rectangle1"
583     shapeType="rectangle"
584     items={items.rectangle1}
585   />
586
587   { /* Rectangle2 appears to the right */}
588   <div
589     style={{
590       position: 'absolute',
591       left: '50%',
592       transform: 'translateX(80px) translateY(-50%)',
593       top: '50%',
594     }}
595   >
596     <FlowchartShape
597       id="rectangle2"
598       shapeType="rectangle"
599       items={items.rectangle2}
600     />
601   </div>
602 </div>
603
604 { /* Continue main vertical line below Rectangle1 */}
605 <div style={{ margin: '2px 0', fontSize: '0.8rem' }}></div>
606 <FlowchartShape

```

```

607         id="parallelogram2"
608         shapeType="parallelogram"
609         items={items.parallelogram2}
610       />
611       <div style={{ margin: '2px 0', fontSize: '0.8rem' }}></div>
612       <FlowchartShape id="oval2" shapeType="oval" items={items.oval2} />
613     </div>
614   </div>
615 </div>
616
617   <DragOverlay>
618     {activeItem ? (
619       <DraggableItem id={activeItem.id} content={activeItem.content} />
620     ) : null}
621   </DragOverlay>
622 </DndContext>
623
624 <hr />
625
626 <div
627   style={{
628     display: 'flex',
629     justifyContent: 'flex-end',
630     gap: '10px',
631     marginTop: '10px',
632   }}
633 >
634   <button onClick={handleReset} className="primary-button" disabled={isSubmitting}>
635     Buat Semula
636   </button>
637   <button onClick={checkAnswer} className="primary-button" disabled={isSubmitting || isCorrect}>
638     {isSubmitting ? 'Menghantar...' : 'Semak Jawapan'}
639   </button>
640   <button
641     onClick={handleNext}
642     className="primary-button"
643     style={{
644       backgroundColor: isCorrect ? '#2ecc71' : '#999',
645       opacity: isCorrect ? 1 : 0.5,
646       cursor: isCorrect ? 'pointer' : 'not-allowed',
647     }}
648     disabled={!isCorrect}
649   >
650     Seterusnya
651   </button>
652 </div>
653 </div>
654 );
655 }
656
657 export default Mission2_Pembinaan;

```

```

1  import React, { useState } from 'react';
2  import { DndContext, DragOverlay, closestCenter } from '@dnd-kit/core';
3  import { SortableContext, arrayMove, verticalListSortingStrategy, useSortable } from '@dnd-kit/sortable';
4  import { useDroppable } from '@dnd-kit/core';
5  import { CSS } from '@dnd-kit/utilities';
6
7  // --- Reusable Sortable Item ---
8  function SortableItem({ id, content }) {
9    const { attributes, listeners, setNodeRef, transform, transition } = useSortable({ id });
10   const style = { transform: CSS.Transform.toString(transform), transition };
11   return (
12     <div
13       ref={setNodeRef}
14       style={style}
15       {...attributes}
16       {...listeners}
17       className="dnd-item"
18     >
19       {content}
20     </div>
21   );
22 }
23
24 // --- Overlay while dragging ---
25 function DraggableItem({ content }) {
26   return <div className="dnd-item-overlay">{content}</div>;
27 }
28
29 // --- Droppable Column ---
30 function DroppableSortableColumn({ id, title, items }) {
31   const { setNodeRef } = useDroppable({ id });
32   return (
33     <div
34       ref={setNodeRef}
35       className={`dnd-column-split ${id.startsWith('available') ? 'available' : ''}`}
36     >
37       <h4>{title}</h4>
38       <SortableContext items={items.map(i => i.id)} strategy={verticalListSortingStrategy}>
39         {items.map(item => (
40           <SortableItem key={item.id} id={item.id} content={item.content} />
41         ))}
42       </SortableContext>
43     </div>
44   );
45 }
46
47 // --- Initial States ---
48 const INITIAL_DATA = {
49   available: [
50     { id: 'd-1', content: 'Nama Pelajar' },
51     { id: 'd-2', content: 'Markah PA' },
52     { id: 'd-3', content: 'Status Lulus' }, // Fixed from 'Syarat Lulus' to match correct logic
53     { id: 'd-4', content: 'Kelas' },
54     { id: 'd-5', content: 'Program' },
55     { id: 'd-6', content: 'Syarat Lulus (≥50%)' },
56     { id: 'd-7', content: 'Markah PB' },
57     { id: 'd-8', content: 'Kehadiran (≥80%)' },
58     { id: 'd-9', content: 'No Telefon' },
59     { id: 'd-10', content: 'Gred Huruf' },
60   ],
61   pending: [],
62 };
63
64 const INITIAL_STEPS = {
65   available_langkah: [
66     { id: '1-1', content: 'Semak Markah PB dan PA ≥50' },
67     { id: '1-2', content: 'Semak Kehadiran ≥80%' },

```



```

68     { id: '1-3', content: 'Tentukan Status: Lulus atau Gagal' },
69   ],
70   ordered_langkah: [],
71 };
72
73 // --- Deep Clone Helper ---
74 const clone = (obj) => JSON.parse(JSON.stringify(obj));
75
76 // --- Main Component ---
77 function Mission2_Pengabstrakan({ onContinue, onFeedback }) {
78   const [dataItems, setDataItems] = useState(() => clone(INITIAL_DATA));
79   const [stepItems, setStepItems] = useState(() => clone(INITIAL_STEPS));
80   const [activeId, setActiveId] = useState(null);
81   const [isCorrect, setIsCorrect] = useState(false);
82
83   // Backend Integration State
84   const [earnedScore, setEarnedScore] = useState(0);
85   const [attempts, setAttempts] = useState(0);
86   const [isSubmitting, setIsSubmitting] = useState(false);
87
88   // --- Helpers ---
89   function findContainer(itemId) {
90     if (!itemId) return null;
91     for (const container of Object.keys(dataItems)) {
92       if (dataItems[container].find(i => i.id === itemId)) return container;
93     }
94     for (const container of Object.keys(stepItems)) {
95       if (stepItems[container].find(i => i.id === itemId)) return container;
96     }
97     return null;
98   }
99
100   function handleDragStart(event) {
101     setActiveId(event.active.id);
102   }
103
104   function handleDragEnd(event) {
105     const { active, over } = event;
106     if (!over) {
107       setActiveId(null);
108       return;
109     }
110
111     const activeContainer = findContainer(active.id);
112     const overContainer = findContainer(over.id) || over.id;
113     if (!activeContainer || !overContainer) {
114       setActiveId(null);
115       return;
116     }
117
118     setIsCorrect(false); // Unlock next button after any drag
119
120     // Determine which state to update
121     if (Object.keys(dataItems).includes(activeContainer) || Object.keys(dataItems).includes(overContainer)) {
122       updateItemsState(dataItems, setDataItems, active, over);
123     } else {
124       updateItemsState(stepItems, setStepItems, active, over);
125     }
126
127     setActiveId(null);
128   }
129
130   function updateItemsState(itemsObj, setItems, active, over) {
131     const activeIdLocal = active.id;
132     const overIdLocal = over.id;
133     const activeContainer = findContainer(activeIdLocal);
134     const overContainer = findContainer(overIdLocal) || overIdLocal;
135
136     if (activeContainer === overContainer) {
137       setItems(prev => {
138         const list = [...prev[activeContainer]];
139         const fromIndex = list.findIndex(i => i.id === activeIdLocal);
140         const toIndex = list.findIndex(i => i.id === overIdLocal);
141         if (fromIndex === -1 || toIndex === -1) return prev;
142         const newList = arrayMove(list, fromIndex, toIndex);
143         return { ...prev, [activeContainer]: newList };
144       });

```

```

145     return;
146   }
147
148   // Moving between containers
149   setItems(prev => {
150     const prevActiveList = prev[activeContainer] || [];
151     const prevOverList = prev[overContainer] || [];
152
153     const movedItem = prevActiveList.find(i => i.id === activeIdLocal);
154     if (!movedItem) return prev;
155
156     const newActiveList = prevActiveList.filter(i => i.id !== activeIdLocal);
157     const overIndex = prevOverList.findIndex(i => i.id === overIdLocal);
158     const newOverList = [...prevOverList];
159     if (overIndex === -1) newOverList.push(movedItem);
160     else newOverList.splice(overIndex, 0, movedItem);
161
162     return { ...prev, [activeContainer]: newActiveList, [overContainer]: newOverList };
163   });
164 }
165
166 // --- Check Answer & SCORING ---
167 const checkAnswer = async () => {
168   // Correct Data logic based on Slide 706/708
169   const correctDataPenting = ['d-2', 'd-6', 'd-7', 'd-8',].sort(); // Markah PA, Syarat, PB, Gred, Status
170   const userDataPenting = dataItems.penting.map(i => i.id).sort();
171   const isDataCorrect = JSON.stringify(correctDataPenting) === JSON.stringify(userDataPenting);
172
173   const correctLangkah = ['1-2', '1-1', '1-3']; // Semak Markah -> Tentukan Status
174   const userLangkah = stepItems.ordered_langkah.map(i => i.id);
175   const isLangkahCorrect = JSON.stringify(correctLangkah) === JSON.stringify(userLangkah);
176
177   const ok = isDataCorrect && isLangkahCorrect;
178
179   if (!ok) {
180     setAttempts(prev => prev + 1);
181     setIsCorrect(false);
182     onFeedback('❌ Salah: Ada maklumat tidak relevan dipilih. (-5 Markah)', 3000, false);
183     return;
184   }
185
186   // Correct: Calculate Score
187   const calculatedScore = Math.max(5, 25 - (attempts * 5));
188
189   setIsSubmitting(true);
190   const token = localStorage.getItem('token');
191
192   try {
193     await fetch('https://algoquest-api.onrender.com/api/mission/submit', {
194       method: 'POST',
195       headers: {
196         'Content-Type': 'application/json',
197         'Authorization': `Bearer ${token}`
198       },
199       body: JSON.stringify({
200         mission: 2,
201         phase: 'pengabstrakan',
202         isCorrect: true,
203         score: calculatedScore,
204         badge: null
205       })
206     });
207
208     setEarnedScore(calculatedScore);
209     setIsCorrect(true);
210     onFeedback('✅ Betul: Hebat! Anda berjaya mengenal pasti semua maklumat penting.', 3000, true);
211
212   } catch (err) {
213     console.error("Error submitting:", err);
214     onFeedback('⚠️ Ralat menghubungi pelayan.', 3000, false);
215   } finally {
216     setIsSubmitting(false);
217   }
218 };
219
220 // --- Reset ---
221 const handleReset = () => {

```

```

222 setDataItems(clone(INITIAL_DATA));
223 setStepItems(clone(INITIAL_STEPS));
224 setActiveId(null);
225 setIsCorrect(false);
226 setAttempts(0); // Reset attempts
227 onFeedback('🔄 Susunan telah direset. Cuba semula dengan teliti.', 2000, null);
228 };
229
230 const allItems = [
231   ...dataItems.available,
232   ...dataItems.penting,
233   ...stepItems.available_langkah,
234   ...stepItems.ordered_langkah,
235 ];
236 const activeItem = activeId ? allItems.find(i => i.id === activeId) : null;
237
238 return (
239   <div>
240     <h3>TAHAP 2: PENGABSTRAKAN</h3>
241     <p>
242       Sistem Peperiksaan perlu mengenal pasti maklumat penting untuk menentukan sama ada pelajar lulus
243     </p>
244     <hr />
245
246     <DndContext
247       onDragStart={handleDragStart}
248       onDragEnd={handleDragEnd}
249       collisionDetection={closestCenter}
250     >
251       <div className="dnd-container-split">
252         <DraggableSortColumn
253           id="available"
254           title="SENARAI DATA"
255           items={dataItems.available}
256         />
257         <DraggableSortColumn
258           id="penting"
259           title="DATA PENTING"
260           items={dataItems.penting}
261         />
262       </div>
263
264       <div className="dnd-container-split">
265         <DraggableSortColumn
266           id="available_langkah"
267           title="LANGKAH MODEL (TIDAK TERSUSUN)"
268           items={stepItems.available_langkah}
269         />
270         <DraggableSortColumn
271           id="ordered_langkah"
272           title="MODEL LOGIK (TERSUSUN)"
273           items={stepItems.ordered_langkah}
274         />
275       </div>
276
277       <DragOverlay>
278         {activeItem ? <DraggableItem content={activeItem.content} /> : null}
279       </DragOverlay>
280     </DndContext>
281
282     <hr />
283
284     <div style={{ display: 'flex', justifyContent: 'flex-end', gap: 10, marginTop: 20 }}>
285       <button
286         className="primary-button"
287         onClick={handleReset}
288         style={{ transition: 'transform 0.2s ease', backgroundColor: '#555' }}
289         disabled={isSubmitting}
290       >
291         Buat Semula
292       </button>
293
294       <button
295         className="primary-button"
296         onClick={checkAnswer}
297         style={{ transition: 'transform 0.2s ease' }}
298         disabled={isSubmitting || isCorrect}

```

```

299     >
300     {isSubmitting ? 'Menghantar...' : 'Semak Jawapan'}
301   </button>
302
303   <button
304     className="primary-button"
305     onClick={isCorrect ? () => onContinue(earnedScore) : undefined}
306     disabled={!isCorrect}
307     style={{
308       backgroundColor: isCorrect ? '#2ecc71' : '#999',
309       opacity: isCorrect ? 1 : 0.5,
310       cursor: isCorrect ? 'pointer' : 'not-allowed',
311       transition: 'transform 0.2s ease, background-color 0.2s ease',
312     }}
313   >
314     Seterusnya
315   </button>
316 </div>
317
318 </div>
319 );
320 }
321
322 export default Mission2_Pengabstrakan;

```

```

1  import React, { useState } from 'react';
2  import { DndContext, DragOverlay, closestCenter } from '@dnd-kit/core';
3  import { SortableContext, arrayMove, verticalListSortingStrategy, useSortable } from '@dnd-kit/sortable';
4  import { useDroppable } from '@dnd-kit/core';
5  import { CSS } from '@dnd-kit/utilities';
6
7  // --- Reusable Sortable Item ---
8  function SortableItem({ id, content }) {
9    const { attributes, listeners, setNodeRef, transform, transition } = useSortable({ id });
10
11    const style = {
12      transform: CSS.Transform.toString(transform),
13      transition,
14    };
15
16    return (
17      <div ref={setNodeRef} style={style} {...attributes} {...listeners} className="dnd-item">
18        {content}
19      </div>
20    );
21  }
22
23  // --- Reusable Draggable Item (for the overlay) ---
24  function DraggableItem({ id, content }) {
25    return <div className="dnd-item-overlay">{content}</div>;
26  }
27
28  // --- Reusable Droppable Column ---
29  function DroppableSortableColumn({ id, title, items }) {
30    const { setNodeRef } = useDroppable({ id });
31
32    return (
33      <div ref={setNodeRef} className={`dnd-column ${id === 'available' ? 'available' : ''}`>
34        <h4>{title.toUpperCase()}</h4>
35        <SortableContext items={items.map(i => i.id)} strategy={verticalListSortingStrategy}>
36          {items.map(item => (
37            <SortableItem key={item.id} id={item.id} content={item.content} />
38          ))}
39        </SortableContext>
40      </div>
41    );
42  }
43
44  // --- Main Component ---
45  function Mission2_Penguraian({ onContinue, onFeedback }) {
46    const [items, setItems] = useState([
47      available: [
48        { id: 'item-1', content: 'Kira Markah Keseluruhan' },
49        { id: 'item-2', content: 'Markah PA' },
50        { id: 'item-3', content: 'Tentukan Gred' },
51        { id: 'item-4', content: 'Gred Huruf' },
52        { id: 'item-5', content: 'Kehadiran (%)' }, // Fixed from 'No Pendaftaran' to match correct logic
53        { id: 'item-6', content: 'Semak Syarat Lulus' },
54        { id: 'item-7', content: 'Status Lulus' },
55        { id: 'item-8', content: 'Markah PB' },
56      ],
57      input: [],
58      proses: [],
59      output: [],
60    ]);
61
62    const [activeId, setActiveId] = useState(null);
63    const [isCorrect, setIsCorrect] = useState(false);
64
65    // Backend Integration State
66    const [earnedScore, setEarnedScore] = useState(0);
67    const [attempts, setAttempts] = useState(0);

```

```

68     const [isSubmitting, setIsSubmitting] = useState(false);
69
70     function findContainer(itemId) {
71         if (!itemId) return null;
72         if (Object.keys(items).includes(itemId)) return itemId;
73         for (const container of Object.keys(items)) {
74             if (items[container].find(item => item.id === itemId)) return container;
75         }
76         return null;
77     }
78
79     function handleDragStart(event) {
80         setActiveId(event.active.id);
81     }
82
83     function handleDragEnd(event) {
84         const { active, over } = event;
85         if (!over) {
86             setActiveId(null);
87             return;
88         }
89
90         const activeContainer = findContainer(active.id);
91         const overContainer = findContainer(over.id);
92
93         if (!activeContainer || !overContainer || !items[activeContainer] || !items[overContainer]) {
94             setActiveId(null);
95             return;
96         }
97
98         // Reset correct state if user moves items after checking
99         setIsCorrect(false);
100
101         if (activeContainer !== overContainer) {
102             setItems(prev => {
103                 const activeItems = prev[activeContainer];
104                 const overItems = prev[overContainer];
105                 const activeIndex = activeItems.findIndex(item => item.id === active.id);
106                 const [movedItem] = activeItems.splice(activeIndex, 1);
107
108                 const overItem = overItems.find(item => item.id === over.id);
109                 const overIndex = overItem ? overItems.findIndex(item => item.id === over.id) : overItems.length;
110
111                 overItems.splice(overIndex, 0, movedItem);
112
113                 return {
114                     ...prev,
115                     [activeContainer]: [...activeItems],
116                     [overContainer]: [...overItems],
117                 };
118             });
119         } else {
120             setItems(prev => {
121                 const activeItems = prev[activeContainer];
122                 const activeIndex = activeItems.findIndex(item => item.id === active.id);
123                 const overIndex = activeItems.findIndex(item => item.id === over.id);
124                 return {
125                     ...prev,
126                     [activeContainer]: arrayMove(activeItems, activeIndex, overIndex),
127                 };
128             });
129         }
130         setActiveId(null);
131     }
132
133     // ✅ Updated to use Backend Scoring
134     const checkAnswer = async () => {
135         const correctInput = ['item-2', 'item-5', 'item-8'].sort();
136         const correctProses = ['item-1', 'item-3', 'item-6'].sort();
137         const correctOutput = ['item-4', 'item-7'].sort();
138
139         const playerInput = items.input.map(i => i.id).sort();
140         const playerProses = items.proses.map(i => i.id).sort();
141         const playerOutput = items.output.map(i => i.id).sort();
142
143         const ok =
144             JSON.stringify(playerInput) === JSON.stringify(correctInput) &&

```

```

145     JSON.stringify(playerProses) === JSON.stringify(correctProses) &&
146     JSON.stringify(playerOutput) === JSON.stringify(correctOutput);
147
148     if (!ok) {
149         setAttempts(prev => prev + 1);
150         setIsCorrect(false);
151         onFeedback('❌ Hmm, semak semula. (-5 Markah)', 3000, false); // Red Glow 'false'
152         return;
153     }
154
155     // Correct: Calculate Score
156     const calculatedScore = Math.max(5, 25 - (attempts * 5));
157
158     setIsSubmitting(true);
159     const token = localStorage.getItem('token');
160
161     try {
162         await fetch('https://algoquest-api.onrender.com/api/mission/submit', {
163             method: 'POST',
164             headers: {
165                 'Content-Type': 'application/json',
166                 'Authorization': `Bearer ${token}`
167             },
168             body: JSON.stringify({
169                 mission: 2,
170                 phase: 'penguraian',
171                 isCorrect: true,
172                 score: calculatedScore,
173                 badge: null
174             })
175         });
176
177         setEarnedScore(calculatedScore);
178         setIsCorrect(true);
179         // Green Glow 'true'
180         onFeedback('✅ Hebat! Anda telah mengenal pasti input, proses dan output dengan betul.', 3000, true);
181
182     } catch (err) {
183         console.error("Error submitting:", err);
184         onFeedback('⚠️ Ralat menghubungi pelayan.', 3000, false);
185     } finally {
186         setIsSubmitting(false);
187     }
188 };
189
190 const reset = () => {
191     setItems({
192         available: [
193             { id: 'item-1', content: 'Kira Markah Keseluruhan' },
194             { id: 'item-2', content: 'Markah PA' },
195             { id: 'item-3', content: 'Tentukan Gred' },
196             { id: 'item-4', content: 'Gred Huruf' },
197             { id: 'item-5', content: 'No Pendaftaran' },
198             { id: 'item-6', content: 'Semak Syarat Lulus' },
199             { id: 'item-7', content: 'Status Lulus' },
200             { id: 'item-8', content: 'Markah PB' },
201         ],
202         input: [],
203         proses: [],
204         output: [],
205     });
206     setIsCorrect(false);
207     setAttempts(0); // Reset attempts
208     onFeedback('🔄 Susunan telah direset. Cuba semula.', 2000, null);
209 };
210
211 const handleNext = () => {
212     if (!isCorrect) return;
213     onContinue(earnedScore); // Pass earned score
214 };
215
216 const activeItem = activeId ? [...items.available, ...items.input, ...items.proses, ...items.output].find(
217
218 return (
219     <div>
220         <h3>TAHAP 1: PENGURAIAN</h3>
221         <p>Sistem Peperiksaan sedang dibangunkan untuk menilai keputusan pelajar secara automatik. <br />Tug

```

```

222
223 <hr />
224 <DndContext onDragStart={handleDragStart} onDragEnd={handleDragEnd} collisionDetection={closestCenter}
225 <div className="dnd-container">
226   {Object.keys(items).map(containerId => (
227     <DraggableSortTableColumn
228       key={containerId}
229       id={containerId}
230       title={containerId === 'available' ? 'Item Tersedia' : containerId}
231       items={items[containerId]}
232     />
233   ))}
234 </div>
235 <DragOverlay>
236   {activeItem && <DraggableItem id={activeItem.id} content={activeItem.content} />}
237 </DragOverlay>
238 </DndContext>
239 <hr />
240
241 {/* Buttons */}
242 <div style={{ display: 'flex', justifyContent: 'flex-end', gap: 10, marginTop: 20 }}>
243   <button className="primary-button" onClick={reset} disabled={isSubmitting}>
244     Buat Semula
245   </button>
246
247   <button className="primary-button" onClick={checkAnswer} disabled={isSubmitting}>
248     {isSubmitting ? 'Menghantar...' : 'Semak Jawapan'}
249   </button>
250
251   <button
252     className="primary-button"
253     style={{
254       backgroundColor: isCorrect ? '#2ecc71' : '#999',
255       cursor: isCorrect ? 'pointer' : 'not-allowed',
256     }}
257     disabled={!isCorrect}
258     onClick={isCorrect ? handleNext : undefined}
259   >
260     Seterusnya
261   </button>
262 </div>
263 </div>
264 );
265 }
266
267 export default Mission2_Penguraian;

```



```

1  import React, { useState, useEffect } from 'react';
2
3  function Mission2_Penyahpepijat({ onContinue, setRobotText, onBadgeEarned, onFeedback }) {
4      const [selectedAnswer, setSelectedAnswer] = useState(null);
5      const [isCorrect, setIsCorrect] = useState(false);
6
7      // Backend Integration State
8      const [earnedScore, setEarnedScore] = useState(0);
9      const [attempts, setAttempts] = useState(0);
10     const [isSubmitting, setIsSubmitting] = useState(false);
11
12     // ✅ Jawapan betul
13     const correctAnswer = 'B';
14
15     // Set initial robot text when component loads
16     useEffect(() => {
17         if (setRobotText) {
18             setRobotText(
19                 'Semak pseudokod dan output tersebut. Kenal pasti ralat logik dan pilih pembetulan yang paling tepat.'
20             );
21         }
22     }, [setRobotText]);
23
24     const handleReset = () => {
25         setSelectedAnswer(null);
26         setIsCorrect(false);
27         setAttempts(0); // Reset attempts
28         if (setRobotText) {
29             setRobotText(
30                 '🔄 Pilihan telah direset. Sila pilih jawapan sekali lagi.'
31             );
32         }
33     };
34
35     const checkAnswer = async () => {
36         if (!selectedAnswer) {
37             if (onFeedback) {
38                 onFeedback('❌ Sila pilih satu jawapan dahulu sebelum menghantar.', 3000, false);
39             }
40             return;
41         }
42
43         const ok = selectedAnswer === correctAnswer;
44
45         if (!ok) {
46             setAttempts(prev => prev + 1);
47             setIsCorrect(false);
48             if (onFeedback) {
49                 onFeedback('❌ "Semak semula logik syarat. Sistem sepatutnya menilai kedua-dua markah." (-5 Ma
50             )
51             return;
52         }
53
54         // Correct: Calculate Score
55         const calculatedScore = Math.max(5, 25 - (attempts * 5));
56
57         setIsSubmitting(true);
58         const token = localStorage.getItem('token');
59
60         try {
61             await fetch('https://algoquest-api.onrender.com/api/mission/submit', {
62                 method: 'POST',
63                 headers: {
64                     'Content-Type': 'application/json',
65                     'Authorization': `Bearer ${token}`
66                 },
67                 body: JSON.stringify({

```

```

68     mission: 2,
69     phase: 'penyahpepijat',
70     isCorrect: true,
71     score: calculatedScore,
72     // Badge awarded if correct on first try
73     badge: attempts === 0 ? 'Master Pemulih Logik' : null
74   })
75   });
76
77   setEarnedScore(calculatedScore);
78   setIsCorrect(true);
79
80   let badgeMsg = '';
81   if (attempts === 0) {
82     badgeMsg = '\n\n🏆 Anda telah memperoleh lencana "Master Pemulih Logik"';
83     if (onBadgeEarned) onBadgeEarned('Master Pemulih Logik');
84   }
85
86   if (onFeedback) {
87     onFeedback(
88       '✅ "Hebat! Anda telah membetulkan ralat logik.${badgeMsg}',
89       3000,
90       true
91     );
92   }
93
94   } catch (err) {
95     console.error("Error submitting:", err);
96     if (onFeedback) onFeedback('⚠️ Ralat menghubungi pelayan.', 3000, false);
97   } finally {
98     setIsSubmitting(false);
99   }
100 };
101
102 const handleNext = () => {
103   if (!isCorrect) return;
104   const badge = attempts === 0 ? 'Master Pemulih Logik' : null;
105   onContinue && onContinue(earnedScore, badge);
106 };
107
108 return (
109   <div>
110     <h3>TAHAP 4: PENYAHPEPIJATAN</h3>
111
112     <p>
113       Sistem peperiksaan mengalami ralat dalam menentukan keputusan pelajar.
114       Robot Algo mengesan bahawa walaupun pelajar mendapat markah tinggi,
115       keputusan masih dipaparkan sebagai <strong>"Gagal"</strong>.
116       Tugas anda ialah membetulkan pseudokod supaya sistem berfungsi dengan betul.
117     </p>
118     <hr />
119
120     <div className="quiz-container" style={{ display: 'flex', gap: '4%', marginTop: '20px' }}>
121       <div style={{ width: '48%' }}>
122         <div className="highlight-box" style={{
123           textAlign: 'center',
124           fontWeight: 'bold',
125           fontSize: '1.1rem',
126           lineHeight: '1.5',
127           textTransform: 'uppercase'
128         }}>
129           PSEUDOKOD
130         </div>
131         <div style={{ textAlign: 'center', marginTop: '8px' }}>
132           <p>Mula</p>
133           <p>Masukkan Markah PB dan Markah PA</p>
134           <p>Jika PB {'>'} 50 dan PA {'<'} 50</p>
135           <p>&nbsp;&nbsp;&nbsp;Status = &quot;Lulus&quot;</p>
136           <p>Jika tidak</p>
137           <p>&nbsp;&nbsp;&nbsp;Status = &quot;Gagal&quot;</p>
138           <p>Cetak Status</p>
139           <p>Tamat</p>
140         </div>
141       </div>

```

```

145 </div>
146
147 { /* Maklumat Output (kanan) */}
148 <div className="quiz-answers" style={{ width: '48%' }}>
149   <div
150     className="highlight-box"
151     style={{
152       textAlign: 'center',
153       fontWeight: 'bold',
154       fontSize: '1.1rem',
155       lineHeight: '1.5',
156       textTransform: 'uppercase'
157     }}
158   >
159     MAKLUMAT OUTPUT
160   </div>
161
162   { /* 2-column layout inside Maklumat Output */}
163   <div
164     className="output-table"
165     style={{
166       display: 'grid',
167       gridTemplateColumns: '1fr 1fr',
168       gap: '8px',
169       marginTop: '10px',
170       textAlign: 'center',
171     }}
172   >
173     { /* Column 1: Input */}
174     <div>
175       <p style={{ fontWeight: 'bold' }}>Input</p>
176       <p>PB = 70, PA = 80</p>
177       <p>PB = 70, PA = 30</p>
178       <p>PB = 40, PA = 60</p>
179       <p>PB = 60, PA = 60</p>
180     </div>
181
182     { /* Column 2: Output Sistem */}
183     <div>
184       <p style={{ fontWeight: 'bold' }}>Output Sistem</p>
185       <p>Status: Gagal ✖ (sepatutnya Lulus)</p>
186       <p>Status: Lulus ✔</p>
187       <p>Status: Gagal ✔</p>
188       <p>Status: Gagal ✖ (sepatutnya Lulus)</p>
189     </div>
190   </div>
191 </div>
192 </div>
193
194
195 <hr />
196 <h4>Apakah punca utama kesilapan logik berdasarkan perbezaan output di atas?</h4>
197
198 { /* Pilihan jawapan - gaya sama seperti sebelum ini */}
199 <div>
200   <button
201     className={`choice-button ${selectedAnswer === 'A' ? 'selected' : ''}`}
202     onClick={() => {
203       setSelectedAnswer('A');
204       setIsCorrect(false);
205     }}
206   >
207     A. Struktur if-else salah ─ arahan "Jika tidak" sepatutnya diletakkan sebelum pernyataan "Lulus".
208   </button>
209
210   <button
211     className={`choice-button ${selectedAnswer === 'B' ? 'selected' : ''}`}
212     onClick={() => {
213       setSelectedAnswer('B');
214       setIsCorrect(false);
215     }}
216   >
217     B. Syarat logik salah ─ sepatutnya guna PB ≥ 50 dan PA ≥ 50 untuk menentukan Lulus.
218   </button>
219
220   <button
221     className={`choice-button ${selectedAnswer === 'C' ? 'selected' : ''}`}

```

```

222     onClick={() => {
223       setSelectedAnswer('C');
224       setIsCorrect(false);
225     }}
226   >
227     C. Arahan "Cetak Status" sepatutnya diletakkan sebelum "Jika tidak".
228   </button>
229
230   <button
231     className={`choice-button ${selectedAnswer === 'D' ? 'selected' : ''}`}
232     onClick={() => {
233       setSelectedAnswer('D');
234       setIsCorrect(false);
235     }}
236   >
237     D. Perbandingan PA {'<'} 50 tidak memberi kesan kepada output.
238   </button>
239 </div>
240
241 <hr />
242
243 /* 3 buttons on the right */
244 <div style={{ display: 'flex', justifyContent: 'flex-end', gap: '10px', marginTop: '10px' }}>
245   <button
246     onClick={handleReset}
247     className="primary-button"
248     disabled={isSubmitting}
249   >
250     Buat Semula
251   </button>
252
253   <button
254     onClick={checkAnswer}
255     className="primary-button"
256     disabled={isSubmitting}
257   >
258     Semak Jawapan
259   </button>
260
261   <button
262     onClick={isCorrect ? handleNext : undefined}
263     className="primary-button"
264     style={{
265       backgroundColor: isCorrect ? '#2ecc71' : '#999',
266       cursor: isCorrect ? 'pointer' : 'not-allowed',
267     }}
268     disabled={!isCorrect}
269   >
270     Hantar
271   </button>
272 </div>
273 </div>
274 );
275 }
276
277 export default Mission2_Penyahpepijat;

```

```
1 import React, { useState } from 'react';
2 import MissionLayout from './MissionLayout';
3 import Mission3_Penguraian from './Mission3_Penguraian';
4 import Mission3_Pengabstrakan from './Mission3_Pengabstrakan';
5 import Mission3_Pembinaan from './Mission3_Pembinaan';
6 import Mission3_Penyahpepijat from './Mission3_Penyahpepijat';
7 import Mission3_Complete from './Mission3_Complete';
8
9 function Mission3({ onMissionComplete }) {
10   const [missionPhase, setMissionPhase] = useState('penguraian');
11   const [totalMissionScore, setTotalMissionScore] = useState(0);
12   const [earnedBadges, setEarnedBadges] = useState([]);
13
14   // Global feedback state for Robot
15   const [feedbackText, setFeedbackText] = useState('');
16   const [isCorrect, setIsCorrect] = useState(null);
17
18   // --- API HELPERS ---
19   const savePhaseScore = async (phaseName, score) => {
20     const token = localStorage.getItem('token');
21     try {
22       await fetch('https://algoquest-api.onrender.com/api/mission/submit', {
23         method: 'POST',
24         headers: { 'Content-Type': 'application/json', 'Authorization': `Bearer ${token}` },
25         body: JSON.stringify({ mission: 3, phase: phaseName, isCorrect: score > 0, score: score, badge: null });
26       });
27     } catch (e) { console.error("Save failed", e); }
28   };
29
30   const saveBadge = async (badgeName) => {
31     const token = localStorage.getItem('token');
32     try {
33       await fetch('https://algoquest-api.onrender.com/api/mission/submit', {
34         method: 'POST',
35         headers: { 'Content-Type': 'application/json', 'Authorization': `Bearer ${token}` },
36         body: JSON.stringify({ mission: 3, phase: missionPhase, isCorrect: true, score: 0, badge: badgeName });
37       });
38     } catch (e) { console.error("Badge save failed", e); }
39   };
40
41   // --- FEEDBACK HANDLER ---
42   const handleFeedback = (message, duration = 3000, correctState = null) => {
43     setFeedbackText(message);
44
45     // Pass true/false/null for robot glow
46     setIsCorrect(correctState);
47
48     setTimeout(() => {
49       setFeedbackText('');
50       setIsCorrect(null);
51     }, duration);
52   };
53
54   // --- PHASE COMPLETION HANDLER ---
55   const handlePhaseComplete = (nextPhase, scoreEarned, badgeEarned = null) => {
56     const isCorrectAnswer = scoreEarned > 0;
57
58     // Update totals and badges
59     setTotalMissionScore(prev => prev + scoreEarned);
60     if (badgeEarned) setEarnedBadges(prev => [...prev, badgeEarned]);
61
62     // Save to backend
63     savePhaseScore(missionPhase, scoreEarned);
64     if (badgeEarned) saveBadge(badgeEarned);
65
66     // Robot glow for phase completion
67     setIsCorrect(isCorrectAnswer);
68   };
69 }
```

```

69     setMissionPhase(nextPhase);
70     setFeedbackText('');
71 };
72
73 const robotTexts = {
74   penguraian: "Pelajar membayar tiga jenis yuran iaitu yuran pengajian, yuran asrama dan yuran aktiviti p
75   pengabstrakan: "Hai! Saya akan bantu anda mengenal pasti maklumat penting dan menyusun langkah logik ba
76   pembinaan: "Bantu Sistem Kewangan menyusun langkah algoritma bagi memproses bayaran yuran setiap pelaja
77   penyahpepijat: "Sistem Kewangan mengesan keputusan yang salah walaupun bayaran telah dibuat penuh. Bant
78   complete: "Tahniah! Anda telah menyelesaikan Unit Kewangan."
79 };
80
81 const renderPhase = () => {
82   switch (missionPhase) {
83     case 'penguraian':
84       return (
85         <Mission3_Penguraian
86           onContinue={({score}) => handlePhaseComplete('pengabstrakan', score)}
87           onFeedback={handleFeedback}
88         />
89       );
90     case 'pengabstrakan':
91       return (
92         <Mission3_Pengabstrakan
93           onContinue={({score}) => handlePhaseComplete('pembinaan', score)}
94           onFeedback={handleFeedback}
95         />
96       );
97     case 'pembinaan':
98       return (
99         <Mission3_Pembinaan
100           onContinue={({score, badge}) => handlePhaseComplete('penyahpepijat', score, badge)}
101           onFeedback={handleFeedback}
102         />
103       );
104     case 'penyahpepijat':
105       return (
106         <Mission3_Penyahpepijat
107           onContinue={({score, badge}) => handlePhaseComplete('complete', score, badge)}
108           onFeedback={handleFeedback}
109         />
110       );
111     case 'complete':
112       return (
113         <Mission3_Complete
114           score={totalMissionScore}
115           badges={earnedBadges}
116           onContinue={() => onMissionComplete(totalMissionScore, earnedBadges)}
117         />
118       );
119     default:
120       return <div>Memuatkan...</div>;
121   }
122 };
123
124 return (
125   <MissionLayout
126     robotText={feedbackText || robotTexts[missionPhase]}
127     isCorrect={isCorrect} //  glow control
128   >
129     {renderPhase()}
130   </MissionLayout>
131 );
132 }
133
134 export default Mission3;

```

```

1  import React from 'react';
2
3  function Mission3_Complete({ score, badges, onContinue }) {
4    // 1. Ensure badges is always an array
5    const displayBadges = Array.isArray(badges) ? badges : [];
6
7    // 2. Fix: Use the score directly (it's already a number)
8    const finalScore = score || 0;
9
10   return (
11     <div
12       style={{
13         width: '100%',
14         height: '100%',
15         display: 'flex',
16         justifyContent: 'center',
17         alignItems: 'center',
18         animation: 'fadeIn 1s ease-in'
19       }}
20     >
21       <div className="complete-screen" style={{ textAlign: 'center', animation: 'fadeInUp 1s ease-out' }}>
22         <h2>Tahniah! Anda Selesai Unit Kewangan! 🎉</h2>
23         <p>Sistem pengiraan yuran pelajar kini berfungsi dengan betul.</p>
24         <hr style={{ width: '80%', margin: '20px auto' }} />
25
26         <h3 style={{ margin: '15px 0' }}>Markah Misi: {finalScore} / 100</h3>
27
28         { /* Badge Section */ }
29         {displayBadges.length > 0 && (
30           <div className="badges-container" style={{ margin: '20px 0' }}>
31             <h4>📋 Lencana Diperolehi:</h4>
32             <div style={{ display: 'flex', flexWrap: 'wrap', gap: '10px', justifyContent: 'center' }}>
33               {displayBadges.map((badge, idx) => (
34                 <div
35                   key={idx}
36                   style={{
37                     position: 'relative',
38                     display: 'inline-flex',
39                     alignItems: 'center',
40                     justifyContent: 'center',
41                     padding: '10px 20px',
42                     borderRadius: '9999px',
43                     fontWeight: 'bold',
44                     fontSize: '1rem',
45                     color: '#fff',
46                     background: 'linear-gradient(135deg, #FFD700, #FFC200, #FFB000)',
47                     overflow: 'hidden',
48                     cursor: 'default',
49                     transition: 'transform 0.3s ease, box-shadow 0.3s ease',
50                     boxShadow: '0 2px 6px rgba(0,0,0,0.3)'
51                   }}
52                   className="badge"
53                 >
54                   <span style={{ display: 'inline-block', width: '16px', height: '16px', marginRight: '8px' }}>
55                     <span style={{ zIndex: 1 }}>{badge}</span>
56                   </span>
57                   <span style={{
58                     position: 'absolute',
59                     top: '-50%',
60                     left: '-50%',
61                     width: '200%',
62                     height: '200%',
63                     background: 'linear-gradient(120deg, rgba(255,255,255,0.2) 0%, rgba(255,255,255,0) 60%',
64                     transform: 'rotate(25deg)',
65                     pointerEvents: 'none',
66                     animation: 'shine 2s infinite',
67                     zIndex: 0

```

```

68         }}
69         <</span>
70     </div>
71     ))}
72 </div>
73 </div>
74 }}
75
76 <hr style={{ width: '80%', margin: '20px auto' }} />
77 <p>Anda telah memulihkan semua sistem di Kampus Digital!</p>
78
79 <button
80   onClick={onContinue}
81   className="primary-button"
82   style={{
83     width: '300px',
84     marginTop: '20px',
85     transition: 'transform 0.2s ease, background-color 0.2s ease'
86   }}
87   onMouseEnter={e => (e.currentTarget.style.transform = 'scale(1.05)'')}
88   onMouseLeave={e => (e.currentTarget.style.transform = 'scale(1)'')}
89 >
90   Kembali ke Kampus Digital
91 </button>
92
93 {/* Inline keyframes */}
94 <style>{`
95   @keyframes shine {
96     0% { transform: rotate(25deg) translateX(-100%); }
97     50% { transform: rotate(25deg) translateX(100%); }
98     100% { transform: rotate(25deg) translateX(-100%); }
99   }
100   @keyframes fadeIn { from { opacity: 0; } to { opacity: 1; } }
101   @keyframes fadeInUp { from { opacity: 0; transform: translateY(20px); } to { opacity: 1; transform: translateY(0); } }
102   .badge:hover { transform: scale(1.1); box-shadow: 0 4px 12px rgba(0,0,0,0.4); }
103 `}</style>
104 </div>
105 </div>
106   );
107 }
108
109 export default Mission3_Complete;

```



```
1 import React, { useState } from 'react';
2 import { DndContext, DragOverlay, closestCenter, useDraggable } from '@dnd-kit/core';
3 import { SortableContext, verticalListSortingStrategy, useSortable } from '@dnd-kit/sortable';
4 import { CSS } from '@dnd-kit/utilities';
5
6 // ----- Sortable Item -----
7 function SortableItem({ id, content, source = 'list' }) {
8   const { attributes, listeners, setNodeRef, transform, transition } = useSortable({ id });
9   const isShape = source === 'shape';
10  const style = {
11    transform: CSS.Transform.toString(transform),
12    transition,
13    marginBottom: isShape ? '0' : '10px',
14    fontSize: isShape ? '0.8rem' : '1rem',
15    padding: isShape ? '4px' : '10px',
16    width: '100%',
17    display: 'flex',
18    justifyContent: 'center',
19    alignItems: 'center',
20    touchAction: 'none',
21    cursor: 'grab',
22
23    // NEW: wrap text inside shapes without stretching the shape
24    whiteSpace: 'normal', // allow line breaks
25    wordBreak: 'break-word', // break long words
26    overflowWrap: 'break-word',
27    textAlign: 'center',
28  };
29
30  return (
31    <div
32      ref={setNodeRef}
33      style={style}
34      {...attributes}
35      {...listeners}
36      className={`dnd-item flowchart-shape-${isShape ? 'rectangle' : 'list'}`}
37    >
38      {content}
39    </div>
40  );
41 }
42
43 // ----- Drag Overlay -----
44 function DraggableItem({ content }) {
45   return (
46     <div
47       className="dnd-item-overlay flowchart-shape-rectangle"
48       style={{
49         boxShadow: '0 5px 15px rgba(0,0,0,0.25)',
50         fontSize: '0.9rem',
51         padding: '10px',
52       }}
53     >
54       {content}
55     </div>
56   );
57 }
58
59 // ----- Droppable Flowchart Shape -----
60 function FlowchartShape({ id, shapeType, items }) {
61   const { setNodeRef, isOver } = useDraggable({ id });
62
63   let shapeClass = 'flowchart-shape-rectangle';
64   if (shapeType === 'oval') shapeClass = 'flowchart-shape-oval';
65   else if (shapeType === 'diamond') shapeClass = 'flowchart-shape-diamond';
66   else if (shapeType === 'parallelogram') shapeClass = 'flowchart-shape-parallelogram';
67 }
```

```

68     return (
69       <div
70         ref={setNodeRef}
71         className={`flowchart-shape ${shapeClass}`}
72         style={{
73           margin: '5px',
74           padding: '4px',
75           width: '140px',           // fixed width
76           height: '50px',          // fixed height
77           borderColor: '#00ffff',
78           backgroundColor: 'rgba(0,0,0,0.1)',
79           borderWidth: '2px',
80           borderStyle: 'solid',
81           outline: isOver ? '2px dashed #00ffff' : 'none',
82           position: 'relative',
83           display: 'flex',
84           flexDirection: 'column',
85           alignItems: 'center',
86           justifyContent: 'center',
87           textAlign: 'center',
88         }}
89       >
90         <SortableContext items={items.map((i) => i.id)} strategy={verticalListSortingStrategy}>
91           {items.length > 0 ? (
92             <SortableItem
93               id={items[0].id}
94               content={items[0].content}
95               source="shape"
96               style={{
97                 fontSize: '0.7rem',           // smaller font
98                 whiteSpace: 'normal',         // allow line breaks
99                 wordBreak: 'break-word',      // break long words
100                overflow: 'hidden',           // prevent overflow
101                textAlign: 'center',
102              }}
103             />
104           ) : (
105             <span style={{
106               opacity: 0.3,
107               fontSize: '0.7rem',
108               pointerEvents: 'none',
109               textAlign: 'center',
110               whiteSpace: 'normal',
111               wordBreak: 'break-word'
112             }}
113             >
114               Seret sini
115             </span>
116           )}
117         </SortableContext>
118       </div>
119     );
120   }
121 }
122
123 // ----- Main Component -----
124 function Mission3_Pembinaan({ onContinue, setRobotText, onBadgeEarned, onFeedback }) {
125   const initialItems = {
126     available: [
127       { id: 's1', content: 'Mula' },
128       { id: 's2', content: 'Masukkan No Pendaftaran' },
129       { id: 's3', content: 'Masukkan Jenis Yuran, Jumlah Yuran dan Jumlah Bayaran' },
130       { id: 's4', content: 'Kira Baki= Jumlah Yuran - Jumlah Bayaran' },
131       { id: 's5', content: 'Jika Baki = 0' },
132       { id: 's6', content: 'Status Bayaran = Lunas' },
133       { id: 's7', content: 'Papar No Pendaftaran, Status Bayaran, Baki Yuran' },
134       { id: 's8', content: 'Ulang pelajar seterusnya?' },
135       { id: 's9', content: 'Tamat' },
136       { id: 's10', content: 'Status Bayaran = Belum Lunas' },
137     ],
138     oval1: [], parallelogram1: [], parallelogram2: [], rectangle1: [],
139     diamond1: [], rectangle2: [], parallelogram3: [], diamond2: [], oval2: [], rectangle3: []
140   };
141
142   const [items, setItems] = useState(initialItems);
143   const [activeId, setActiveId] = useState(null);
144   const [isCorrect, setIsCorrect] = useState(false);

```

```

145
146 // Backend Integration State
147 const [earnedScore, setEarnedScore] = useState(0);
148 const [attempts, setAttempts] = useState(0);
149 const [isSubmitting, setIsSubmitting] = useState(false);
150
151 function findContainer(itemId) {
152   if (!itemId) return null;
153   if (Object.prototype.hasOwnProperty.call(items, itemId)) return itemId;
154   for (const container of Object.keys(items)) {
155     if (items[container].find((item) => item.id === itemId)) return container;
156   }
157   return null;
158 }
159
160 const handleDragStart = (event) => setActiveId(event.active.id);
161
162 const handleDragEnd = (event) => {
163   const { active, over } = event;
164   if (!over) { setActiveId(null); return; }
165
166   const activeContainer = findContainer(active.id);
167   const overContainer = findContainer(over.id);
168
169   if (!activeContainer || !overContainer) { setActiveId(null); return; }
170   if (activeContainer === overContainer && active.id === over.id) { setActiveId(null); return; }
171
172   // Reset correct state on drag
173   setIsCorrect(false);
174
175   setItems(prev => {
176     const newState = { ...prev };
177     const activeItems = [...prev[activeContainer]];
178     const overItems = [...prev[overContainer]];
179     const activeIndex = activeItems.findIndex(i => i.id === active.id);
180     if (activeIndex === -1) return prev;
181     const [movedItem] = activeItems.splice(activeIndex, 1);
182
183     // All shapes accept only 1 item
184     if (overItems.length === 0) newState[overContainer] = [movedItem];
185     else activeItems.splice(activeIndex, 0, movedItem); // reject drop
186
187     newState[activeContainer] = activeItems;
188     return newState;
189   });
190
191   setActiveId(null);
192 };
193
194 const handleReset = () => {
195   setItems(initialItems);
196   setIsCorrect(false);
197   setAttempts(0);
198   onFeedback?.('🔄 Carta alir telah direset. Cuba susun semula langkah yang betul.', null);
199 };
200
201 const checkAnswer = async () => {
202   const correctOrder = {
203     oval1: 'Mula',
204     parallelogram1: 'Masukkan No Pendaftaran',
205     parallelogram2: 'Masukkan Jenis Yuran, Jumlah Yuran dan Jumlah Bayaran',
206     rectangle1: 'Kira Baki= Jumlah Yuran - Jumlah Bayaran',
207     diamond1: 'Jika Baki = 0',
208     rectangle2: 'Status Bayaran = Lunas',
209     parallelogram3: 'Papar No Pendaftaran, Status Bayaran, Baki Yuran',
210     diamond2: 'Ulang pelajar seterusnya?',
211     oval2: 'Tamat',
212     rectangle3: 'Status Bayaran = Belum Lunas'
213   };
214
215   const isRight = Object.entries(correctOrder).every(([slot, text]) =>
216     items[slot][0]?.content === text
217   );
218
219   if (!isRight) {
220     setAttempts(prev => prev + 1);
221     setIsCorrect(false);

```

```

222     onFeedback?.('✗ Jawaban Salah (Susunan atau Formula Salah) "Masih ada kesalahan logik! (-5 Markah
223     return;
224 }
225
226 // Correct: Calculate Score
227 const calculatedScore = Math.max(5, 25 - (attempts * 5));
228
229 setIsSubmitting(true);
230 const token = localStorage.getItem('token');
231
232 try {
233     // Call Backend API
234     const response = await fetch('https://algoquest-api.onrender.com/api/mission/submit', {
235         method: 'POST',
236         headers: {
237             'Content-Type': 'application/json',
238             'Authorization': `Bearer ${token}`
239         },
240         body: JSON.stringify({
241             mission: 3,
242             phase: 'pembinaan',
243             isCorrect: true,
244             score: calculatedScore,
245             // Badge awarded if correct on first try
246             badge: attempts === 0 ? 'Master Algoritma' : null
247         })
248     });
249
250     const data = await response.json();
251
252     setEarnedScore(calculatedScore);
253     setIsCorrect(true);
254
255     let badgeMsg = '';
256     if (attempts === 0) {
257         badgeMsg = '\n\n🏆 Anda telah memperoleh lencana "Master Algoritma".';
258         onBadgeEarned?.('Master Algoritma');
259     }
260
261     onFeedback?.('✅ Jawaban Betul! "Hebat! Struktur algoritma kewangan kamu lengkap.${badgeMsg}`, 3000,
262
263     } catch (err) {
264         console.error("Error submitting:", err);
265         if (onFeedback) onFeedback('⚠️ Ralat menghubungi pelayan.', 3000, false);
266     } finally {
267         setIsSubmitting(false);
268     }
269 };
270
271 const handleNext = () => {
272     if (!isCorrect) return;
273     // Pass score AND badge name (if earned) to parent
274     const badge = attempts === 0 ? 'Master Algoritma' : null;
275     onContinue?.(earnedScore, badge);
276 };
277
278 const allItems = Object.values(items).flat();
279 const activeItem = activeId ? allItems.find(i => i.id === activeId) : null;
280
281 return (
282     <div>
283         <h3>TAHAP 3: PEMBINAAN ALGORITMA</h3>
284         <p>Sistem Kewangan Kampus Digital kini sedang membina algoritma lengkap untuk memproses bayaran yuran
285         Bantu sistem menyusun langkah pseudokod dalam urutan yang betul dan padankan setiap langkah dengan simbol </p>
286     </div>
287     <hr />
288
289     <DndContext onDragStart={handleDragStart} onDragEnd={handleDragEnd} collisionDetection={closestCenter>
290         <div style={{ display: 'flex', gap: '100px', justifyContent: 'center', alignItems: 'stretch', margin:
291             /* Pseudocode */
292             <div style={{ width: '320px', backgroundColor: 'rgba(255,255,255,0.05)', padding: '15px', border:
293                 <h4 style={{ marginTop: 10, textAlign: 'center', color: '#00ffff' }}>PSEUDOKOD</h4>
294                 <SortableContext items={items.available.map(i => i.id)} strategy={verticalListSortingStrategy}>
295                     <div style={{ flexGrow: 1 }}>
296                         {items.available.map(item => <SortableItem key={item.id} id={item.id} content={item.content}
297                     </div>
298                 </SortableContext>

```

```

299     </div>
300
301     { /* Flowchart */}
302 <div
303   style={{
304     display: 'flex',
305     flexDirection: 'column',
306     justifyContent: 'center',
307     position: 'relative',
308     transform: 'scale(0.9)',
309     transformOrigin: 'top center',
310   }}
311 >
312   <h4 style={{ marginTop: 25, textAlign: 'center', color: '#00ffff' }}>
313     CARTA ALIR
314   </h4>
315   <div
316     style={{
317       display: 'flex',
318       flexDirection: 'column',
319       alignItems: 'center',
320       position: 'relative',
321     }}
322   >
323     <FlowchartShape id="oval1" shapeType="oval" items={items.oval1} />
324     <div style={{ fontSize: '0.8rem' }}></div>
325
326     <FlowchartShape id="parallelogram1" shapeType="parallelogram" items={items.parallelogram1} />
327     <div style={{ fontSize: '0.8rem' }}></div>
328
329     <FlowchartShape id="parallelogram2" shapeType="parallelogram" items={items.parallelogram2} />
330     <div style={{ fontSize: '0.8rem' }}></div>
331
332     <FlowchartShape id="rectangle1" shapeType="rectangle" items={items.rectangle1} />
333     <div style={{ fontSize: '0.8rem' }}></div>
334
335     <FlowchartShape id="diamond1" shapeType="diamond" items={items.diamond1} />
336     <div style={{ fontSize: '0.8rem' }}>Ya</div>
337
338     { /* Rectangle2 & Rectangle3 */}
339     <div style={{ position: 'relative', width: '100%', display: 'flex', justifyContent: 'center' }}>
340       <FlowchartShape id="rectangle2" shapeType="rectangle" items={items.rectangle2} />
341       <div style={{ position: 'absolute', left: '50%', transform: 'translateX(100px) translateY(0)' }}>
342         <FlowchartShape id="rectangle3" shapeType="rectangle" items={items.rectangle3} />
343       </div>
344     </div>
345
346     <div style={{ fontSize: '0.8rem' }}></div>
347     <FlowchartShape id="parallelogram3" shapeType="parallelogram" items={items.parallelogram3} />
348     <div style={{ fontSize: '0.8rem' }}></div>
349
350     <FlowchartShape id="diamond2" shapeType="diamond" items={items.diamond2} />
351     <div style={{ fontSize: '0.8rem' }}>Tidak</div>
352     <FlowchartShape id="oval2" shapeType="oval" items={items.oval2} />
353
354     { /* === Arrows === */}
355     {(() => {
356       // Base offset for tweaking entire arrow position
357       const arrowOffsetX = 100; // moves arrows horizontally
358       const arrowOffsetY = 200; // moves arrows vertically
359
360       // Arrow configurations (adjust top, horizontal, vertical as needed)
361       const arrows = {
362         arrow1: {
363           // Diamond1 → Rectangle3
364           top: 220 + arrowOffsetY,
365           horizontal: 100,
366           vertical: 50,
367           label: 'Tidak',
368           labelOffset: -15,
369         },
370         arrow2: {
371           // Rectangle3 → Parallelogram3
372           top: 370 + arrowOffsetY,
373           horizontal: 100,
374           vertical: 50,
375           label: '',

```

```

376   },
377   arrow3: {
378     // Diamond2 → Parallelogram1 (left/up/right)
379     top: 500 + arrowOffsetY,
380     horizontal: 20,
381     vertical: 580,
382     label: 'Ya',
383     labelOffset: -15,
384   },
385 };
386
387 // Arrowhead size
388 const arrowHeadSize = 6;
389
390 // Helper for arrowhead
391 const ArrowHead = ({ top, left, direction }) => {
392   const rotation = {
393     down: '0deg',
394     up: '180deg',
395     left: '-90deg',
396     right: '90deg',
397   }[direction];
398   return (
399     <div
400       style={{
401         position: 'absolute',
402         top: `${top}px`,
403         left: `${left}px`,
404         width: 0,
405         height: 0,
406         borderLeft: `${arrowHeadSize}px solid transparent`,
407         borderRight: `${arrowHeadSize}px solid transparent`,
408         borderBottom: `${arrowHeadSize}px solid #ffffff`,
409         transform: `rotate(${rotation})`,
410       }}
411     />
412   );
413 };
414
415 return (
416   <>
417     { /* Arrow 1: Diamond1 → Rectangle3 */ }
418     <div
419       style={{
420         position: 'absolute',
421         top: `${arrows.arrow1.top}px`,
422         left: `calc(50% + ${arrowOffsetX}px)`,
423         width: `${arrows.arrow1.horizontal}px`,
424         height: '1px',
425         backgroundColor: '#ffffff',
426       }}
427     />
428     <div
429       style={{
430         position: 'absolute',
431         top: `${arrows.arrow1.top}px`,
432         left: `calc(50% + ${arrowOffsetX} + arrows.arrow1.horizontal}px)`,
433         width: '1px',
434         height: `${arrows.arrow1.vertical}px`,
435         backgroundColor: '#ffffff',
436       }}
437     />
438     <ArrowHead
439       top={arrows.arrow1.top + arrows.arrow1.vertical}
440       left={70 + arrowOffsetX + arrows.arrow1.horizontal} // 50% base
441       direction="up"
442     />
443     <span
444       style={{
445         position: 'absolute',
446         top: `${arrows.arrow1.top + arrows.arrow1.labelOffset}px`,
447         left: `calc(50% + ${arrowOffsetX} + arrows.arrow1.horizontal / 4}px)`,
448         fontSize: '0.7rem',
449         color: '#ffffff',
450       }}
451     >
452       {arrows.arrow1.label}

```

```

453 </span>
454
455 {/* Arrow 2: Rectangle3 → Parallelogram3 */}
456 <div
457   style={{
458     position: 'absolute',
459     top: `${arrows.arrow2.top}px`,
460     left: `calc(50% + ${arrowOffsetX + arrows.arrow2.horizontal}px)`,
461     width: '1px',
462     height: `${arrows.arrow2.vertical}px`,
463     backgroundColor: 'rgba(0,0,0,0.1)',
464   }}
465 >
466 <div
467   style={{
468     position: 'absolute',
469     top: `${arrows.arrow2.top + arrows.arrow2.vertical}px`,
470     left: `calc(50% + ${arrowOffsetX}px)`,
471     width: `${arrows.arrow2.horizontal}px`,
472     height: '1px',
473     backgroundColor: 'rgba(0,0,0,0.1)',
474   }}
475 >
476 <ArrowHead
477   top={-2 + arrows.arrow2.top + arrows.arrow2.vertical}
478   left={70 + arrowOffsetX} // end of horizontal line
479   direction="left"
480 >
481
482 {/* Arrow 3: Diamond2 → Parallelogram1 (left/up/right) */}
483 <div
484   style={{
485     position: 'absolute',
486     top: `${arrows.arrow3.top}px`,
487     left: `calc(50% - ${arrowOffsetX}px)`,
488     width: `${arrows.arrow3.horizontal}px`,
489     height: '1px',
490     backgroundColor: 'rgba(0,0,0,0.1)',
491   }}
492 >
493 <div
494   style={{
495     position: 'absolute',
496     top: `${arrows.arrow3.top - arrows.arrow3.vertical}px`,
497     left: `calc(50% - ${arrowOffsetX}px)`,
498     width: '1px',
499     height: `${arrows.arrow3.vertical}px`,
500     backgroundColor: 'rgba(0,0,0,0.1)',
501   }}
502 >
503 <div
504   style={{
505     position: 'absolute',
506     top: `${arrows.arrow3.top - arrows.arrow3.vertical}px`,
507     left: `calc(38% - ${arrowOffsetX - arrows.arrow3.horizontal}px)`,
508     width: `${arrows.arrow3.horizontal}px`,
509     height: '1px',
510     backgroundColor: 'rgba(0,0,0,0.1)',
511   }}
512 >
513 <ArrowHead
514   top={-2 + arrows.arrow3.top - arrows.arrow3.vertical}
515   left={70 - arrowOffsetX + arrows.arrow3.horizontal} // adjust for horizontal end
516   direction="right"
517 >
518 <span
519   style={{
520     position: 'absolute',
521     top: `${arrows.arrow3.top + arrows.arrow3.labelOffset}px`,
522     left: `calc(50% - ${arrowOffsetX - arrows.arrow3.horizontal / 2}px)`,
523     fontSize: '0.7rem',
524     color: 'black',
525   }}
526 >
527   {arrows.arrow3.label}
528 </span>
529 </div>

```

```

530     );
531   })()}
532
533   </div>
534 </div>
535
536
537   </div>
538
539   <DragOverlay>{activeItem && <DraggableItem content={activeItem.content} />}</DragOverlay>
540 </DndContext>
541
542   <hr />
543   <div style={{ display:'flex', justifyContent:'flex-end', gap:'10px', marginTop:'10px' }}>
544     <button onClick={handleReset} className="primary-button" disabled={isSubmitting}>Buat Semula</button>
545     <button onClick={checkAnswer} className="primary-button" disabled={isSubmitting || isCorrect}>
546       {isSubmitting ? 'Menghantar...' : 'Semak Jawapan'}
547   </button>
548   <button
549     onClick={handleNext}
550     className="primary-button"
551     style={{ backgroundColor: isCorrect ? '#2ecc71' : '#999', opacity: isCorrect ? 1 : 0.5, cursor: isCorrect ? 'default' : 'not-allowed' }}
552     disabled={!isCorrect}
553   >
554     Seterusnya
555   </button>
556 </div>
557 </div>
558   );
559 }
560
561 export default Mission3_Pembinaan;

```



```

1  import React, { useState } from 'react';
2  import { DndContext, DragOverlay, closestCenter } from '@dnd-kit/core';
3  import { SortableContext, arrayMove, verticalListSortingStrategy, useSortable } from '@dnd-kit/sortable';
4  import { useDroppable } from '@dnd-kit/core';
5  import { CSS } from '@dnd-kit/utilities';
6
7  // --- Reusable Sortable Item ---
8  function SortableItem({ id, content }) {
9    const { attributes, listeners, setNodeRef, transform, transition } = useSortable({ id });
10   const style = { transform: CSS.Transform.toString(transform), transition };
11   return (
12     <div
13       ref={setNodeRef}
14       style={style}
15       {...attributes}
16       {...listeners}
17       className="dnd-item"
18     >
19       {content}
20     </div>
21   );
22 }
23
24 // --- Draggable Overlay Item ---
25 function DraggableItem({ content }) {
26   return <div className="dnd-item-overlay">{content}</div>;
27 }
28
29 // --- Droppable Column ---
30 function DroppableSortableColumn({ id, title, items }) {
31   const { setNodeRef } = useDroppable({ id });
32
33   return (
34     <div ref={setNodeRef} className={`dnd-column-split ${id.startsWith('available') ? 'available' : ''}`>
35       <h4>{title.toUpperCase()}</h4>
36       <SortableContext items={items.map(i => i.id)} strategy={verticalListSortingStrategy}>
37         {items.map(item => (
38           <SortableItem key={item.id} id={item.id} content={item.content} />
39         ))}
40       </SortableContext>
41     </div>
42   );
43 }
44
45 // --- Initial States ---
46 const INITIAL_DATA = {
47   available_data: [
48     { id: 'd-1', content: 'Nama Pelajar' },
49     { id: 'd-2', content: 'No Pendaftaran' },
50     { id: 'd-3', content: 'Jenis Yuran (Pengajian, Asrama, Aktiviti)' },
51     { id: 'd-4', content: 'Jumlah Yuran' },
52     { id: 'd-5', content: 'Kaedah Bayaran (FPX / Tunai)' },
53     { id: 'd-6', content: 'No Telefon' },
54     { id: 'd-7', content: 'Jumlah Bayaran' },
55     { id: 'd-8', content: 'Tarikh Bayaran' },
56     { id: 'd-9', content: 'Status Bayaran (Lunas / Belum Lunas)' },
57     { id: 'd-10', content: 'Nama Pegawai Kewangan' },
58   ],
59   penting: [],
60 };
61
62 const INITIAL_STEPS = {
63   available_langkah: [
64     { id: '1-1', content: 'Masukkan Jumlah Bayaran' },
65     { id: '1-2', content: 'Papar Baki Yuran' },
66     { id: '1-3', content: 'Masukkan Jumlah Yuran' },
67     { id: '1-4', content: 'Tentukan Status Bayaran (Lunas/Belum Lunas)' },

```

```

68     { id: '1-5', content: 'Masukkan No Pendaftaran' },
69     { id: '1-6', content: 'Kira Baki = Jumlah Yuran - Jumlah Bayaran' },
70 ],
71 ordered_langkah: [],
72 };
73
74 const clone = (obj) => JSON.parse(JSON.stringify(obj));
75
76 // --- Main Component ---
77 function Mission3_Pengabstrakan({ onContinue, onFeedback }) {
78     const [dataItems, setDataItems] = useState(() => clone(INITIAL_DATA));
79     const [stepItems, setStepItems] = useState(() => clone(INITIAL_STEPS));
80     const [activeId, setActiveId] = useState(null);
81     const [isCorrect, setIsCorrect] = useState(false);
82
83     // Scoring State
84     const [earnedScore, setEarnedScore] = useState(0);
85     const [attempts, setAttempts] = useState(0);
86     const [isSubmitting, setIsSubmitting] = useState(false);
87
88     function findContainer(itemId) {
89         if (!itemId) return null;
90         if (Object.keys(dataItems).includes(itemId)) return itemId;
91         if (Object.keys(stepItems).includes(itemId)) return itemId;
92         for (const container of Object.keys(dataItems)) if (dataItems[container].find(item => item.id === itemId)) return container;
93         for (const container of Object.keys(stepItems)) if (stepItems[container].find(item => item.id === itemId)) return container;
94         return null;
95     }
96
97     function handleDragStart(event) {
98         setActiveId(event.active.id);
99     }
100
101     function handleDragEnd(event) {
102         const { active, over } = event;
103         if (!over) { setActiveId(null); return; }
104
105         const activeContainer = findContainer(active.id);
106         const overContainer = findContainer(over.id);
107         if (!activeContainer || !overContainer) { setActiveId(null); return; }
108
109         setIsCorrect(false); // Reset state
110
111         if (Object.keys(dataItems).includes(activeContainer)) {
112             moveItem(dataItems, setDataItems, active, over);
113         } else {
114             moveItem(stepItems, setStepItems, active, over);
115         }
116
117         setActiveId(null);
118     }
119
120     function moveItem(items, setItems, active, over) {
121         const activeId = active.id;
122         const overId = over.id;
123         const activeContainer = findContainer(activeId);
124         const overContainer = findContainer(overId) || overId;
125
126         if (activeContainer !== overContainer) {
127             setItems(prev => {
128                 const activeItems = prev[activeContainer];
129                 const overItems = prev[overContainer];
130                 const index = activeItems.findIndex(item => item.id === activeId);
131                 const [moved] = activeItems.splice(index, 1);
132                 const overIndex = overItems.findIndex(item => item.id === overId);
133                 if (overIndex === -1) overItems.push(moved);
134                 else overItems.splice(overIndex, 0, moved);
135                 return { ...prev, [activeContainer]: [...activeItems], [overContainer]: [...overItems] };
136             });
137         } else {
138             setItems(prev => {
139                 const list = prev[activeContainer];
140                 const fromIndex = list.findIndex(i => i.id === activeId);
141                 const toIndex = list.findIndex(i => i.id === overId);
142                 return { ...prev, [activeContainer]: arrayMove(list, fromIndex, toIndex) };
143             });
144         }
145     }
146 }

```

```

145 }
146
147 // --- CHECK Answer ---
148 const checkAnswer = async () => {
149   // Based on slide 989 logic (implied) or your previous code:
150   // Correct Data: No Pendaftaran, Jenis Yuran, Jumlah Yuran, Jumlah Bayaran, Status Bayaran
151   const correctData = ['d-2', 'd-3', 'd-4', 'd-7', 'd-9'].sort();
152   const playerData = dataItems.penting.map(i => i.id).sort();
153
154   // Correct Steps Order: Input No -> Input Yuran -> Input Bayaran -> Kira Baki -> Tentukan Status -> Pap
155   const correctLangkah = ['1-5', '1-3', '1-1', '1-6', '1-4', '1-2'];
156   const playerLangkah = stepItems.ordered_langkah.map(i => i.id);
157
158   const ok =
159     JSON.stringify(correctData) === JSON.stringify(playerData) &&
160     JSON.stringify(correctLangkah) === JSON.stringify(playerLangkah);
161
162   if (!ok) {
163     setAttempts(prev => prev + 1);
164     setIsCorrect(false);
165     onFeedback('❌ Salah "Urutan ini belum logik. Fikir semula urutan sebenar dalam proses bayaran pel
166     return;
167   }
168
169   // Correct
170   const calculatedScore = Math.max(5, 25 - (attempts * 5));
171
172   setIsSubmitting(true);
173   const token = localStorage.getItem('token');
174
175   try {
176     await fetch('https://algoquest-api.onrender.com/api/mission/submit', {
177       method: 'POST',
178       headers: {
179         'Content-Type': 'application/json',
180         'Authorization': `Bearer ${token}`
181       },
182       body: JSON.stringify({
183         mission: 3,
184         phase: 'pengabstrakan',
185         isCorrect: true,
186         score: calculatedScore,
187         badge: null
188       })
189     });
190
191     setEarnedScore(calculatedScore);
192     setIsCorrect(true);
193     onFeedback('✅ "Hebat! Anda telah menyusun logik sistem kewangan Kampus Digital dengan tepat. Kini s
194
195   } catch (err) {
196     console.error("Error submitting:", err);
197     onFeedback('⚠️ Ralat menghubungi pelayan.', 3000, false);
198   } finally {
199     setIsSubmitting(false);
200   }
201 };
202
203 const handleReset = () => {
204   setDataItems(clone(INITIAL_DATA));
205   setStepItems(clone(INITIAL_STEPS));
206   setActiveId(null);
207   setIsCorrect(false);
208   setAttempts(0); // Reset attempts
209   onFeedback('🔄 Susunan telah direset. Cuba semula dengan teliti.', 2000, null);
210 };
211
212 const allItems = [...dataItems.available_data, ...dataItems.penting, ...stepItems.available_langkah, ...s
213 const activeItem = activeId ? allItems.find(i => i.id === activeId) : null;
214
215 return (
216   <div>
217     <h3>TAHAP 2: PENGABSTRAKAN</h3>
218     <p>
219       Sistem Kewangan Kampus Digital kini sedang dibangunkan untuk mengurus bayaran yuran pelajar seca
220     </p>
221     <hr />

```

```

222
223 <DndContext onDragStart={handleDragStart} onDragEnd={handleDragEnd} collisionDetection={closestCenter}
224 <div className="dnd-container-split">
225 <DraggableSortTableColumn id="available_data" title="SENARAI DATA" items={dataItems.available_data} />
226 <DraggableSortTableColumn id="penting" title="DATA PENTING" items={dataItems.penting} />
227 </div>
228 <div className="dnd-container-split">
229 <DraggableSortTableColumn id="available_langkah" title="LANGKAH MODEL (TIDAK TERSUSUN)" items={stepItems.available_langkah} />
230 <DraggableSortTableColumn id="ordered_langkah" title="MODEL LOGIK (TERSUSUN)" items={stepItems.ordered_langkah} />
231 </div>
232 <DragOverlay>
233 {activeItem} && <DraggableItem content={activeItem.content} />
234 </DragOverlay>
235 </DndContext>
236
237 <hr />
238
239 <div style={{ display: 'flex', justifyContent: 'flex-end', gap: '10px', marginTop: '10px' }}>
240 <button
241   onClick={handleReset}
242   className="primary-button"
243   disabled={isSubmitting}
244 >
245   Buat Semula
246 </button>
247
248 <button
249   onClick={checkAnswer}
250   className="primary-button"
251   disabled={isSubmitting}
252 >
253   Semak Jawapan
254 </button>
255
256 <button
257   onClick={isCorrect ? () => onContinue(earnedScore) : undefined}
258   className="primary-button"
259   style={{
260     backgroundColor: isCorrect ? '#2ecc71' : '#999',
261     cursor: isCorrect ? 'pointer' : 'not-allowed',
262   }}
263   disabled={!isCorrect}
264 >
265   Seterusnya
266 </button>
267 </div>
268 </div>
269 );
270 }
271
272 export default Mission3_Pengabstrakan;

```

```

1  import React, { useState, useEffect } from 'react';
2  import { DndContext, DragOverlay, closestCenter } from '@dnd-kit/core';
3  import {
4    SortableContext,
5    arrayMove,
6    verticalListSortingStrategy,
7    useSortable
8  } from '@dnd-kit/sortable';
9  import { useDroppable } from '@dnd-kit/core';
10 import { CSS } from '@dnd-kit/utilities';
11
12 // --- Reusable Sortable Item ---
13 function SortableItem({ id, content }) {
14   const { attributes, listeners, setNodeRef, transform, transition } = useSortable({ id });
15   const style = { transform: CSS.Transform.toString(transform), transition };
16   return (
17     <div ref={setNodeRef} {...attributes} {...listeners} style={style} className="dnd-item">
18       {content}
19     </div>
20   );
21 }
22
23 // --- Draggable Overlay Item ---
24 function DraggableItem({ content }) {
25   return <div className="dnd-item-overlay">{content}</div>;
26 }
27
28 // --- Droppable Column ---
29 function DroppableSortableColumn({ id, title, items }) {
30   const { setNodeRef } = useDroppable({ id });
31   return (
32     <div ref={setNodeRef} className={`dnd-column ${id === 'available' ? 'available' : ''}`>
33       <h4>{title.toUpperCase()}</h4>
34       <SortableContext items={items.map(i => i.id)} strategy={verticalListSortingStrategy}>
35         {items.map(item => (
36           <SortableItem key={item.id} id={item.id} content={item.content} />
37         ))}
38       </SortableContext>
39     </div>
40   );
41 }
42
43 // --- Initial Items for "Student Fees" Scenario ---
44 const INITIAL_ITEMS = {
45   available: [
46     { id: 'item-1', content: 'Tentukan Status Bayaran (Lunas / Belum Lunas)' },
47     { id: 'item-2', content: 'Jenis Yuran (Pengajian, Asrama, Aktiviti)' },
48     { id: 'item-3', content: 'Kira Jumlah Yuran - Jumlah Bayaran' },
49     { id: 'item-4', content: 'Cetak Resit Pembayaran' },
50     { id: 'item-5', content: 'Jumlah Bayaran' },
51     { id: 'item-6', content: 'No Pendaftaran' },
52     { id: 'item-7', content: 'Papar Baki Yuran' },
53     { id: 'item-8', content: 'Jumlah Yuran' },
54     { id: 'item-9', content: 'Status Bayaran (Lunas/Belum Lunas)' },
55     { id: 'item-10', content: 'Ulang setiap pelajar' },
56     { id: 'item-11', content: 'Kaedah Bayaran (FPX / Tunai)' },
57   ],
58   input: [],
59   proses: [],
60   output: [],
61 };
62
63 const clone = (obj) => JSON.parse(JSON.stringify(obj));
64
65 // --- Mission 3 Penguraian ---
66 function Mission3_Penguraian({ onContinue, setRobotText, onFeedback, isCorrect: propIsCorrect }) {
67   const [items, setItems] = useState(clone(INITIAL_ITEMS));

```

```

68   const [activeId, setActiveId] = useState(null);
69   const [isCorrect, setIsCorrect] = useState(false);
70
71   // Scoring State
72   const [earnedScore, setEarnedScore] = useState(0);
73   const [attempts, setAttempts] = useState(0);
74   const [isSubmitting, setIsSubmitting] = useState(false);
75
76   useEffect(() => {
77     if (setRobotText) {
78       setRobotText('Kenal pasti Input, Proses, dan Output untuk sistem yuran pelajar.');
```

```

79     }
80   }, [setRobotText]);
81
82   const findContainer = (itemId) => {
83     if (!itemId) return null;
84     for (const container of Object.keys(items)) {
85       if (items[container].find(i => i.id === itemId)) return container;
86     }
87     return null;
88   };
89
90   const handleDragStart = (event) => setActiveId(event.active.id);
91
92   const handleDragEnd = (event) => {
93     const { active, over } = event;
94     if (!over) { setActiveId(null); return; }
95     const activeContainer = findContainer(active.id);
96     const overContainer = findContainer(over.id) || over.id;
97     if (!activeContainer || !overContainer) { setActiveId(null); return; }
98
99     // Reset correctness state on any drag
100    setIsCorrect(false);
101
102    if (activeContainer === overContainer) {
103      setItems(prev => {
104        const list = [...prev[activeContainer]];
105        const from = list.findIndex(i => i.id === active.id);
106        const to = list.findIndex(i => i.id === over.id);
107        if (from === -1 || to === -1) return prev;
108        return { ...prev, [activeContainer]: arrayMove(list, from, to) };
109      });
110    } else {
111      setItems(prev => {
112        const source = [...prev[activeContainer]];
113        const target = [...prev[overContainer]];
114        const moved = source.find(i => i.id === active.id);
115        if (!moved) return prev;
116        const newSource = source.filter(i => i.id !== active.id);
117        const overIndex = target.findIndex(i => i.id === over.id);
118        if (overIndex === -1) target.push(moved);
119        else target.splice(overIndex, 0, moved);
120        return { ...prev, [activeContainer]: newSource, [overContainer]: target };
121      });
122    }
123
124    setActiveId(null);
125  };
126
127  // --- CHECK Answer & SCORE ---
128  const checkAnswer = async () => {
129    // Correct answers for Student Fees
130    const correctInput = ['item-6', 'item-2', 'item-8', 'item-5', 'item-11'].sort();
131    const correctProses = ['item-3', 'item-1', 'item-10'].sort();
132    const correctOutput = ['item-7', 'item-9', 'item-4'].sort();
133
134    const playerInput = items.input.map(i => i.id).sort();
135    const playerProses = items.proses.map(i => i.id).sort();
136    const playerOutput = items.output.map(i => i.id).sort();
137
138    const ok =
139      JSON.stringify(playerInput) === JSON.stringify(correctInput) &&
140      JSON.stringify(playerProses) === JSON.stringify(correctProses) &&
141      JSON.stringify(playerOutput) === JSON.stringify(correctOutput);
142
143    if (!ok) {
144      setAttempts(prev => prev + 1);

```

```

145     setIsCorrect(false);
146     if (onFeedback) onFeedback('⚠️ Semak semula. Semua data pelajar seperti nombor pendaftaran adalah
147     return;
148 }
149
150 // Correct: Calculate Score
151 const calculatedScore = Math.max(5, 25 - (attempts * 5));
152
153 setIsSubmitting(true);
154 const token = localStorage.getItem('token');
155
156 try {
157     const response = await fetch('https://algoquest-api.onrender.com/api/mission/submit', {
158       method: 'POST',
159       headers: {
160         'Content-Type': 'application/json',
161         'Authorization': `Bearer ${token}`
162       },
163       body: JSON.stringify({
164         mission: 3,
165         phase: 'penguraian',
166         isCorrect: true,
167         score: calculatedScore,
168         badge: null
169       })
170     });
171
172     setEarnedScore(calculatedScore);
173     setIsCorrect(true);
174     if (onFeedback) onFeedback('✅ Bagus! Anda telah mengenal pasti komponen utama sistem kewangan.', 3000, null);
175
176   } catch (err) {
177     console.error("Error submitting:", err);
178     if (onFeedback) onFeedback('⚠️ Ralat menghubungi pelayan.', 3000, false);
179   } finally {
180     setIsSubmitting(false);
181   }
182 };
183
184 const handleReset = () => {
185   setItems(clone(INITIAL_ITEMS));
186   setActiveId(null);
187   setIsCorrect(false);
188   setAttempts(0);
189   if (onFeedback) onFeedback('Susunan telah direset. Cuba semula dengan teliti.', 2000, null);
190 };
191
192 const handleNext = () => {
193   if (!isCorrect) return;
194   onContinue(earnedScore);
195 };
196
197 const activeItem = activeId
198   ? [...items.available, ...items.input, ...items.proses, ...items.output].find(i => i.id === activeId)
199   : null;
200
201 return (
202   <div>
203     <h3>TAHAP 1: PENGURAIAN</h3>
204     <p>
205       Dalam Sistem Kewangan Kampus Digital, setiap pelajar perlu membayar beberapa jenis yuran seperti pe
206
207     </p>
208     <hr />
209     <DndContext onDragStart={handleDragStart} onDragEnd={handleDragEnd} collisionDetection={closestCenter}>
210       <div className="dnd-container">
211         {Object.keys(items).map(id => (
212           <DraggableSortColumn key={id} id={id} title={id === 'available' ? 'Item Tersedia' : id} ite
213         ))}
214       </div>
215       <DragOverlay>
216         {activeItem && <DraggableItem content={activeItem.content} />}
217       </DragOverlay>
218     </DndContext>
219     <hr />
220
221     <div style={{ display: 'flex', justifyContent: 'flex-end', gap: '10px', marginTop: '10px' }}>

```

```

222     <button
223         onClick={handleReset}
224         className="primary-button"
225         disabled={isSubmitting}
226     >
227         Buat Semula
228     </button>
229
230     <button
231         onClick={checkAnswer}
232         className="primary-button"
233         disabled={isSubmitting}
234     >
235         Semak Jawapan
236     </button>
237
238     <button
239         onClick={isCorrect ? handleNext : undefined}
240         className="primary-button"
241         style={{
242             backgroundColor: isCorrect ? '#2ecc71' : '#999',
243             cursor: isCorrect ? 'pointer' : 'not-allowed',
244         }}
245         disabled={!isCorrect}
246     >
247         Seterusnya
248     </button>
249 </div>
250 </div>
251 );
252 }
253
254 export default Mission3_Penguraian;

```



```
1 import React, { useState, useEffect } from 'react';
2
3 function Mission3_Penyahpepijat({ onContinue, onFeedback, onBadgeEarned, setRobotText }) {
4   const [selectedAnswer, setSelectedAnswer] = useState(null);
5   const [isCorrect, setIsCorrect] = useState(false);
6   const correctAnswer = 'D';
7
8   // Backend Integration State
9   const [earnedScore, setEarnedScore] = useState(0);
10  const [attempts, setAttempts] = useState(0);
11  const [isSubmitting, setIsSubmitting] = useState(false);
12
13  // Set initial robot text when component loads
14  useEffect(() => {
15    if (setRobotText) {
16      setRobotText(
17        'Sistem Kewangan mengesan keputusan yang salah walaupun bayaran telah dibuat penuh. Bantu saya ken
18      );
19    }
20  }, [setRobotText]);
21
22  const checkAnswer = async () => {
23    if (!selectedAnswer) {
24      if (onFeedback) onFeedback('❌ Sila pilih satu jawapan dahulu.', 3000, false);
25      return;
26    }
27
28    const ok = selectedAnswer === correctAnswer;
29
30    if (!ok) {
31      setAttempts(prev => prev + 1);
32      setIsCorrect(false);
33      if (onFeedback) {
34        onFeedback('❌ Salah. Semak semula logik perbandingan dalam pseudokod. (-5 Markah)', 3000, false);
35      }
36      return;
37    }
38
39    // Correct: Calculate Score
40    const calculatedScore = Math.max(5, 25 - (attempts * 5));
41
42    setIsSubmitting(true);
43    const token = localStorage.getItem('token');
44
45    try {
46      // Call Backend API
47      const response = await fetch('https://algoquest-api.onrender.com/api/mission/submit', {
48        method: 'POST',
49        headers: {
50          'Content-Type': 'application/json',
51          'Authorization': `Bearer ${token}`
52        },
53        body: JSON.stringify({
54          mission: 3,
55          phase: 'penyahpepijat',
56          isCorrect: true,
57          score: calculatedScore,
58          // Badge awarded if correct on first try
59          badge: attempts === 0 ? 'Master Pemulih Logik' : null
60        })
61      });
62
63      const data = await response.json();
64      setEarnedScore(calculatedScore);
65      setIsCorrect(true);
66
67      let badgeMsg = '';
```

[illegible]

```

145         <td style={{ border: '1px solid #79f8f8', padding: '5px' }}>Status = Belum Lunas ✖</td>
146         <td style={{ border: '1px solid #79f8f8', padding: '5px' }}>Status = Lunas ✔</td>
147     </tr>
148     <tr>
149         <td style={{ border: '1px solid #79f8f8', padding: '5px' }}>Jumlah Yuran = 1500, Jumlah Bay
150         <td style={{ border: '1px solid #79f8f8', padding: '5px' }}>Status = Belum Lunas ✔</td>
151         <td style={{ border: '1px solid #79f8f8', padding: '5px' }}>Status = Belum Lunas ✔</td>
152     </tr>
153     <tr>
154         <td style={{ border: '1px solid #79f8f8', padding: '5px' }}>Jumlah Yuran = 1200, Jumlah Bay
155         <td style={{ border: '1px solid #79f8f8', padding: '5px' }}>Status = Belum Lunas ✖</td>
156         <td style={{ border: '1px solid #79f8f8', padding: '5px' }}>Status = Lunas ✔</td>
157     </tr>
158 </tbody>
159 </table>
160 </div>
161 </div>
162
163 <hr />
164 <h4>Apakah punca utama kesilapan logik berdasarkan pseudokod di atas?</h4>
165
166 <div style={{ display: 'flex', flexDirection: 'column', gap: '10px' }}>
167     <button
168         className={`choice-button ${selectedAnswer === 'A' ? 'selected' : ''}`}
169         onClick={() => { setSelectedAnswer('A'); setIsCorrect(false); }}
170     >
171         A. Formula baki tidak dikira dengan betul.
172     </button>
173     <button
174         className={`choice-button ${selectedAnswer === 'B' ? 'selected' : ''}`}
175         onClick={() => { setSelectedAnswer('B'); setIsCorrect(false); }}
176     >
177         B. Paparan output dilakukan sebelum proses penentuan status.
178     </button>
179     <button
180         className={`choice-button ${selectedAnswer === 'C' ? 'selected' : ''}`}
181         onClick={() => { setSelectedAnswer('C'); setIsCorrect(false); }}
182     >
183         C. Arahan “Cetak Status” sepatutnya diletakkan sebelum “Jika tidak”.
184     </button>
185     <button
186         className={`choice-button ${selectedAnswer === 'D' ? 'selected' : ''}`}
187         onClick={() => { setSelectedAnswer('D'); setIsCorrect(false); }}
188     >
189         D. Syarat logik tidak tepat — perlu guna “≥” untuk status lunas.
190     </button>
191 </div>
192
193 <hr />
194 <div style={{ display: 'flex', justifyContent: 'flex-end', gap: '10px' }}>
195     <button onClick={handleReset} className="primary-button" disabled={isSubmitting}>
196         Buat Semula
197     </button>
198
199     <button onClick={checkAnswer} className="primary-button" disabled={isSubmitting}>
200         Semak Jawapan
201     </button>
202
203     <button
204         onClick={isCorrect ? handleNext : undefined}
205         className="primary-button"
206         style={{
207             backgroundColor: isCorrect ? '#2ecc71' : '#999',
208             cursor: isCorrect ? 'pointer' : 'not-allowed',
209         }}
210         disabled={!isCorrect}
211     >
212         Hantar
213     </button>
214 </div>
215 </div>
216 </div>
217 </div>
218 </div>
219
220 export default Mission3_Penyahpepijat;

```



```
1 import React from 'react';
2
3 function MissionLayout({ children, robotText, isCorrect }) {
4
5   // ★ Dynamic glow style (static part)
6   const glowColor =
7     isCorrect === true
8     ? 'rgba(46, 204, 113, 0.9)' // Green
9     : isCorrect === false
10    ? 'rgba(231, 76, 60, 0.9)' // Red
11    : 'transparent';
12
13   return (
14     <div className="mission-layout">
15
16       <div className="robot-sidebar">
17
18         /* ★ Robot Glow with pulse animation */
19         
24
25         <div className="robot-chat-bubble">
26           <p>{robotText}</p>
27         </div>
28
29       </div>
30
31       <div className="mission-content">
32         {children}
33       </div>
34
35       /* ★ CSS injected directly for convenience */
36       <style>{`
37         .robot-img {
38           width: 200px;
39           transition: filter 0.2s ease;
40         }
41
42         /* GREEN GLOW */
43         .glow-green {
44           animation: pulse-green 2.5s ease-out forwards;
45         }
46
47         /* RED GLOW */
48         .glow-red {
49           animation: pulse-red 2.5s ease-out forwards;
50         }
51
52         /* ★ Pulse + fade animations */
53         @keyframes pulse-green {
54           0% {
55             filter: drop-shadow(0 0 0px rgba(46, 204, 113, 0));
56           }
57           40% {
58             filter: drop-shadow(0 0 40px rgba(46, 204, 113, 1));
59           }
60           100% {
61             filter: drop-shadow(0 0 0px rgba(46, 204, 113, 0));
62           }
63         }
64
65         @keyframes pulse-red {
66           0% {
67             filter: drop-shadow(0 0 0px rgba(231, 76, 60, 0));
68           }
```

```
69         40% {
70             filter: drop-shadow(0 0 40px rgba(231, 76, 60, 1));
71         }
72         100% {
73             filter: drop-shadow(0 0 0px rgba(231, 76, 60, 0));
74         }
75     }
76 }</style>
77 </div>
78 );
79 }
80
81 export default MissionLayout;
```

```
1 import React, { useState, useEffect } from 'react';
2 import { Link } from 'react-router-dom';
3
4 // Define audio object outside the component for efficiency
5 const hoverAudio = new Audio('/hover.mp3');
6 hoverAudio.volume = 0.4;
7
8 const playHoverSound = () => {
9   try {
10     hoverAudio.currentTime = 0;
11     hoverAudio.play();
12   } catch (e) { }
13 };
14
15 const INITIAL_MOTIVATION = "Sila cipta akaun anda. Jom pulihkan Kampus Digital!";
16 const SUCCESS_MESSAGE = "Akaun berjaya didaftarkan! Sila log masuk.";
17
18 function Register() {
19   const [name, setName] = useState('');
20   const [email, setEmail] = useState('');
21   const [password, setPassword] = useState('');
22   const [robotMessage, setRobotMessage] = useState(INITIAL_MOTIVATION);
23
24   useEffect(() => {
25     setRobotMessage(INITIAL_MOTIVATION);
26   }, []);
27
28   const handleSubmit = async (e) => {
29     e.preventDefault();
30     setRobotMessage("Memproses pendaftaran akaun..."); // Message while loading
31
32     try {
33       const response = await fetch('https://algoquest-api.onrender.com/api/register', {
34         method: 'POST',
35         headers: {
36           'Content-Type': 'application/json',
37         },
38         body: JSON.stringify({
39           name,
40           email,
41           password,
42         }),
43       });
44
45       const data = await response.json();
46
47       if (data.id) { // Check for successful registration by looking for user ID
48         // Show success message from the robot
49         setRobotMessage(SUCCESS_MESSAGE);
50
51         // Wait 2 seconds, then navigate
52         setTimeout(() => {
53           window.location.href = '/login';
54         }, 2000);
55       } else {
56         setRobotMessage("Pendaftaran gagal. Emel mungkin sudah wujud.");
57         setTimeout(() => setRobotMessage(INITIAL_MOTIVATION), 4000);
58       }
59     } catch (err) {
60       console.error(err);
61       setRobotMessage("Gagal menghubungi pelayan. Sila semak sambungan anda.");
62       setTimeout(() => setRobotMessage(INITIAL_MOTIVATION), 4000);
63     }
64   };
65
66   return (
```

```

69     <div className="auth-form">
70       {/* 1. Robot Chat Bubble Structure */}
71       <div className="robot-speech-container">
72         <div className="robot-speech-bubble">
73           {/* 2. Apply the simple CSS fade-in animation */}
74           <span className="message-reveal">{robotMessage}</span>
75         </div>
76       </div>
77
78       <h2>Cipta Akaun Baharu</h2>
79       <form onSubmit={handleSubmit}>
80         <div>
81           <label>Nama: </label>
82           <input type="text" placeholder="Masukkan nama anda" value={name} onChange={(e) => setName(e.target.value)} />
83         </div>
84         <div>
85           <label>Emel: </label>
86           <input type="email" placeholder="Masukkan emel anda" value={email} onChange={(e) => setEmail(e.target.value)} />
87         </div>
88         <div>
89           <label>Kata Laluan: </label>
90           <input type="password" placeholder="Masukkan kata laluan" value={password} onChange={(e) => setPassword(e.target.value)} />
91         </div>
92         <button type="submit" onMouseEnter={playHoverSound}>Daftar Akaun</button>
93       </form>
94       <p>Sudah ada akaun? <Link to="/login">Log masuk di sini</Link></p>
95     </div>
96   );
97 }
98
99 export default Register;

```



```

1  import React, { useState, useEffect } from 'react';
2
3  function RobotMessage({ message }) {
4      const [displayedText, setDisplayedText] = useState('');
5      const [isTyping, setIsTyping] = useState(true);
6
7      useEffect(() => {
8          if (!message) return;
9
10         setDisplayedText(''); // Reset text when new message arrives
11         setIsTyping(true);
12
13         const textLength = message.length;
14         const duration = Math.min(textLength * 50, 2500); // 50ms per character, max 2.5s
15
16         // 1. Set up the typing animation using CSS variables for keyframe steps
17         const typingStyle = document.createElement('style');
18         typingStyle.innerHTML = `
19             .typewriter {
20                 width: ${textLength}ch;
21                 animation: typing ${duration}ms steps(${textLength}, end), blink 0.5s step-end infinite;
22                 animation-fill-mode: forwards;
23                 border-right: 2px solid rgba(121, 248, 248, 0.7);
24             }
25             @keyframes typing {
26                 from { width: 0 }
27                 to { width: ${textLength}ch }
28             }
29         `;
30         document.head.appendChild(typingStyle);
31
32         // 2. Clear typing state after animation duration
33         const typingTimer = setTimeout(() => {
34             setIsTyping(false);
35         }, duration);
36
37         return () => {
38             clearTimeout(typingTimer);
39             document.head.removeChild(typingStyle);
40         };
41     }, [message]);
42
43     if (!message) return null; // Hide if no message is set
44
45     return (
46         <div className="robot-speech-container">
47             <div className="robot-speech-bubble">
48                 {/* 3. The typewriter class controls the animation */}
49                 <p className={isTyping ? 'typewriter' : ''} style={{ whiteSpace: 'normal', width: 'auto', border: '1px solid #ccc', padding: '5px 10px 5px 10px' }}>
50                     {message}
51                 </p>
52             </div>
53         </div>
54     );
55 }
56
57 export default RobotMessage;

```

```

1  import React from 'react';
2
3  // Added new props: isMuted and onToggleMute
4  function SettingsMenu({ onClose, onLogout, onReturnToHub, isMuted, onToggleMute }) {
5
6      // Simplifies the function call for returning to the Hub
7      const handleGoToHub = () => {
8          onClose();
9          // Calls the function from App.js which contains the navigation logic
10         if (onReturnToHub) onReturnToHub();
11     };
12
13     // Logout handler
14     const handleLogoutClick = () => {
15         onClose();
16         onLogout();
17     };
18
19     return (
20         <div className="settings-backdrop" onClick={onClose}>
21             { /* Modal Box */ }
22             <div className="settings-menu" onClick={(e) => e.stopPropagation()}>
23                 <h3 style={{ color: '#79f8f8', marginBottom: '20px' }}>TETAPAN</h3>
24
25                 { /* 1. SAMBUNG PERMAINAN / CLOSE */ }
26                 <button
27                     type="button"
28                     onClick={onClose}
29                     className="primary-button"
30                     style={{ backgroundColor: '#2ecc71' }}
31                 >
32                     SAMBUNG PERMAINAN
33                 </button>
34
35                 { /* 2. MUTE/UNMUTE MUSIC BUTTON (New) */ }
36                 <button
37                     type="button"
38                     onClick={onToggleMute}
39                     className="primary-button"
40                     style={{
41                         // Red background when muted, blue when unmuted/playing
42                         backgroundColor: isMuted ? '#cc4444' : '#1e90ff'
43                     }}
44                 >
45                     {isMuted ? '🔴 MUZIK : TIADA' : '🔵 MUZIK : ADA'}
46                 </button>
47
48                 { /* 4. LOG KELUAR */ }
49                 <button
50                     type="button"
51                     onClick={handleLogoutClick}
52                     className="primary-button"
53                     style={{ backgroundColor: '#ff6b6b' }}
54                 >
55                     LOG KELUAR
56                 </button>
57             </div>
58         </div>
59     );
60 }
61
62 export default SettingsMenu;

```

```

1  import React, { useState, useEffect } from 'react';
2
3  const SplashScreen = ({ onFinish }) => {
4    const [fadeOut, setFadeOut] = useState(false);
5    const [messageIndex, setMessageIndex] = useState(0);
6
7    const loadingMessages = [
8      "Menghubungkan ke pangkalan data...",
9      "Memuatkan profil pemain...",
10     "Sediakan modul latihan...",
11     "Menjana dunia Kampus Digital..."
12   ];
13
14   // Rotate messages
15   useEffect(() => {
16     const interval = setInterval(() => {
17       setMessageIndex((i) => (i + 1) % loadingMessages.length);
18     }, 1200);
19     return () => clearInterval(interval);
20   }, []);
21
22   // Handle fade-out then notify parent
23   const triggerFinish = () => {
24     setFadeOut(true);
25     setTimeout(() => onFinish && onFinish(), 800);
26   };
27
28   // Auto fade-out after 3 seconds
29   useEffect(() => {
30     const timer = setTimeout(() => triggerFinish(), 3000);
31     return () => clearTimeout(timer);
32   }, []);
33
34   return (
35     <div
36       style={{
37         ...containerStyle,
38         opacity: fadeOut ? 0 : 1,
39         transition: "opacity 0.8s ease-out"
40       }}
41       onClick={triggerFinish} // Tap anywhere to continue
42     >
43       <div style={particleContainerStyle}>
44         {Array.from({ length: 15 }).map((_, i) => (
45           <div key={i} className="particle" />
46         ))}
47       </div>
48
49       <div style={contentStyle}>
50         
55
56         <p style={subtitleStyle}>Kampus Digital Menanti Anda</p>
57
58         <div style={spinnerStyle}></div>
59
60         <p style={loadingTextStyle}>{loadingMessages[messageIndex]}</p>
61       </div>
62
63       <p style={versionStyle}>Versi 1.0.0</p>
64
65       { /* --- INTERNAL CSS --- */ }
66       <style>
67         {
68           @keyframes fadeIn {

```

```

69         from { opacity: 0; transform: translateY(20px); }
70         to { opacity: 1; transform: translateY(0); }
71     }
72
73     @keyframes spin {
74         0% { transform: rotate(0deg); }
75         100% { transform: rotate(360deg); }
76     }
77
78     @keyframes pulse {
79         0% { transform: scale(1); }
80         50% { transform: scale(1.03); }
81         100% { transform: scale(1); }
82     }
83
84     @keyframes shine {
85         0% { filter: drop-shadow(0 0 5px #f1c40f); }
86         50% { filter: drop-shadow(0 0 15px #ffe066); }
87         100% { filter: drop-shadow(0 0 5px #f1c40f); }
88     }
89
90     @keyframes bgMove {
91         0% { background-position: 0% 50%; }
92         100% { background-position: 100% 50%; }
93     }
94
95     @keyframes floatUp {
96         from { transform: translateY(0); opacity: 0.6; }
97         to { transform: translateY(-40px); opacity: 0; }
98     }
99
100     .particle {
101         position: absolute;
102         width: 4px;
103         height: 4px;
104         background: rgba(255,255,255,0.15);
105         border-radius: 50%;
106         animation: floatUp 4s infinite ease-in-out;
107     }
108
109     .particle:nth-child(odd) {
110         animation-duration: 5s;
111     }
112     `}
113     </style>
114     </div>
115     );
116 };
117
118 // -----
119 // STYLES
120 // -----
121
122 const containerStyle = {
123     position: "fixed",
124     top: 0,
125     left: 0,
126     width: "100vw",
127     height: "100vh",
128     background: "linear-gradient(135deg, #1a1a2e, #16213e, #1e2746)",
129     backgroundSize: "200% 200%",
130     animation: "bgMove 6s ease-in-out infinite alternate",
131     display: "flex",
132     justifyContent: "center",
133     alignItems: "center",
134     zIndex: 9999,
135     fontFamily: "'Segoe UI', Tahoma, Geneva, Verdana, sans-serif",
136     overflow: "hidden",
137     color: "white"
138 };
139
140 const particleContainerStyle = {
141     position: "absolute",
142     width: "100%",
143     height: "100%"
144 };
145

```

```

146   const contentStyle = {
147     textAlign: "center",
148     animation: "fadeIn 1.2s ease-out"
149   };
150
151   const logoImageStyle = {
152     width: "340px",
153     height: "auto",
154     marginBottom: "20px",
155     animation: "pulse 2s infinite ease-in-out, shine 3s infinite ease-in-out"
156   };
157
158   const subtitleStyle = {
159     fontSize: "1.1rem",
160     opacity: 0.85,
161     marginBottom: "25px"
162   };
163
164   const spinnerStyle = {
165     width: "40px",
166     height: "40px",
167     border: "4px solid rgba(255, 255, 255, 0.2)",
168     borderTop: "4px solid #f1c40f",
169     borderRadius: "50%",
170     margin: "0 auto",
171     animation: "spin 1s linear infinite"
172   };
173
174   const loadingTextStyle = {
175     marginTop: "15px",
176     fontSize: "0.9rem",
177     letterSpacing: "2px",
178     textTransform: "uppercase",
179     opacity: 0.7
180   };
181
182   const versionStyle = {
183     position: "absolute",
184     bottom: 20,
185     right: 20,
186     opacity: 0.4,
187     fontSize: "0.8rem"
188   };
189
190   export default SplashScreen;

```

```

1  import React from 'react';
2  import { playHoverSound } from '../audioUtils'; // 1. Import the universal hover function
3
4  function Welcome({ user, onContinue }) {
5
6      const userName = user ? user.name : 'Wira AlgoQuest';
7
8      return (
9          // 1. Swapped the order: Image is now first (on the left)
10         <div className="content-screen">
11
12             {/* Robot Image (now on the left) */}
13             <div className="content-image">
14                 
15             </div>
16
17             {/* Text Content (now on the right) */}
18             <div className="content-text">
19
20                 {/* 2. Added the new "welcome-highlight" block at the top */}
21                 <div className="welcome-highlight super-bold-large">
22                     <p>👋 Selamat datang ke AlgoQuest, <strong>{userName}</strong>!</p>
23                     {/* 3. Apply the override class */}
24                     <p className="normal-welcome-text">Mulakan pengembaraan interaktif untuk menguji dan mengasah Kema
25                 </div>
26
27                 <hr style={{borderColor: '#555'}} />
28
29                 {/* 4. Robot's introduction is now at the bottom */}
30                 <div className="white-text">
31                     <h2>🤖 Hai! Saya <span style={{color: '#79f8f8'}}>Robot Algo</span>, pembantu digital kampus ini.
32                     <p>Bersiap sedia melaksanakan misi simulasi tugas sebenar di kampus.</p>
33                     <p>🕒 Setiap misi akan menguji kemahiran DAAD anda!</p>
34                     <p>Ikuti saya untuk menyelesaikan misi DAAD dan capai lencana tertinggi!</p>
35                 </div>
36
37                 {/* 5. Apply onMouseEnter handler */}
38                 <button onClick={onContinue} onMouseEnter={playHoverSound} className="primary-button">Seterusnya</b
39             </div>
40
41         </div>
42     );
43 }
44
45 export default Welcome;

```

frontend\client\src\hooks\useDeviceType.js

13 lines

0.40 KB

JavaScript

```
1 import { useState, useEffect } from 'react';
2
3 export default function useDeviceType() {
4   const [isMobile, setIsMobile] = useState(window.innerWidth <= 768);
5
6   useEffect(() => {
7     const onResize = () => setIsMobile(window.innerWidth <= 768);
8     window.addEventListener('resize', onResize);
9     return () => window.removeEventListener('resize', onResize);
10  }, []);
11
12  return { isMobile };
13 }
```

```
1  /* Global Reset */
2  * {
3    box-sizing: border-box; /* Makes layout math more intuitive */
4  }
5
6  html, body, #root {
7    margin: 0;
8    padding: 0;
9    width: 100%;
10   height: 100%;
11   font-family: -apple-system, BlinkMacSystemFont, 'Segoe UI', 'Roboto', 'Oxygen',
12     'Ubuntu', 'Cantarell', 'Fira Sans', 'Droid Sans', 'Helvetica Neue',
13     sans-serif;
14   -webkit-font-smoothing: antialiased;
15   -moz-osx-font-smoothing: grayscale;
16 }
```


frontend\client\src\index.js

13 lines

0.37 KB

JavaScript

```
1  import React from 'react';
2  import ReactDOM from 'react-dom/client';
3  import './index.css';
4  import App from './App';
5  import { BrowserRouter } from 'react-router-dom'; // 1. Import BrowserRouter
6
7  const root = ReactDOM.createRoot(document.getElementById('root'));
8  root.render(
9    // 2. Wrap your App component in <BrowserRouter>
10   <BrowserRouter>
11     <App />
12   </BrowserRouter>
13 );
```